

# A Controlled Derivation of the Full Conservative Second Post-Newtonian Sector from the Unified 4D Toy Model

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## Abstract

We present a controlled derivation of the full conservative two-body second post-Newtonian (2PN) sector from the unified 4D toy model. Carrying forward the exact projection / Poisson-hook / worldtube reduction logic and the already-fixed Newtonian+1PN ledger, we close the missing one-body 2PN response slot by a minimal DtN-invariant denominator that preserves the frozen 1PN coefficient and reproduces the isotropic Schwarzschild test-mass target through 2PN. Minimal Bernoulli exactification and quadratic local mass scaling then supply the self/static 2PN seed, with the pure-static isotropic gate fixing  $\lambda_p = \frac{1}{2}$ . The remaining comparable-mass mismatch against the generic-frame ADM target collapses to a finite algebraic residual with a unique solution for the added cross-sector coefficients  $\{q_i, t_i, s_1\}$ . After assembly, the full reduced two-body Lagrangian Legendre-transforms exactly to the standard generic-frame conservative ADM Hamiltonian through 2PN. The paper therefore gives a full conservative 2PN derivation within a declared closure hierarchy, while the appendices preserve the broader throat-response constructions and the remaining dynamic DtN/PDE observables.

## 1 Introduction and scope

### 1.1 Motivation: from full 1PN assembly to full conservative 2PN assembly

The earlier papers in this series established two pieces of the story that are now taken as the starting point rather than the destination. First, the unified 4D toy model supplies the exact projection, Poisson-hook, and worldtube dictionary from which a brane-facing particle mechanics is meant to descend. Second, the companion 1PN assembly paper showed that, once that dictionary is frozen and a short list of closures is declared explicitly, the full conservative two-body 1PN ledger can be recovered in a referee-auditable way. The purpose of the present paper is to take the next step without relaxing that bookkeeping discipline: we ask whether the same program closes at 2PN.

By “full conservative 2PN” we mean the complete near-zone two-body ledger through order  $c^{-4}$ . Concretely, the target includes the free-particle sextic terms, the one-body/self corrections tied to the test-mass limit, the comparable-mass quartic velocity sector at order  $G/r$ , the quadratic velocity sector at order  $G^2/r^2$ , and the static cubic term at order  $G^3/r^3$ . Our goal is not merely to reproduce one orbit-level consequence or one coordinate-specialized formula. The goal is to derive a reduced two-body Lagrangian whose full 2PN content Legendre-transforms exactly to the standard generic-frame ADM Hamiltonian target.

At 2PN the problem changes character. At 1PN, much of the work was to identify and fix the relevant coefficients sector by sector. At 2PN, two additional demands become unavoidable. First, the already-frozen Newtonian+1PN ledger must survive unchanged; a new closure is acceptable only if it repairs the missing 2PN slot without spoiling the settled 1PN coefficients. Second, the

constructive throat-response record becomes much larger than the proof itself. The project now contains tensor-wake decompositions, mouth-port reconstructions, support-minus-closure EFT packaging, minimal throat operators, PDE-side DtN scaffolds, and Family-1 wall reductions. All of that structure matters, but putting all of it in the main line would make the first theorem harder to read rather than easier to trust.

This paper therefore has a deliberately narrower purpose than the full 2PN research record and a stronger purpose than a notebook-level target match. It is narrower because the main text is organized around the smallest proof chain needed for the conservative 2PN claim. It is stronger because, within a stated closure hierarchy, that chain now runs from the carried-forward 4D reduction framework to the full two-body 2PN target:

1. exact carry-forward of the projection / Poisson-hook / worldtube reduction logic and of the frozen Newtonian+1PN ledger,
2. closure of the missing one-body 2PN slot by a minimal DtN-invariant conservative denominator that preserves the settled 1PN coefficient,
3. construction of the self/static 2PN seed from minimal Bernoulli exactification and quadratic local mass scaling,
4. a unique comparable-mass residual solve against the standard generic-frame ADM target, and
5. exact assembly of the full conservative two-body 2PN Lagrangian within the declared hierarchy.

A second purpose of the paper is bookkeeping clarity. In the 4D framework, projection is not a harmless relabeling; it naturally produces exchange, response, and closure terms. That makes the separation between exact carry-forward facts, reduced-regime statements, protocol-dependent closures, and final assembled equalities especially important at 2PN. The reader should be able to see, line by line, which parts of the derivation are inherited exactly, which parts depend on a declared low-frequency or small-body hierarchy, and which parts represent the minimal new closure input required to finish the conservative proof. That separation is part of the result.

This paper is also downstream of the corrected Paper 7 / Paper 8 matter-gauge ontology. Electric-charge sign is the fixed topological puncture orientation  $\eta_Q = \pm 1$ , the microscopic branch coupling is  $q_* = \eta_Q e_*$  with  $e_* > 0$ , and the canonically normalized brane charge is  $q_{\text{eff}} = q_*/\sqrt{Z_{\text{int}}} = \eta_Q e_{\text{eff}}$ , where  $Z_{\text{int}} \equiv \int Z(w) dw$ , with  $e_{\text{eff}} = e_*/\sqrt{Z_{\text{int}}}$ . Quantized circulation belongs to the magnetic/vortical sector rather than the electric-charge dictionary, and the zero-mode Maxwell reduction ( $A_w \approx 0$ ,  $\partial_w A_\mu \approx 0$ ,  $J^w \approx 0$ ,  $F_{\mu w} \approx 0$ ) is used here only as a controlled far-field brane limit. Separately, the scalar Newtonian mass-dressing coefficient historically denoted  $q$  in pre-4D gravity bookkeeping is not electric charge; in the present downstream notation it is written  $\kappa_\rho$ , fixed by Newtonian matching to  $\kappa_\rho = 1$ .

## 1.2 What this paper claims

The main claims are the following.

1. The exact 4D projection / Poisson-hook / worldtube framework from the earlier papers can be carried forward unchanged as the reduction backbone of the present 2PN derivation.
2. Within a minimal DtN-invariant conservative closure, the missing one-body 2PN response slot is fixed by a denominator that preserves the frozen 1PN coefficient and exactly reproduces the isotropic Schwarzschild test-mass target through 2PN.

3. Minimal Bernoulli exactification together with quadratic local mass scaling yields the self/static 2PN seed used in the assembly, with the pure-static isotropic gate selecting  $\lambda_p = 1/2$ .
4. Once the frozen Newtonian+1PN block and the one-body/self/static 2PN seed are fixed, the remaining comparable-mass residual is solved uniquely in a compact invariant basis, giving an added cross block whose Legendre transform matches the generic-frame ADM target exactly.
5. Combining these ingredients yields the full conservative two-body 2PN Lagrangian within the stated closure hierarchy, with the status of each ingredient recorded explicitly as exact, controlled, protocol-dependent, or full assembly.
6. A constructive throat-response interpretation of the new 2PN cross content already exists: the carried-forward dipole wake survives from 1PN, the genuinely new conservative response lives in a small  $P_0 \oplus P_2$  layer together with a geometry-closure channel, and the appendix program records the corresponding DtN, PDE, and Family-1 realizations.

Stated differently, the point of the paper is not that the 4D model can be adjusted until a 2PN-looking polynomial appears. The point is that, once the derivation is organized sector by sector, the full conservative two-body 2PN structure can be recovered with a short list of named and auditable ingredients. The constructive throat-response material then becomes an interpretation and continuation of that proof rather than a substitute for it.

### 1.3 What this paper does not claim

The negative side of the scope is equally important.

First, this paper does *not* claim an assumption-free theorem that begins with a fully solved moving-throat bulk background and ends automatically with the complete point-particle 2PN dynamics. As at 1PN, the particle limit used here is a controlled reduction inside a declared closure hierarchy. The proof in the main text is therefore a reduced-theory derivation, not a finished collective-coordinate theorem for the full bulk PDE.

Second, this paper does *not* claim a first-principles dynamical derivation of every new 2PN response datum. The one-body conservative denominator is fixed here within a minimal DtN-invariant packaging, and the final comparable-mass cross block is fixed here by an exact residual solve once the lower-order ledger and the declared seed are frozen. But the detailed dynamical origin of the pole scales and of the pure geometry-closure channel belongs to the appendix-side throat program and to later work. Those quantities should still be read as inner-throat observables, not as already-completed PDE theorems.

Third, this paper is restricted to the conservative sector. It does not address radiation reaction, dissipative leakage, spin couplings, or genuinely non-adiabatic internal response. It also does not attempt a full many-body dynamical completion beyond the conservative local-potential consequences already implied by the static scaling and throat-response packaging.

Fourth, the appendices should not be read as optional leftovers, but neither should they be confused with the load-bearing proof of the main theorem. They preserve the constructive record of how the 2PN cross content can be packaged in mouth-port, support-minus-closure, DtN, and Family-1 language. That record matters for the next paper, but the present paper's main claim is the conservative 2PN proof itself.

These non-claims are not qualifications to be hidden. They are part of the correct interpretation of the result: this paper establishes a full conservative two-body 2PN derivation *within a stated closure hierarchy*.

## 1.4 Claim taxonomy

To keep the argument referee-safe, we use the same claim taxonomy as in the 1PN assembly paper.

### Exact

identities that follow directly from the declared reduction framework, symmetry bookkeeping, exact algebraic transformations, or exact residual matching once the relevant frozen ledger and closures have been stated.

### Controlled reduction

steps that require an explicit regime assumption, such as the carried-forward small-body hierarchy, near-zone/low-frequency reduction, quasi-static bookkeeping, or the reduced DtN packaging used to close the one-body slot.

### Protocol closure

coefficients that are fixed only within a declared response closure rather than by a fully solved moving-throat derivation. In the present paper, the central example is the minimal conservative packaging used to close the one-body response denominator.

### Full assembly

the final exact equality between the assembled reduced two-body theory and the standard generic-frame ADM 2PN target once all earlier ingredients are inserted.

This taxonomy is not cosmetic. At 2PN it prevents three logically distinct moves from being blurred together: importing exact 4D structure, declaring the smallest new closures needed to finish the conservative ledger, and performing the exact algebraic solve and final target match once those closures are frozen.

## 1.5 Roadmap

Section 2 gives a front-end ledger of the main coefficients, proof steps, and open observables. Section 3 then records the minimal carry-forward dictionary from the earlier papers: the imported projection logic, the frozen Newtonian+1PN two-body ledger, and the small closure hierarchy used in the present 2PN assembly. Section 4 closes the missing one-body 2PN response slot and compares the resulting denominator directly to the isotropic test-mass target. Section 5 derives the self/static seed used in the assembly and states the finite-size gate that prevents over-claiming the static sector. Section 6 performs the comparable-mass residual solve against the standard ADM target, and Section 7 assembles the full conservative two-body 2PN Lagrangian and states the exact match. Sections 8–10 then record the fixed-vs-open status ledger, the verification logic tied to the symbolic harnesses, and the concluding interpretation.

The appendices collect the larger constructive record that is important but not required for the main proof: one-body DtN algebra, static-scaling details, the full ADM residual solve, tensor-wake and mouth-port decompositions, support-minus-closure EFT packaging, minimal throat operators, PDE/DtN scaffolding, Family-1 soft-wall and traction-balance reductions, geometry-breathing closures, and the final coupled response module. They are used here to keep the main theorem readable without discarding the broader 2PN program that motivated it.

## 2 Executive summary of results

This paper is the point at which the 4D program becomes a full conservative second post-Newtonian derivation rather than a partial closure story. The earlier 4D papers supply the exact projection dictionary, exact balance laws, near-zone Poisson hook, and worldtube reduction needed to define the brane-facing Newtonian sector, while the companion 1PN assembly paper

freezes the complete Newtonian+1PN two-body ledger. The present paper then carries those ingredients through the smallest additional chain needed to close the conservative 2PN problem: a DtN-invariant one-body denominator, a self/static seed from Bernoulli exactification and local mass scaling, and a unique comparable-mass lift of the remaining residual against the standard generic-frame ADM target.

**Headline result.** Within the stated closure hierarchy, the toy model reproduces the full conservative two-body 2PN ledger through order  $c^{-4}$ . The proof is not a single coefficient fit but a short triangulated chain. The carried Newtonian+1PN structure is held fixed, the missing one-body 2PN slot is closed in a way that preserves the settled 1PN coefficient, the pure-static isotropic gate selects  $\lambda_p = 1/2$ , and the remaining comparable-mass mismatch then collapses to a finite algebraic residual with a unique solution. The final reduced two-body Lagrangian Legendre-transforms exactly to the standard generic-frame ADM Hamiltonian through 2PN.

Table 1 lists the quantities proved in the paper and the logic by which each is obtained.

Table 1: Headline results established in the 2PN paper. The table is deliberately front-loaded: it is meant to function as a referee’s map of what is carried exactly, what is closed within the declared reduction hierarchy, and what is only claimed after full assembly.

| Quantity                                 | Value                                                                                | Source of derivation                                                     | Claim class                                               |
|------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------|
| Reduction backbone                       | exact projection /<br>Poisson-hook / worldtube<br>dictionary                         | carried forward from the earlier<br>4D papers                            | <b>Exact/<br/>Controlled<br/>reduction</b>                |
| Frozen lower-order<br>ledger             | full Newtonian+1PN<br>two-body block with exact<br>EIH match                         | carried forward from the 1PN<br>assembly paper                           | <b>Exact/ Full<br/>assembly</b>                           |
| One-body denominator                     | $D_{\text{eff}}(u) =$<br>$C_s(u) \left[ 1 + \frac{32768}{3249} (G(u) - 1)^2 \right]$ | minimal DtN-invariant closure<br>preserving the frozen 1PN slot          | <b>Controlled<br/>reduction/<br/>Protocol<br/>closure</b> |
| Test-mass 2PN target                     | $D_{\text{eff}}(u) =$<br>$1 - 4u + 8u^2 + O(u^3)$                                    | exact isotropic Schwarzschild<br>match through 2PN                       | <b>Full<br/>assembly</b>                                  |
| Static scaling coefficient               | $\lambda_p = \frac{1}{2}$                                                            | pure-static isotropic target gate                                        | <b>Controlled<br/>reduction</b>                           |
| Self/static 2PN seed                     | minimal Bernoulli<br>exactification + quadratic<br>local mass scaling                | reduced one-body/self/static<br>sector                                   | <b>Controlled<br/>reduction</b>                           |
| Comparable-mass<br>residual coefficients | unique solution for<br>$\{q_i, t_i, s_1\}$                                           | compact invariant-basis solve<br>against the generic-frame ADM<br>target | <b>Exact</b>                                              |
| Full conservative<br>two-body 2PN block  | exact generic-frame ADM<br>match after assembly                                      | full two-body reduction and<br>Legendre check                            | <b>Full<br/>assembly</b>                                  |

**How the ledger closes.** There are three nested stories in the proof. The first is the one-body slot. Writing the conservative denominator as

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \alpha (G - 1) + \beta (G - 1)^2 \right],$$

preservation of the already-frozen 1PN coefficient forces the linear correction to vanish,

$$\alpha = 0.$$

The first admissible new term is therefore quadratic in the geometry invariant. With the carried throat slope

$$G(u) = 1 + \frac{57}{64}u + O(u^2),$$

matching the exact isotropic Schwarzschild test-mass target through 2PN fixes

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \frac{32768}{3249}(G(u) - 1)^2 \right] = 1 - 4u + 8u^2 + O(u^3).$$

So the missing one-body conservative 2PN slot is closed without disturbing the settled 1PN ledger.

The second story is the self/static seed. Minimal Bernoulli exactification supplies the free-particle sextic term and the self-sector  $G/r$  velocity dressing, while quadratic local mass scaling supplies the static cubic hierarchy. The pure-static isotropic gate selects

$$\lambda_\rho = \frac{1}{2},$$

so the reduced self/static 2PN seed entering the assembly is

$$L_{2,\text{self+static}} = \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{7Gm_A m_B}{8r} (v_A^4 + v_B^4) + \frac{2G^2 m_A m_B}{r^2} (m_B v_A^2 + m_A v_B^2) + \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}, \quad (1)$$

with the overall  $c^{-4}$  factor understood.

The third story is the comparable-mass lift. The key simplification is that for

$$L = L_0 + \epsilon L_1 + \epsilon^2 L_2, \quad \epsilon \equiv c^{-2},$$

with quadratic  $L_0$ , the perturbative Legendre transform through 2PN collapses to

$$H_1 = -L_1(v_0), \quad H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0,$$

where  $v_0 = p/m$  and  $A_0 = (\partial L_1 / \partial v)|_{v=v_0}$ . Because the Newtonian+1PN block is already frozen, any genuinely new 2PN Lagrangian term enters the Hamiltonian only through the explicit  $-L_{2,\text{new}}(v_0)$  piece. The remaining mismatch against the standard generic-frame ADM target is therefore a finite algebraic residual in a compact invariant basis, and that residual solves uniquely.

**Solved comparable-mass coefficients.** The exact residual solve yields

$$q_1 = -\frac{7}{4}, \quad q_2 = -\frac{1}{4}, \quad q_3 = \frac{11}{8}, \quad q_4 = \frac{1}{4}, \quad q_5 = -\frac{5}{8}, \quad q_6 = \frac{3}{2}, \quad q_7 = \frac{3}{8},$$

$$t_1 = 0, \quad t_2 = \frac{11}{8}, \quad t_3 = -\frac{15}{4}, \quad t_4 = 0, \quad t_5 = 0, \quad t_6 = \frac{15}{8}, \quad s_1 = \frac{5}{4}.$$

These are the coefficients of the unique added comparable-mass cross block required to upgrade the self/static candidate to the full 2PN target.

**Normalization warning.** In the executive formulas below,  $L_{2,\text{full}}$  denotes the coefficient of  $c^{-4}$  in the total two-body Lagrangian. The Newtonian and 1PN blocks are carried forward unchanged from the earlier paper; only the new 2PN block is displayed explicitly.

**Final assembled result.** Using the shorthand

$$v_{AB} = \mathbf{v}_A \cdot \mathbf{v}_B, \quad v_{An} = \mathbf{v}_A \cdot \mathbf{n}, \quad v_{Bn} = \mathbf{v}_B \cdot \mathbf{n},$$

the new 2PN block inserted into the carried Newtonian+1PN ledger is

$$\begin{aligned} L_{2,\text{full}} = & \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{G m_A m_B}{r} \left[ \frac{7}{8} (v_A^4 + v_B^4) - \frac{7}{4} v_{AB} (v_A^2 + v_B^2) - \frac{1}{4} v_{An} v_{Bn} (v_A^2 + v_B^2) \right. \\ & \left. + \frac{11}{8} v_A^2 v_B^2 + \frac{1}{4} v_{AB}^2 - \frac{5}{8} (v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2} v_{AB} v_{An} v_{Bn} + \frac{3}{8} v_{An}^2 v_{Bn}^2 \right] \\ & + \frac{G^2 m_A m_B}{r^2} \left[ \left( 2m_B + \frac{11}{8} m_A \right) v_A^2 + \left( 2m_A + \frac{11}{8} m_B \right) v_B^2 - \frac{15}{4} (m_A + m_B) v_{AB} \right. \\ & \left. + \frac{15}{8} (m_A v_{An}^2 + m_B v_{Bn}^2) \right] + \frac{G^3 m_A m_B}{4r^3} (m_A^2 + 5m_A m_B + m_B^2). \end{aligned} \quad (2)$$

The claim of the paper is that this is not merely ADM-like: once the carried and newly solved ingredients are inserted, the Legendre transform of the full reduced two-body theory agrees identically with the standard generic-frame conservative ADM target through 2PN.

**How to read the claim classes.** The table mixes four logically different kinds of statements. The carried reduction backbone and the exact residual solve are **Exact** statements once the relevant ledger and closures have been declared. The one-body denominator and the self/static seed are **Controlled reduction** statements because they depend on an explicit reduced-regime hierarchy, and the one-body denominator is also a **Protocol closure** statement because it is fixed within a declared conservative DtN packaging rather than by a fully solved moving-throat PDE theorem. The final equality to the generic-frame ADM target is the central **Full assembly** statement of the paper.

**What is fixed and what remains open.** Within the scope of the present proof, the values

$$\lambda_\rho = \frac{1}{2}, \quad D_{\text{eff}}(u) = C_s(u) \left[ 1 + \frac{32768}{3249} (G(u) - 1)^2 \right], \quad \{q_i, t_i, s_1\} \text{ as listed above,}$$

should be read as derived outputs rather than free knobs. The appendix-side constructive program also fixes a nontrivial zero-frequency throat ledger,

$$R_{1\perp} = \frac{7}{2}, \quad R_{10} = 4, \quad R_0 = R_{20} = R_{21} = R_{22} = 1,$$

$$J = \left( \frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right), \quad \Delta_{\text{geom}} = \frac{281}{80},$$

where  $J$  is an appendix-side throat-response source-weight vector rather than an electromagnetic current. This ledger packages the new conservative cross content into a carried odd dipole backbone, a new even  $P_0 \oplus P_2$  support layer, and a pure- $U$  geometry-closure channel. What remains open are the dynamical pole scales

$$\Omega_{1\perp}, \Omega_{10}, \Omega_0, \Omega_{20}, \Omega_{21}, \Omega_{22}, \Omega_g,$$

and, more broadly, a first-principles moving-throat PDE derivation of the same low-frequency response data. The paper therefore claims a full conservative two-body 2PN derivation within a stated closure hierarchy, not an assumption-free theorem of the fully solved bulk dynamics.

**Appendix map.** The appendices preserve the larger constructive record that is important for the program but not load-bearing for the proof. They collect the one-body DtN algebra, the static-scaling and finite-size details, the full ADM residual solve, tensor-wake and mouth-port decompositions, support-minus-closure EFT packaging, minimal irreducible-channel throat operators, DtN/PDE scaffolding, Family-1 soft-wall and traction-balance reductions, geometry-breathing closures, and the final coupled response module. The main text proves the conservative 2PN theorem; the appendices document how that theorem is embedded in the broader throat-response program.

### 3 Minimal carry-forward dictionary and frozen 1PN ledger

This paper does not reopen the derivation of the Newtonian sector or of the full conservative two-body 1PN ledger. Its job is narrower and more rigid: to state, in one place, exactly what has already been fixed, exactly what notation will be used below, and exactly which new ingredients a genuine 2PN proof is still allowed to introduce. The present section therefore plays a bookkeeping role rather than a constructive one. It freezes the reduction backbone inherited from the earlier 4D papers, records the already-settled Newtonian+1PN two-body dynamics, and makes explicit the small closure hierarchy that will be used in the rest of the paper.

The reason for doing this early is simple. At 2PN one can easily blur together three logically different moves: importing exact or previously-validated lower-order structure, declaring the minimal new closures needed to repair the missing 2PN slots, and performing the exact algebraic assembly once those closures are frozen. The point of the present paper is to keep those moves separate. In particular, the 2PN calculation is not permitted to retune the already-proved 1PN answer. Any acceptable 2PN completion must reduce to the same Newtonian and 1PN ledger when the  $c^{-4}$  block is switched off.

The notation carried forward here is also downstream of the corrected matter-gauge stack. Electric charge does not enter the main conservative 2PN proof; when it is named explicitly it follows the corrected  $(\eta_Q, q_*, q_{\text{eff}})$  ontology above. The residual-basis coefficients  $q_i$  used later are simply ADM-lift coefficients, and the historical gravity-side scalar coefficient once written as bare  $q$  is written here as  $\kappa_\rho$ , fixed by Newtonian matching to  $\kappa_\rho = 1$ .

#### 3.1 What is imported without rederivation

Only a narrow subset of the earlier 4D program is imported into the present paper, but that subset is structural and should be kept conceptually visible.

First, we carry forward the distinction between *projection* and *reduction*. Projection defines what a brane observer measures: it is the operational map from bulk fields to brane-facing observables. Reduction is a second step, performed only after a regime has been declared, in which those projected observables are approximated by an effective particle description. The present 2PN paper uses exactly the same logic as the 1PN assembly paper: projection identities are inherited from the parent 4D description, whereas the particle mechanics emerges only after the same carried near-zone and small-body hierarchy has been adopted.

Second, the exact reduction backbone imported from the earlier papers is unchanged. The bulk-to-brane projection map, the exact projected continuity law, and the exact longitudinal identity are treated as settled structural inputs. In the quasi-static near-zone regime, the same Poisson-hook logic is then carried forward to identify the brane-facing Newtonian potential. Nothing in the present paper modifies that route. The 2PN calculation begins only *after* those lower-level identities have already done their work.

Third, we carry forward the coherent-defect / small-body worldtube reduction that turns a localized object moving in a smooth external field into an effective particle source. As at 1PN, this is a controlled reduction rather than an exact theorem of a fully solved moving-throat bulk

configuration. Its purpose here is not to be rederived but to fix the meaning of the particle variables  $m_A$ ,  $m_B$ ,  $\mathbf{x}_A$ ,  $\mathbf{x}_B$ ,  $\mathbf{v}_A$ , and  $\mathbf{v}_B$  that appear throughout the rest of the paper.

Finally, the full conservative two-body 1PN result is imported as frozen data. That means more than importing a few coefficients. It means that the already-assembled Newtonian+1PN Lagrangian, together with its exact Einstein–Infeld–Hoffmann match, is treated as part of the fixed backbone of the 2PN derivation. The present paper is allowed to add the order- $c^{-4}$  block; it is not allowed to revise what has already been proven at order  $c^{-2}$ .

**Claim class.** *Exact for the projection dictionary, exact projected continuity, and exact longitudinal identity inherited from the earlier 4D papers. Controlled reduction for the quasi-static Poisson hook and the coherent-defect / small-body worldtube reduction that turns those exact structures into an effective particle mechanics.*

### 3.2 Minimal dictionary of variables and invariants

We work throughout with two bodies  $A$  and  $B$  in a generic frame. Their positions, velocities, and masses are denoted

$$\mathbf{x}_A, \quad \mathbf{x}_B, \quad \mathbf{v}_A = \dot{\mathbf{x}}_A, \quad \mathbf{v}_B = \dot{\mathbf{x}}_B, \quad m_A, \quad m_B.$$

The relative separation vector, its magnitude, and the corresponding unit direction are

$$\mathbf{r} \equiv \mathbf{x}_A - \mathbf{x}_B, \quad r \equiv |\mathbf{r}|, \quad \mathbf{n} \equiv \frac{\mathbf{r}}{r}. \quad (3)$$

All main-text formulas are written in this generic frame; no center-of-mass specialization is used in the 2PN residual solve.

The velocity invariants that appear repeatedly are

$$v_A^2 \equiv \mathbf{v}_A \cdot \mathbf{v}_A, \quad v_B^2 \equiv \mathbf{v}_B \cdot \mathbf{v}_B, \quad v_{AB} \equiv \mathbf{v}_A \cdot \mathbf{v}_B, \quad (4)$$

with longitudinal projections

$$v_{An} \equiv \mathbf{v}_A \cdot \mathbf{n}, \quad v_{Bn} \equiv \mathbf{v}_B \cdot \mathbf{n}. \quad (5)$$

These are the only kinematic invariants needed in the main text. Every comparable-mass 2PN cross monomial used later can be written in terms of  $v_A^2$ ,  $v_B^2$ ,  $v_{AB}$ ,  $v_{An}$ , and  $v_{Bn}$ .

It is also convenient to define the local Newtonian pair potentials

$$U_A \equiv \frac{Gm_B}{r}, \quad U_B \equiv \frac{Gm_A}{r}. \quad (6)$$

In the strict one-body reduction, where a heavy source of mass  $M$  is treated as fixed, we write

$$U \equiv \frac{GM}{r}, \quad u \equiv \frac{U}{c^2}, \quad (7)$$

so that the one-body denominator problem can be expressed as a weak-field series in the single dimensionless variable  $u$ .

The rest of the paper uses one additional bookkeeping convention repeatedly. The conservative two-body Lagrangian is expanded as

$$L_{\text{cons}} = L_0 + \frac{1}{c^2} L_1 + \frac{1}{c^4} L_2 + O(c^{-6}), \quad (8)$$

where  $L_0$  is Newtonian,  $L_1$  is the full frozen 1PN block, and  $L_2$  is what the present paper must determine.

### 3.3 Frozen Newtonian + 1PN ledger

The Newtonian block is

$$L_0 = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 + \frac{Gm_A m_B}{r}. \quad (9)$$

Nothing in the present paper alters [equation \(9\)](#). All genuinely new content begins at  $L_2$ .

The full carried 1PN block is likewise frozen:

$$L_1 = \frac{1}{8}(m_A v_A^4 + m_B v_B^4) + \frac{Gm_A m_B}{r} \left[ \frac{3}{2}(v_A^2 + v_B^2) - \frac{7}{2}v_{AB} - \frac{1}{2}v_{An}v_{Bn} \right] - \frac{G^2 m_A m_B (m_A + m_B)}{2r^2}. \quad (10)$$

By construction, [equation \(10\)](#) is not a new ansatz introduced for the present paper. It is the already-proved outcome of the 1PN assembly and therefore the fixed lower-order input to every 2PN statement below.

It is useful to record explicitly which lower-order facts are now being treated as settled. Newtonian matching fixed the scalar mass-dressing coefficient to  $\kappa_\rho = 1$  (historically  $q = 1$  in earlier gravity-side bookkeeping). Weak-field optical consistency fixed the barotropic exponent to  $n = 5$ . Wave-supported free kinematics fixed the quartic kinetic coefficient to  $1/8$ . The scalar/self sector fixed the pair coefficient of  $v_A^2 + v_B^2$  to  $+3/2$ . The carried vector-wake sector fixed the cross coefficients of  $v_{AB}$  and  $v_{An}v_{Bn}$  to  $-7/2$  and  $-1/2$ , respectively. The static nonlinear 1PN term fixed the coefficient of  $-G^2 m_A m_B (m_A + m_B)/(2r^2)$ . On the response side, the topology ledger fixed  $\kappa_{\text{add}} = 1/2$ , the declared adiabatic breathing closure fixed  $\kappa_{\text{PV}} = 3/2$ , and therefore the already-frozen self-sector/precession ledger is

$$\beta_{1\text{PN}} = 1 + \frac{1}{2} + \frac{3}{2} = 3. \quad (11)$$

These values are no longer free parameters. They are carry-forward data.

Equally important is what this means operationally. In the remainder of the paper the 2PN construction is not allowed to shift any coefficient in [equation \(10\)](#), nor is it allowed to reopen the lower-order fitting logic that produced them. The one-body denominator closure must preserve the frozen 1PN slot. The self/static 2PN seed must reduce to the already-fixed static regression at lower order. And the comparable-mass residual solve is performed with  $L_0$  and  $L_1$  held exactly fixed from the start.

**Claim class.** *Controlled reduction for the carried small-body reduction and for the lower-order fixing of  $\kappa_\rho = 1$  and  $n = 5$ . Protocol closure for the already-declared adiabatic response input  $\kappa_{\text{PV}} = 3/2$ , and therefore for the carried value  $\beta_{1\text{PN}} = 3$ . Full assembly for the statement that  $L_0 + c^{-2}L_1$  is already exactly the standard conservative two-body EIH Lagrangian..*

### 3.4 Closure hierarchy used in the present 2PN paper

With the lower-order ledger frozen, the rest of the paper adds only the smallest set of new ingredients needed to complete the conservative 2PN proof.

First, the one-body sector introduces a low-frequency conservative DtN-invariant denominator packaging for the missing test-mass 2PN response slot. Its job is tightly constrained: it must repair the 2PN one-body target without changing the already-fixed 1PN coefficient.

Second, the self sector is extended by a minimal Bernoulli exactification. This is used only to generate the conservative one-body/self 2PN seed required by the main assembly. The paper does not claim that this exhausts the full moving-throat response problem.

Third, the static sector is extended by quadratic local mass scaling. In the main text, only the load-bearing consequence is needed: the pure-static isotropic gate selects

$$\lambda_\rho = \frac{1}{2}. \quad (12)$$

The induced higher-body static hierarchy and related finite-size caveats are recorded in appendices rather than in the proof line.

Fourth, once the frozen lower-order ledger and the declared one-body/self/static seed are inserted, the remaining comparable-mass problem is treated as an exact algebraic residual solve against the standard generic-frame ADM target. This is the step that closes all genuinely new conservative two-body cross content at 2PN.

Everything else belongs to constructive interpretation rather than to the minimal proof. The tensor-wake rebuild, mouth-port decomposition, support-minus-closure EFT packaging, minimal irreducible-channel throat operators, PDE-side DtN scaffolding, Family-1 soft-wall reductions, and the final coupled throat-response module are all preserved in the appendices. They are important for understanding *why* the solved 2PN answer admits a compact throat-response interpretation, but they are not required to state or verify the main conservative theorem.

A useful way to read the rest of the paper is therefore triangular. The bottom layer is the exact and already-carried Newtonian+1PN backbone. On top of that sits the smallest new closure stack needed to define the 2PN seed. Once those inputs are frozen, the final assembly and ADM comparison are exact algebra. That is the sense in which the paper claims a full conservative 2PN derivation within a stated closure hierarchy, rather than a first-principles solution of the fully dynamical inner throat.

## 4 One-body 2PN closure and the test-mass target

The first genuinely new ingredient at 2PN already appears in the strict one-body limit. Once the Newtonian+1PN ledger has been frozen, the problem is no longer to rediscover the lower-order coefficients but to repair one specific missing response slot without disturbing anything that is already fixed. In the present paper that slot is the order- $u^2$  correction to the effective one-body velocity denominator, with

$$u \equiv \frac{U}{c^2}, \quad U \equiv \frac{GM}{r},$$

for a light body moving in the field of a fixed heavy source of mass  $M$ .

This section isolates that one-body problem and closes it in the smallest possible way. The exact isotropic Schwarzschild test-mass target tells us what the completed one-body ledger must look like. The low-frequency mouth / DtN data tell us which invariant building blocks are naturally available on the toy-model side. The answer is then fixed by one simple requirement: any new 2PN correction must preserve the already-frozen 1PN coefficient.

### 4.1 Exact isotropic Schwarzschild test-mass target through 2PN

In isotropic coordinates, the exact test-mass Lagrangian for a particle of mass  $m$  in the Schwarzschild field of a fixed source  $M$  is

$$\frac{L_{\text{Schw}}}{m} = -c^2 \sqrt{\left(\frac{1-u/2}{1+u/2}\right)^2 - (1+u/2)^4 \frac{v^2}{c^2}}. \quad (13)$$

Expanding [equation \(13\)](#) through total post-Newtonian order  $c^{-4}$  gives

$$\begin{aligned} \frac{L_{\text{Schw}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3 \right) + O(c^{-6}). \end{aligned} \quad (14)$$

Equation (14) is the exact one-body target that the carried toy-model reduction must reproduce within the stated closure hierarchy.

For the present section, one feature of equation (14) is the key point. In the carried optical/self packaging, the free  $v^6$  term and the  $Uv^4$  term are already produced by the minimal Bernoulli exactification, while the static  $U^3$  term belongs to the static sector treated in the next section. The genuinely missing one-body response datum is therefore the  $U^2v^2$  coefficient. The exact Schwarzschild target requires that coefficient to be

$$2. \quad (15)$$

A convenient way to see the resulting denominator target is to write the reduced one-body kinetic packaging in the form

$$L_{\text{red,1-body}} = -mc^2(1-u)\sqrt{1 - \frac{v^2/c^2}{D(u)}}, \quad D(u) = 1 - 4u + d_2u^2 + O(u^3). \quad (16)$$

Expanding equation (16) gives

$$\begin{aligned} \frac{L_{\text{red,1-body}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + \left( 6 - \frac{d_2}{2} \right) U^2v^2 \right) + O(c^{-6}). \end{aligned} \quad (17)$$

Matching the exact target coefficient equation (15) therefore forces

$$d_2 = 8, \quad \text{i.e.} \quad D_{\text{target}}(u) = 1 - 4u + 8u^2 + O(u^3). \quad (18)$$

This is the precise one-body datum that the DtN-side closure must reproduce.

## 4.2 Low-frequency DtN invariants and the carried freeze

The low-frequency monopole mouth operator is written as

$$Z_{00}(\omega; u) = Z_2(u)\omega^2 + Z_4(u)\omega^4 + O(\omega^6). \quad (19)$$

From it we form the normalized invariant combinations

$$C_s(u) \equiv \frac{Z_4/Z_2^3}{(Z_4/Z_2^3)_0}, \quad G(u) \equiv \frac{Z_4/Z_2^2}{(Z_4/Z_2^2)_0}. \quad (20)$$

These are the two dimensionless one-body observables that survive the low-frequency normalization and therefore provide the natural conservative building blocks for the present closure.

On the cylinder / Neumann-bottom unit-test branch, the leading coefficients take the form

$$Z_2 = -\frac{L}{c_s^2}, \quad Z_4 = -\frac{L^3}{3c_s^4}, \quad (21)$$

so the invariants reduce exactly to

$$C_s(u) = \frac{c_s^2(u)}{c_{s0}^2}, \quad G(u) = \frac{L(u)}{L_0}. \quad (22)$$

Under the current conservative freeze, the Bernoulli continuation of the already-fixed  $n = 5$  barotropic sector gives

$$C_s(u) = 1 - 4u, \quad (23)$$

while the carried 1PN breathing slope implies

$$G(u) = 1 + g_1u + O(u^2), \quad g_1 = \frac{57}{64}. \quad (24)$$

Only the linear coefficient  $g_1$  is needed for the 2PN one-body match in the main text; higher-order series data are recorded in the appendices.

### 4.3 Minimal denominator ansatz that preserves the frozen 1PN slot

The carried 1PN ledger fixes the coefficient of  $u$  in the effective velocity denominator once and for all. Therefore any admissible 2PN correction must begin at order  $u^2$ . Within the minimal conservative DtN-invariant packaging, the most general correction built from the one-body invariants and truncated at quadratic order in  $G - 1$  is

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \alpha (G - 1) + \beta (G - 1)^2 \right]. \quad (25)$$

This is the smallest ansatz that can both preserve the already-fixed 1PN slot and supply a genuinely new 2PN correction.

**Claim 1** (Minimal one-body 2PN closure). *Within the conservative ansatz [equation \(25\)](#), preservation of the frozen 1PN coefficient forces*

$$\alpha = 0, \quad (26)$$

and the exact isotropic Schwarzschild test-mass target through 2PN uniquely fixes

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \mu (G(u) - 1)^2 \right], \quad \mu = \frac{8}{g_1^2} = \frac{32768}{3249}, \quad (27)$$

so that

$$D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3). \quad (28)$$

The derivation is immediate. Insert

$$G(u) = 1 + g_1 u + g_2 u^2 + O(u^3), \quad C_s(u) = 1 - 4u + O(u^2), \quad (29)$$

into [equation \(25\)](#). One finds

$$D_{\text{eff}}(u) = 1 + (-4 + \alpha g_1)u + O(u^2). \quad (30)$$

Because the coefficient of  $u$  is already frozen by the carried 1PN ledger and because  $g_1 = 57/64 \neq 0$ , the linear correction must vanish, which proves [equation \(26\)](#). The first admissible new contribution is therefore quadratic:

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \mu (G(u) - 1)^2 \right]. \quad (31)$$

Using [equation \(24\)](#), this expands to

$$D_{\text{eff}}(u) = 1 - 4u + \mu g_1^2 u^2 + O(u^3). \quad (32)$$

Matching the exact target denominator [equation \(18\)](#) therefore requires

$$\mu g_1^2 = 8, \quad (33)$$

which gives precisely [equation \(27\)](#).

Several features of this result are worth emphasizing. First, the closure is minimal: the linear geometry correction is forbidden by the frozen 1PN slot, so the first allowed new term is exactly quadratic in  $G - 1$ . Second, the match is exact at the level required here: once the current slope  $g_1 = 57/64$  is imported, the resulting coefficient  $\mu = 32768/3249$  reproduces the exact one-body denominator target through 2PN. Third, the construction is conservative and invariant-based: nothing in the argument depends on a coordinate-specialized comparable-mass fit or on a re-tuning of the already-settled lower-order ledger.

**Claim class.** *Controlled reduction* for the low-frequency DtN packaging and for the import of  $C_s(u) = 1 - 4u$  and  $g_1 = 57/64$  from the carried Bernoulli / breathing ledger. **Protocol closure** for the minimal conservative ansatz  $D_{\text{eff}} = C_s[1 + \alpha(G - 1) + \beta(G - 1)^2]$ . **Full assembly** for the exact equality  $D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3)$  with the isotropic Schwarzschild test-mass target through 2PN..

#### 4.4 Scope of the one-body result

This section closes the one-body missing 2PN response slot, but it is important to state exactly what that means.

What has been established is the following. Within the declared conservative low-frequency DtN packaging, the frozen 1PN coefficient forces the geometry correction to begin quadratically in  $G - 1$ , and the exact isotropic Schwarzschild target then fixes that quadratic coefficient uniquely. As a result, the one-body velocity denominator is no longer an open slot in the 2PN assembly.

What has *not* been claimed is equally important. This section does not derive the quadratic packaging [equation \(27\)](#) from a fully solved moving-throat PDE problem. It also does not by itself determine the static  $U^3$  coefficient or any comparable-mass 2PN cross term. Those belong, respectively, to the self/static sector of the next section and to the exact ADM residual solve that follows later. The appendix records the longer constructive throat-response story—including the relation to the earlier raw resonance-proxy fit and the PDE-side DtN scaffolding— but none of that extra structure is needed for the proof line established here.

### 5 Minimal self/static 2PN sector

The previous section closed the one-body 2PN response slot by fixing the missing denominator correction needed to reproduce the exact isotropic Schwarzschild test-mass target. The present section records the complementary piece of the proof: the smallest self/static sector that can be carried into the full two-body assembly without disturbing the frozen Newtonian+1PN ledger.

There are really two ingredients here. The first is the *self* sector: a minimal Bernoulli exactification of the one-body worldline that preserves the already-fixed 1PN coefficients while generating the universal 2PN free and optical terms. The second is the *static* sector: quadratic local mass scaling, treated by the same pair-counting logic used at lower order, whose two-body coefficient is then fixed by the pure-static isotropic test-mass gate. Once those two ingredients are frozen and the denominator repair from [section 4](#) is inserted, the resulting self/static block is ready to be fed into the exact comparable-mass residual solve of later sections.

#### 5.1 Minimal Bernoulli exactification of the self sector

The carried  $n = 5$  Bernoulli continuation fixes the sound-speed factor

$$C_s(u) = 1 - 4u, \quad u \equiv \frac{U}{c^2}, \quad (34)$$

with  $U = GM/r$  in the strict one-body limit. If one exactifies only the optical denominator while keeping the already-frozen linear mass dressing, the minimal one-body self worldline is

$$L_{\text{self},\min} = -mc^2(1 - u)\sqrt{1 - \frac{v^2/c^2}{1 - 4u}}. \quad (35)$$

This is the smallest Bernoulli continuation that leaves the lower-order bookkeeping intact: it preserves the frozen 1PN self coefficient and introduces no new static term by hand.

Expanding [equation \(35\)](#) through total post-Newtonian order  $c^{-4}$  gives

$$\begin{aligned} \frac{L_{\text{self,min}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 6U^2v^2 \right) + O(c^{-6}). \end{aligned} \quad (36)$$

Several features of [equation \(36\)](#) are load-bearing. First, the carried 1PN self coefficient remains exactly  $+\frac{3}{2}$  on the  $Uv^2$  term, as required by the frozen ledger. Second, the universal free kinetic coefficient is fixed to  $1/16$ . Third, the minimal exactification generates the candidate one-body  $Uv^4$  coefficient  $7/8$ . Finally, before any further response repair is inserted, the raw self sector places coefficient 6 on the  $U^2v^2$  term.

That last coefficient is precisely why the previous section was needed. Comparing [equation \(36\)](#) with the exact isotropic Schwarzschild target [equation \(14\)](#), one sees that the raw minimal self sector already gets the free  $v^6$  and mixed  $Uv^4$  pieces right, but it overshoots the required  $U^2v^2$  coefficient by

$$6 - 2 = 4. \quad (37)$$

The role of [section 4](#) was exactly to repair this residual by replacing the raw optical denominator  $1 - 4u$  with the DtN-completed denominator  $D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3)$ .

It is also worth recording what *is not* done here. One might try to Bernoulli-exactify the mass prefactor as well, replacing  $(1 - u)$  by a fully Bernoulli-dressed factor. But that more aggressive step immediately spoils the frozen lower-order static ledger: it generates a  $+\frac{3}{2}U^2/c^2$  static term at 1PN, whereas the carried pair-counted static sector is already fixed to  $-\frac{1}{2}U^2/c^2$ . Thus only the optical denominator is exactified in the self sector. This is the sense in which the present construction is *minimal*.

**Claim class.** *Controlled reduction for the Bernoulli continuation [equation \(35\)](#), which preserves the frozen 1PN self ledger while generating the universal raw 2PN self coefficients  $1/16$ ,  $7/8$ , and  $6$ .*

## 5.2 Quadratic local mass scaling and the pure-static gate

The static sector is treated by the same local-potential logic used at lower order, but extended to quadratic order in the local field. For each body,

$$m_A^{\text{eff}} = m_A \left( 1 - \frac{U_A}{c^2} + \lambda_\rho \frac{U_A^2}{c^4} \right), \quad m_B^{\text{eff}} = m_B \left( 1 - \frac{U_B}{c^2} + \lambda_\rho \frac{U_B^2}{c^4} \right), \quad (38)$$

with the local Newtonian pair potentials

$$U_A = \frac{Gm_B}{r}, \quad U_B = \frac{Gm_A}{r}. \quad (39)$$

The pair-counted static contribution is then

$$L_{\text{static}} = -\frac{1}{2}(m_A^{\text{eff}}\Phi_A + m_B^{\text{eff}}\Phi_B), \quad \Phi_A = -U_A, \quad \Phi_B = -U_B. \quad (40)$$

Expanding [equation \(40\)](#) gives, through 2PN,

$$L_{\text{static}} = \frac{Gm_Am_B}{r} - \frac{G^2m_Am_B(m_A + m_B)}{2c^2r^2} + \frac{\lambda_\rho G^3m_Am_B(m_A^2 + m_B^2)}{2c^4r^3} + O(c^{-6}). \quad (41)$$

Thus the two-body static 2PN coefficient is simply  $\lambda_\rho/2$ .

The one-body meaning of this coefficient is immediate. In the strict test-mass limit, [equation \(41\)](#) contributes

$$\frac{L_{\text{static},1\text{-body}}}{m} = U - \frac{1}{2c^2}U^2 + \frac{\lambda_\rho}{2c^4}U^3 + O(c^{-6}). \quad (42)$$

The exact isotropic Schwarzschild target of [equation \(14\)](#) requires the  $U^3$  coefficient to be  $+\frac{1}{4}$ . Therefore the pure-static test-mass gate fixes

$$\lambda_\rho = \frac{1}{2}. \quad (43)$$

This is the unique value compatible with the already-frozen 1PN static term and with the exact isotropic 2PN static coefficient.

**Claim 2** (Minimal static 2PN closure). *Within quadratic local mass scaling and the carried pair-counting rule, matching the pure-static one-body target through 2PN uniquely sets*

$$\lambda_\rho = \frac{1}{2}. \quad (44)$$

Equivalently, the two-body static 2PN block is fixed to

$$L_{2,\text{static}} = \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}, \quad (45)$$

where, as throughout,  $L_2$  denotes the coefficient of  $c^{-4}$  in the expansion  $L_{\text{cons}} = L_0 + c^{-2}L_1 + c^{-4}L_2 + O(c^{-6})$ .

At the level needed for the main proof, [equation \(43\)](#) is the only new static datum that must be carried forward. The same quadratic local mass scaling also induces a higher-body cubic-static hierarchy. Those terms are important for the broader constructive program and are recorded in the appendices, but they are not needed for the conservative two-body 2PN theorem of the present paper.

**Claim class.** *Controlled reduction for quadratic local mass scaling and pair counting. Full assembly for the pure-static isotropic one-body gate  $\lambda_\rho = \frac{1}{2}$ .*

### 5.3 The frozen self/static seed used in the two-body assembly

It is useful to separate the raw self/static seed from its final one-body repair. Combining [equation \(36\)](#) with [equation \(42\)](#) gives the strict one-body self/static candidate

$$\begin{aligned} \frac{L_{\text{self+static},1\text{-body}}^{\text{raw}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 6U^2v^2 + \frac{\lambda_\rho}{2}U^3 \right) + O(c^{-6}). \end{aligned} \quad (46)$$

After imposing  $\lambda_\rho = \frac{1}{2}$ , the pure-static target is matched exactly, and one is left with the sole residual  $4U^2v^2$  already isolated in [equation \(37\)](#). That residual is what the denominator closure of [section 4](#) removes.

Accordingly, once the one-body DtN repair is inserted, the one-body self/static seed becomes

$$\begin{aligned} \frac{L_{\text{self+static},1\text{-body}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3 \right) + O(c^{-6}), \end{aligned} \quad (47)$$

which is now identical to the exact target [equation \(14\)](#).

The corresponding generic-frame two-body self/static block is then obtained by the usual symmetric promotion of the one-body monomials:

$$L_{2,\text{self+static}} = \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{7Gm_A m_B}{8r} (v_A^4 + v_B^4) + \frac{2G^2 m_A m_B}{r^2} (m_B v_A^2 + m_A v_B^2) + \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}. \quad (48)$$

This is the entire self/static input used later in the comparable-mass ADM lift. No quartic cross-velocity monomial appears in [equation \(48\)](#); there is no term proportional to  $v_{AB}(v_A^2 + v_B^2)$ ,  $v_A^2 v_B^2$ ,  $v_{AB}^2$ , or  $v_{An} v_{Bn}$ -type structures. Likewise, at order  $G^2/r^2$  only the direct one-body-descended terms ( $m_B v_A^2 + m_A v_B^2$ ) are present. The necessity of the missing comparable-mass cross structures is therefore not hidden inside the self/static sector; it will emerge cleanly as a residual in the next-stage ADM comparison.

**Claim 3** (Frozen self/static 2PN seed). *Within the closure hierarchy of the present paper, the full self/static contribution to  $L_2$  is exactly [equation \(48\)](#). In the strict one-body limit it reduces to [equation \(47\)](#), which matches the exact isotropic Schwarzschild test-mass expansion through 2PN.*

**Claim class.** ***Controlled reduction** for the minimal Bernoulli self exactification and quadratic local mass scaling. **Full assembly** for the statement that the repaired one-body self/static seed matches the exact isotropic Schwarzschild target through 2PN..*

## 5.4 Scope and regime of use

Two cautions should be kept visible.

First, [equation \(48\)](#) is not yet the full conservative two-body 2PN answer. It is only the self/static seed. Once inserted into the exact Legendre-transform comparison, it leaves a nonzero comparable-mass residual. The whole point of the next section is to solve that residual in a symmetry-respecting invariant basis and thereby determine the genuinely new 2PN cross block.

Second, the present sector is derived under the same controlled small-body logic as the rest of the effective particle reduction. In particular, universal 2PN terms must dominate over explicit finite-size corrections. The appendix derives the sharp gate

$$\frac{F_{\text{finite-size}}^{(2)}}{F_{\text{universal}}^{(2)}} \sim \left( \frac{ac^2}{GM} \right)^2, \quad (49)$$

so the relevant regime is not merely  $a \ll r$ , but the stronger universal-2PN condition  $a \ll GM/c^2$ . This caveat does not alter the algebraic two-body proof once the regime is declared, but it is part of the honest scope statement of the paper.

## 6 Comparable-mass 2PN ADM lift

The previous two sections fixed everything that can be read off from the one-body sector and from the minimal self/static continuation of the carried toy-model ledger. But that still does not prove the full conservative two-body 2PN result. Once the repaired one-body denominator and the static gate  $\lambda_\rho = 1/2$  have been imposed, there can still remain genuinely comparable-mass terms that vanish in the strict test-mass limit and therefore cannot be seen by one-body matching alone.

This section isolates exactly those missing terms and solves for them by matching to the standard generic-frame ADM Hamiltonian through 2PN. The key point is that, once the Newtonian+1PN ledger is frozen, the passage from the Lagrangian to the Hamiltonian becomes

algebraically simple enough that every new 2PN cross monomial enters linearly. The comparable-mass lift is therefore not an open-ended fit: it is an exact residual solve in a compact invariant basis.

### 6.1 Perturbative Legendre transform through 2PN

Let

$$L = L_0 + \epsilon L_1 + \epsilon^2 L_2, \quad \epsilon \equiv c^{-2}, \quad (50)$$

with  $L_0$  quadratic in the velocities,

$$L_0 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{G m_A m_B}{r}. \quad (51)$$

Introduce the six-vector notation  $v = (\mathbf{v}_A, \mathbf{v}_B)$ ,  $p = (\mathbf{p}_A, \mathbf{p}_B)$ , and the diagonal mass matrix

$$M \equiv \text{diag}(m_A I_3, m_B I_3). \quad (52)$$

Then

$$p = \frac{\partial L}{\partial v} = Mv + \epsilon \frac{\partial L_1}{\partial v} + \epsilon^2 \frac{\partial L_2}{\partial v}. \quad (53)$$

Writing the perturbative inverse as

$$v = v_0 + \epsilon v_1 + \epsilon^2 v_2 + O(\epsilon^3), \quad v_0 = M^{-1} p = \left( \frac{\mathbf{p}_A}{m_A}, \frac{\mathbf{p}_B}{m_B} \right), \quad (54)$$

and defining

$$A_0 \equiv \left. \frac{\partial L_1}{\partial v} \right|_{v=v_0}, \quad (55)$$

the Legendre transform through 2PN collapses to the standard compact form

$$H_1 = -L_1(v_0), \quad H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0. \quad (56)$$

The derivation is routine and is deferred to Appendix E; the only point needed in the main text is the consequence. Once the full carried 1PN block is frozen, the quadratic correction  $\frac{1}{2} A_0^T M^{-1} A_0$  is already fixed. Therefore any *new* 2PN Lagrangian block contributes to the Hamiltonian only through

$$\Delta H_{2,\text{new}} = -L_{2,\text{new}}(v_0). \quad (57)$$

This is the algebraic reason the comparable-mass lift is tractable: after the lower-order ledger is frozen, the remaining 2PN matching problem is linear in the unknown cross coefficients.

**Claim class.** *Exact for the perturbative Legendre-transform identity [equation \(56\)](#)..*

### 6.2 Residual structure with no new cross sector

Now insert the full frozen Newtonian+1PN ledger together with the repaired self/static 2PN seed from [equation \(48\)](#). By construction, the 1PN piece still Legendre-transforms exactly to the standard EIH/ADM target at order  $c^{-2}$ . The only possible mismatch therefore sits at order  $c^{-4}$ .

If one includes *no* new comparable-mass 2PN cross sector beyond the self/static seed, the Legendre-transformed Hamiltonian does *not* yet equal the standard generic-frame ADM target. But the mismatch has an important and very clean form: every missing term is a genuinely comparable-mass structure. In particular, the residual contains only three classes of monomials:

1. quartic cross-velocity terms proportional to  $G/r$ ,

2. quadratic-velocity comparable-mass terms proportional to  $G^2/r^2$ ,
3. one static cross mass polynomial proportional to  $G^3 m_A^2 m_B^2 / r^3$ .

No new free  $v^6$  term appears. No new one-body  $Uv^4$ ,  $U^2v^2$ , or  $U^3$  term appears. Those sectors were already fixed by the minimal self/static seed and by the one-body denominator closure of [section 4](#). What remains is therefore purely a comparable-mass lift.

This observation is more than a classification. It means that the residual vanishes in the strict test-mass limit with the heavy source held at rest. So the yet-missing 2PN content is not hiding in the one-body closure; it is exactly the additional two-body cross data needed to lift the repaired one-body answer to the full comparable-mass problem.

### 6.3 Compact invariant basis and unique solve

To solve the residual, we use the smallest symmetry-respecting generic-frame basis built from the carried invariants  $v_A^2$ ,  $v_B^2$ ,  $v_{AB}$ ,  $v_{An}$ , and  $v_{Bn}$ . For the  $G/r$  quartic sector we take

$$\begin{aligned} Q_1 &= v_{AB}(v_A^2 + v_B^2), & Q_2 &= v_{An}v_{Bn}(v_A^2 + v_B^2), & Q_3 &= v_A^2v_B^2, \\ Q_4 &= v_{AB}^2, & Q_5 &= v_A^2v_{Bn}^2 + v_B^2v_{An}^2, & Q_6 &= v_{AB}v_{An}v_{Bn}, & Q_7 &= v_{An}^2v_{Bn}^2. \end{aligned} \quad (58)$$

For the  $G^2/r^2$  quadratic-velocity sector we take

$$\begin{aligned} T_1 &= m_Bv_A^2 + m_Av_B^2, & T_2 &= m_Av_A^2 + m_Bv_B^2, & T_3 &= (m_A + m_B)v_{AB}, \\ T_4 &= (m_A + m_B)v_{An}v_{Bn}, & T_5 &= m_Bv_{An}^2 + m_Av_{Bn}^2, & T_6 &= m_Av_{An}^2 + m_Bv_{Bn}^2. \end{aligned} \quad (59)$$

The static comparable-mass slot is spanned by the single monomial

$$S_1 \equiv \frac{G^3 m_A^2 m_B^2}{r^3}. \quad (60)$$

Accordingly, the most general added comparable-mass block in this compact basis is

$$L_{2,\text{cross}}^{\text{ans}} = \frac{Gm_Am_B}{r} \sum_{i=1}^7 q_i Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{i=1}^6 t_i T_i + s_1 S_1. \quad (61)$$

Because of [equation \(57\)](#), matching the residual Hamiltonian to the standard generic-frame ADM target is now a linear algebra problem for the unknown coefficients  $q_i$ ,  $t_i$ , and  $s_1$ .

**Claim 4** (Unique comparable-mass residual solve). *In the compact invariant basis [equations \(58\)](#) to [\(60\)](#), the residual solve is unique and fixes the added comparable-mass 2PN coefficients to*

$$\begin{aligned} q_1 &= -\frac{7}{4}, & q_2 &= -\frac{1}{4}, & q_3 &= \frac{11}{8}, & q_4 &= \frac{1}{4}, \\ q_5 &= -\frac{5}{8}, & q_6 &= \frac{3}{2}, & q_7 &= \frac{3}{8}, \end{aligned} \quad (62)$$

$$t_1 = 0, \quad t_2 = \frac{11}{8}, \quad t_3 = -\frac{15}{4}, \quad t_4 = 0, \quad t_5 = 0, \quad t_6 = \frac{15}{8}, \quad (63)$$

$$s_1 = \frac{5}{4}. \quad (64)$$

Three of the quadratic basis coefficients therefore vanish automatically,

$$t_1 = t_4 = t_5 = 0. \quad (65)$$

This is useful structurally as well as algebraically. It shows that the missing comparable-mass information does *not* require every symmetry-allowed monomial. The exact ADM lift lands in a still smaller subspace than the original basis makes available.

**Claim class.** *Exact for the residual solve once the frozen lower-order ledger and the declared self/static 2PN seed are inserted..*

## 6.4 Added comparable-mass cross block

Substituting the solved coefficients into [equation \(61\)](#) gives the required added comparable-mass 2PN block,

$$\begin{aligned}
L_{2,\text{cross}} = & \frac{Gm_A m_B}{r} \left[ -\frac{7}{4} v_{AB}(v_A^2 + v_B^2) - \frac{1}{4} v_{An} v_{Bn}(v_A^2 + v_B^2) + \frac{11}{8} v_A^2 v_B^2 + \frac{1}{4} v_{AB}^2 \right. \\
& \left. - \frac{5}{8} (v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2} v_{AB} v_{An} v_{Bn} + \frac{3}{8} v_{An}^2 v_{Bn}^2 \right] \\
& + \frac{G^2 m_A m_B}{r^2} \left[ \frac{11}{8} (m_A v_A^2 + m_B v_B^2) - \frac{15}{4} (m_A + m_B) v_{AB} + \frac{15}{8} (m_A v_{An}^2 + m_B v_{Bn}^2) \right] \\
& + \frac{5G^3 m_A^2 m_B^2}{4r^3}.
\end{aligned} \tag{66}$$

As throughout, the overall post-Newtonian factor  $c^{-4}$  is understood. Equation [equation \(66\)](#) is one of the central equations of the paper: it is the unique extra comparable-mass content that must be added to the repaired self/static seed in order to lift the one-body-complete 2PN answer to the full generic-frame conservative two-body target.

Several features of [equation \(66\)](#) should be noted immediately. First, its entire quartic  $G/r$  sector is genuinely cross-coupled; no term reduces to another copy of a one-body monomial. Second, in the  $G^2/r^2$  sector, only the  $T_2$ ,  $T_3$ , and  $T_6$  structures survive the exact solve. Third, the static cross term is positive and precisely large enough to upgrade the self/static mass polynomial from  $(m_A^2 + m_B^2)$  to  $(m_A^2 + 5m_A m_B + m_B^2)$  once the full 2PN block is assembled in the next section.

## 6.5 Sanity checks to report in the main text

Three quick checks should be stated explicitly before moving on to the final assembly.

First, the frozen 1PN block still transforms exactly to the standard generic-frame ADM Hamiltonian at order  $c^{-2}$ . The present solve does not repair a hidden lower-order error; it addresses only a genuine 2PN comparable-mass residual.

Second, the full added cross block [equation \(66\)](#) vanishes in the strict test-mass limit with the heavy source held at rest. The quartic  $G/r$  terms all require cross kinematics and therefore disappear immediately. The  $G^2/r^2$  and static pieces are suppressed by at least one additional power of the light mass, so after dividing by the test mass they vanish as well. This is exactly as it should be: the one-body closure had already fixed the entire strict test-mass sector.

Third, the static mass polynomial comes out correctly only after the comparable-mass lift is included. The repaired self/static seed contributed

$$\frac{G^3 m_A m_B}{4r^3} (m_A^2 + m_B^2), \tag{67}$$

whereas the added cross block contributes

$$\frac{5G^3 m_A^2 m_B^2}{4r^3} = \frac{G^3 m_A m_B}{4r^3} (5m_A m_B). \tag{68}$$

Together these give the exact ADM-compatible polynomial

$$\frac{G^3 m_A m_B}{4r^3} (m_A^2 + 5m_A m_B + m_B^2). \tag{69}$$

That upgrade is a useful one-line check that the comparable-mass residual solve is not merely qualitative; it lands on the exact mass structure required by the standard Hamiltonian target.

The next section simply assembles [equations \(48\)](#) and [\(66\)](#) into the full two-body 2PN Lagrangian and records the exact equality to the standard generic-frame ADM result.

## 7 Full two-body 2PN Lagrangian and exact ADM match

The previous sections isolated the proof into logically distinct parts. The carry-forward Newtonian+1PN ledger was frozen. The one-body missing 2PN response slot was closed by the denominator repair of [section 4](#). The minimal self/static 2PN seed was then fixed in [section 5](#). Finally, [section 6](#) showed that the remaining comparable-mass discrepancy is a purely algebraic residual and solved it uniquely in a compact invariant basis.

This section simply assembles those ingredients and states the final theorem in the same style as the 1PN EIH-match section. The result is a complete conservative two-body 2PN Lagrangian within the stated closure hierarchy, together with exact equality of its Legendre transform to the standard generic-frame ADM Hamiltonian target.

### 7.1 Derived two-body Lagrangian through 2PN

By definition,

$$L_{\text{cons}}^{\text{derived}} = L_0 + \frac{1}{c^2} L_1 + \frac{1}{c^4} L_{2,\text{full}} + O(c^{-6}), \quad (70)$$

with the frozen lower-order blocks

$$L_0 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{G m_A m_B}{r}, \quad (71)$$

$$L_1 = \frac{1}{8} (m_A v_A^4 + m_B v_B^4) + \frac{G m_A m_B}{r} \left[ \frac{3}{2} (v_A^2 + v_B^2) - \frac{7}{2} v_{AB} - \frac{1}{2} v_{An} v_{Bn} \right] - \frac{G^2 m_A m_B (m_A + m_B)}{2r^2}, \quad (72)$$

and with the full 2PN block obtained by adding the frozen self/static seed [equation \(48\)](#) to the uniquely solved comparable-mass cross block [equation \(66\)](#):

$$L_{2,\text{full}} = \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{G m_A m_B}{r} \left[ \frac{7}{8} (v_A^4 + v_B^4) - \frac{7}{4} v_{AB} (v_A^2 + v_B^2) - \frac{1}{4} v_{An} v_{Bn} (v_A^2 + v_B^2) \right. \\ \left. + \frac{11}{8} v_A^2 v_B^2 + \frac{1}{4} v_{AB}^2 - \frac{5}{8} (v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2} v_{AB} v_{An} v_{Bn} + \frac{3}{8} v_{An}^2 v_{Bn}^2 \right] \\ + \frac{G^2 m_A m_B}{r^2} \left[ \left( 2m_B + \frac{11}{8} m_A \right) v_A^2 + \left( 2m_A + \frac{11}{8} m_B \right) v_B^2 - \frac{15}{4} (m_A + m_B) v_{AB} \right. \\ \left. + \frac{15}{8} (m_A v_{An}^2 + m_B v_{Bn}^2) \right] + \frac{G^3 m_A m_B}{4r^3} (m_A^2 + 5m_A m_B + m_B^2). \quad (73)$$

Equation (73) is the final conservative two-body 2PN Lagrangian block derived in this paper.

Written out in one line, the full conservative two-body Lagrangian through 2PN is therefore

$$L_{\text{cons}}^{\text{derived}} = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{G m_A m_B}{r} \\ + \frac{1}{c^2} \left\{ \frac{1}{8} (m_A v_A^4 + m_B v_B^4) + \frac{G m_A m_B}{r} \left[ \frac{3}{2} (v_A^2 + v_B^2) - \frac{7}{2} v_{AB} - \frac{1}{2} v_{An} v_{Bn} \right] \right. \\ \left. - \frac{G^2 m_A m_B (m_A + m_B)}{2r^2} \right\} + \frac{1}{c^4} L_{2,\text{full}} + O(c^{-6}). \quad (74)$$

Several structural features are worth recording explicitly. First, the free kinetic hierarchy now reads  $\frac{1}{8} m v^4$  at 1PN and  $\frac{1}{16} m v^6$  at 2PN, exactly as required by the carried Bernoulli/self packaging. Second, the repaired one-body/self/static data are visible in the  $\frac{7}{8}$ , 2, and  $\frac{1}{4}$

coefficients that survive in the strict test-mass reduction of [equation \(73\)](#). Third, all genuinely comparable-mass new content sits in the cross structures that were absent from the self/static seed and were solved uniquely by the ADM residual lift.

A useful bookkeeping remark is that every coefficient in [equation \(73\)](#) now has a named provenance. The coefficients  $1/16$  and  $7/8$  descend from the minimal Bernoulli exactification of the self sector. The coefficients  $2$  and  $1/4$  are the repaired one-body  $U^2v^2$  and pure-static slots after the denominator closure and static gate are imposed. The remaining quartic cross-velocity coefficients, the nontrivial  $G^2/r^2$  comparable-mass coefficients, and the  $+5/4 m_A^2 m_B^2$  static cross upgrade come from the exact residual solve of [section 6](#). So the final answer is not a single omnibus ansatz; it is an assembly of independently audited sectors.

## 7.2 Standard generic-frame ADM target and exact equality

Let

$$H_{\text{cons}}^{\text{derived}} = H_0 + \frac{1}{c^2} H_1 + \frac{1}{c^4} H_2 + O(c^{-6}) \quad (75)$$

be the perturbative Legendre transform of [equation \(70\)](#). The standard conservative generic-frame two-body ADM Hamiltonian through 2PN is written in the corresponding expansion

$$H_{\text{cons}}^{\text{ADM}} = H_0^{\text{ADM}} + \frac{1}{c^2} H_{1\text{PN}}^{\text{ADM}} + \frac{1}{c^4} H_{2\text{PN}}^{\text{ADM}} + O(c^{-6}). \quad (76)$$

The Newtonian and 1PN equalities were already frozen earlier:

$$H_0 = H_0^{\text{ADM}}, \quad H_1 = H_{1\text{PN}}^{\text{ADM}}. \quad (77)$$

Thus the only nontrivial content left is the order- $c^{-4}$  block.

**Claim 5** (Full conservative two-body 2PN assembly). *After restoring the overall factor  $c^{-4}$ , the full assembled two-body 2PN Lagrangian [equation \(74\)](#) Legendre-transforms exactly to the standard conservative generic-frame ADM Hamiltonian target through 2PN. Equivalently,*

$$H_2 = H_{2\text{PN}}^{\text{ADM}}, \quad \text{so that} \quad \Delta H_2 \equiv H_2 - H_{2\text{PN}}^{\text{ADM}} = 0. \quad (78)$$

The proof is now short because all of the hard work has already been separated out. By [equation \(56\)](#), once the frozen 1PN ledger is fixed, every new 2PN Lagrangian term enters the Hamiltonian only through its direct value at  $v_0 = p/m$ . The self/static block [equation \(48\)](#) supplies the repaired one-body content but leaves a nonzero comparable-mass residual. In [section 6](#), that residual was solved exactly in the compact invariant basis [equations \(58\) to \(60\)](#), yielding the unique added cross block [equation \(66\)](#). Substituting that block into the Legendre transform removes the residual identically, which is precisely the statement of [equation \(78\)](#).

It is worth stating explicitly what kind of equality this is. The present paper does not merely show that the assembled two-body Lagrangian has the right test-mass limit or that its coefficients can be gauge-shifted into a familiar shape. It shows something stronger: within the declared closure hierarchy, the concrete assembled Lagrangian [equation \(74\)](#) has a perturbative Legendre transform whose 2PN block is exactly the standard generic-frame ADM target. So the entire conservative two-body 2PN coefficient set is closed.

A few short sanity checks should be kept visible here. First, the frozen 1PN Lagrangian still transforms to the standard ADM  $H_{1\text{PN}}$  target exactly, so the new 2PN completion does not spoil any lower-order result. Second, the added comparable-mass cross block vanishes in the strict test-mass limit with body  $B$  held at rest, confirming that no one-body coefficient has been silently retuned. Third, the static mass polynomial in the assembled 2PN block is upgraded from  $(m_A^2 + m_B^2)$  in the raw self/static seed to  $(m_A^2 + 5m_A m_B + m_B^2)$ , exactly the mass structure required by the Hamiltonian-side match.

**Claim class.** *Full assembly for the exact equality [equation \(78\)](#), with **Exact** algebra and the previously declared **Controlled reduction** / **Protocol closure** inputs inherited from the one-body closure and self/static seed..*

### 7.3 Compact statement of the proof chain

The logical chain can now be summarized in one paragraph. The exact projection / Poisson-hook / worldtube backbone supplies the reduced particle variables and the already-frozen Newtonian+1PN ledger. The one-body DtN-invariant denominator closure fixes the missing test-mass 2PN  $U^2v^2$  slot without altering the carried 1PN coefficient. Minimal Bernoulli exactification plus quadratic local mass scaling then fix the self/static seed, with the pure-static isotropic gate selecting  $\lambda_\rho = 1/2$ . Once those inputs are frozen, the remaining mismatch with the standard generic-frame ADM target is a purely comparable-mass residual. That residual is solved uniquely in a compact invariant basis, and adding the resulting cross block yields the full assembled Lagrangian [equation \(74\)](#). The final equality [equation \(78\)](#) is therefore not a global fit; it is the end-point of an auditable sector-by-sector proof chain.

One immediate consequence is worth stating outright. Once the closure hierarchy of the present paper is accepted, there are no remaining free dimensionless coefficients in the full conservative two-body 2PN sector. What remains open is not the algebraic assembly, but the deeper constructive explanation of some of its new comparable-mass coefficients from the inner-throat response side.

### 7.4 Short interpretive pointer to the appendices

The main proof stops at the exact ADM match, because that is the smallest readable route to the conservative 2PN theorem. But the assembled cross block is not an opaque remainder term. The appendix program shows that it already admits a compact constructive packaging: the frozen odd dipole wake from 1PN, a genuinely new even  $P_0 \oplus P_2$  support layer, and a pure-potential geometry-closure term. Those appendix developments are important for understanding *why* the solved coefficients organize so economically, but they are not needed for the main proof that the assembled Lagrangian matches the standard conservative generic-frame ADM target exactly.

## 8 Fixed versus symbolic quantities

One of the main purposes of the present paper is to make the coefficient status of the 2PN toy-model program transparent. Earlier stages of the project necessarily mixed together several logically different objects:

1. quantities already fixed by exact identities, controlled reductions, or exact algebraic matching,
2. quantities fixed only after a specific conservative response hierarchy is declared,
3. appendix-side throat data that are now fixed at zero frequency by the solved conservative 2PN cross target, and
4. quantities that should still remain symbolic because they belong to dynamic DtN/PDE observables, finite-size structure, or sectors beyond the conservative near-zone 2PN problem.

Now that the full conservative two-body 2PN assembly has been completed, it is useful to record that distinction explicitly. The point of the present section is not to soften the main theorem, but to sharpen it. Within the closure hierarchy adopted in this paper, the conservative two-body 2PN sector contains no remaining free *dimensionless* coefficients. What remains symbolic belongs either to the declared response packaging itself, to appendix-side dynamic throat observables, or to physics beyond the scope of the present conservative derivation.

## 8.1 Quantities fixed by the present derivation chain

The following quantities should now be regarded as fixed carry-forward values or fixed outputs of the present 2PN derivation.

**Carried lower-order backbone.** The lower-order ledger remains frozen throughout the paper. In particular,

$$\kappa_\rho = 1, \quad n = 5, \quad \kappa_{\text{add}} = \frac{1}{2}, \quad (79)$$

while the conservative parity-even wake coefficients remain

$$C_{\parallel} = -\frac{7}{2}, \quad C_L = -\frac{1}{2}, \quad (80)$$

and the full Newtonian+1PN two-body Lagrangian is already fixed to equal the standard conservative EIH target exactly. These are no longer tunable inputs. They are the frozen backbone on which the 2PN calculation is built.

**One-body 2PN denominator closure.** The missing test-mass response slot is fixed by the minimal invariant denominator closure of [section 4](#):

$$\alpha = 0, \quad \mu = \frac{32768}{3249}, \quad D_{\text{eff}}(u) = C_s(u) \left[ 1 + \frac{32768}{3249} (G(u) - 1)^2 \right]. \quad (81)$$

Equivalently,

$$D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3), \quad (82)$$

so the exact isotropic Schwarzschild one-body target is reproduced through 2PN.

**Static 2PN gate.** Quadratic local mass scaling plus the pure-static isotropic test-mass gate fix

$$\lambda_\rho = \frac{1}{2}, \quad (83)$$

so the two-body static 2PN block is

$$L_{2,\text{static}} = \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}. \quad (84)$$

In the one-body reduction this is exactly the statement that the  $U^3$  coefficient is fixed to  $+\frac{1}{4}$ .

**Comparable-mass residual solve.** Once the repaired one-body/self/static seed is frozen, the remaining comparable-mass 2PN discrepancy is fixed uniquely by the exact ADM residual solve of [section 6](#). The solved coefficients are

$$\begin{aligned} q_1 &= -\frac{7}{4}, & q_2 &= -\frac{1}{4}, & q_3 &= \frac{11}{8}, & q_4 &= \frac{1}{4}, \\ q_5 &= -\frac{5}{8}, & q_6 &= \frac{3}{2}, & q_7 &= \frac{3}{8}, \end{aligned} \quad (85)$$

$$t_1 = 0, \quad t_2 = \frac{11}{8}, \quad t_3 = -\frac{15}{4}, \quad t_4 = 0, \quad t_5 = 0, \quad t_6 = \frac{15}{8}, \quad (86)$$

$$s_1 = \frac{5}{4}. \quad (87)$$

Thus the full added comparable-mass cross block is no longer an ansatz. It is a uniquely solved object.

**Full conservative two-body 2PN sector.** After assembly, the complete derived two-body Lagrangian [equation \(74\)](#) satisfies

$$H_2 = H_{2\text{PN}}^{\text{ADM}}, \quad \Delta H_2 \equiv H_2 - H_{2\text{PN}}^{\text{ADM}} = 0. \quad (88)$$

So, within the declared closure hierarchy, the entire conservative two-body 2PN coefficient ledger is fixed. There are no remaining free dimensionless coefficients in the main theorem.

## 8.2 Quantities fixed only within the declared response hierarchy

A smaller set of quantities is also fixed in the present paper, but only after one explicit conservative response hierarchy is declared. These should be treated as *protocol-fixed*, not as assumption-free theorems of the fully dynamical moving-throat bulk problem.

**Carried breathing/pressure-volume response.** The 1PN closure already fixed

$$\kappa_{\text{PV}} = \frac{3}{2}, \quad \beta_{1\text{PN}} = \kappa_\rho + \kappa_{\text{add}} + \kappa_{\text{PV}} = 3, \quad (89)$$

within the adiabatic one-degree-of-freedom response protocol. These values are carried unchanged into the present 2PN paper. Likewise, the same closure fixes the breathing slope

$$g_1 = \frac{57}{64}, \quad (90)$$

which is the only geometry-slope datum needed in the main-text one-body 2PN denominator match.

**Minimal conservative one-body packaging.** The specific invariant ansatz

$$D_{\text{eff}}(u) = C_s(u)[1 + \alpha(G - 1) + \beta(G - 1)^2] \quad (91)$$

is itself part of the declared low-frequency conservative packaging. Within that ansatz, preserving the frozen 1PN slot forces  $\alpha = 0$ , and the exact isotropic one-body target then fixes the quadratic coefficient uniquely. But the ansatz is still a declared closure move rather than a direct theorem of the full moving-throat PDE.

**Minimal self/static continuation.** The self-sector Bernoulli exactification and the quadratic local mass-scaling rule are also part of the stated closure hierarchy. They are the smallest conservative choices that preserve the frozen lower-order ledger while generating the universal 2PN one-body/self/static seed. The exact ADM match proves that they are sufficient for the conservative two-body 2PN theorem, but it does not by itself prove that no richer inner-throat response exists beyond this minimal packaging.

The cleanest way to summarize the status is therefore the following. The main conservative coefficient ledger is fixed, but some of the response-side *explanatory* machinery remains protocol-dependent until the inner-throat DtN/PDE problem is solved directly.

## 8.3 Appendix-side throat data already fixed at zero frequency

Although the main proof stops at the exact ADM match, the appendix program already shows that the solved 2PN cross block admits a compact throat-response packaging. Within that packaging, several zero-frequency throat data are no longer free.

**Odd-channel static residues.** The conservative odd dipole sector fixes

$$R_{1\perp} = \frac{7}{2}, \quad R_{10} = 4. \quad (92)$$

These are the zero-frequency transverse and longitudinal dipole residues carried by the minimal dressed odd channel.

**Even support residues.** The conservative even support sector fixes the canonical zero-frequency support residues to

$$R_0 = R_{20} = R_{21} = R_{22} = 1. \quad (93)$$

Equivalently, the even support metric is canonically normalized at zero frequency in the basis used by the appendices.

**Scalar source vector and geometry-closure deficit.** The solved even  $P_0 \oplus P_2$  operator fixes the source vector

$$J = \left( \frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right), \quad (94)$$

so that

$$J \cdot J = \frac{381}{80}. \quad (95)$$

Here  $J$  is the appendix-side throat-response source-weight vector, not a Maxwell current. Because the exact solved static cross coefficient is only  $5/4$ , the direct closure term is fixed to the deficit

$$\Delta_{\text{geom}} = \frac{281}{80}. \quad (96)$$

Thus the appendix-side zero-frequency even operator is no longer phenomenological: both its support/source data and its required pure- $U$  closure subtraction are fixed by the solved conservative 2PN cross target.

These quantities are not needed to state the main theorem, but they matter for the constructive interpretation of the result. They show that the full conservative 2PN cross sector can already be packaged as carried odd dipole wake data, a minimal even support/source layer, and one direct geometry-closure term.

## 8.4 Open DtN/PDE observables

What remains open on the inner-throat side is now sharply constrained. The conservative 2PN algebra fixes the zero-frequency data, but it does not yet fix the dynamic pole scales of the minimal axisymmetric low-frequency DtN completion. Those open observables are

$$\Omega_{1\perp}, \quad \Omega_{10}, \quad \Omega_0, \quad \Omega_{20}, \quad \Omega_{21}, \quad \Omega_{22}, \quad \Omega_g. \quad (97)$$

Equivalently, in a low-frequency Taylor expansion one may write

$$Y_0(\omega) = 1 + \chi_0 \omega^2 + \cdots, \quad Y_2(\omega) = 1 + \chi_2 \omega^2 + \cdots, \quad (98)$$

$$Y_{1\perp}(\omega) = \frac{7}{2} + \chi_{1\perp} \omega^2 + \cdots, \quad Y_{10}(\omega) = 4 + \chi_{10} \omega^2 + \cdots, \quad (99)$$

and the coefficients  $\chi$  are precisely the first genuinely new inner-throat dynamic observables beyond the conservative zero-frequency 2PN match.

A near-spherical support simplification is still available if desired,

$$\Omega_{20} = \Omega_{21} = \Omega_{22}, \quad (100)$$

but this is optional and is not required anywhere in the main proof. The important point is that the remaining PDE problem is now narrow and explicit: one no longer needs to guess the conservative zero-frequency operator, but only to derive its dynamical completion.

## 8.5 Other quantities that remain symbolic

Beyond the open pole data above, several broader structures should still be treated as symbolic or unsolved.

**Absolute dimensional normalizations.** The present paper fixes dimensionless conservative coefficients, but it does not by itself fix the absolute dimensional normalization of the parent medium. In particular, in

$$P(\rho) = K_{\text{EOS}} \rho^n, \quad n = 5, \quad (101)$$

the exponent  $n$  is fixed, but the dimensional constant  $K_{\text{EOS}}$  is not. Likewise,  $G$  and  $c$  enter the reduced description as measured effective constants; the present paper fixes how they appear in the dimensionless ledger, not their numerical values from first principles.

**Projection/localization profiles and finite-size moments.** The exact projected identities used by the reduction do not require explicit closed forms for the detailed projection and localization profiles,

$$W(w), \quad Z(w), \quad (102)$$

and the 2PN proof likewise does not fix explicit finite-size or higher-multipole data such as worldtube quadrupoles. These remain profile-dependent and belong to a more detailed finite-size theory.

**General dynamic response beyond the conservative low-frequency closure.** Outside the specific conservative low-frequency packaging adopted here, the internal response should remain symbolic. In particular, one should expect more general data such as damping, phase lag, compliance, multi-mode mixing, or driven/open-system couplings, none of which is fixed by the present conservative near-zone 2PN derivation. The same caution applies to the detailed Family-1 soft-wall / endcap inertia completion: those constructions strongly constrain the dynamic interpretation, but they are not part of the main algebraic theorem.

**Beyond-conservative-2PN sectors.** The present paper does not solve radiation reaction, dissipative leakage, spin couplings, higher post-Newtonian orders, or strong-field non-perturbative completion. Likewise, the paper does not claim a first-principles solution of the fully dynamical moving-throat bulk PDE for arbitrary motion. The worldtube/particle reduction remains a controlled effective description rather than a complete direct bulk solution.

## 8.6 Compact status ledger

For convenience, [table 2](#) summarizes the status of the main quantities entering the present 2PN paper.

## 8.7 Interpretation

The cleanest way to summarize the paper is therefore the following.

Within the declared closure hierarchy, the conservative two-body 2PN sector of the toy model is fully fixed and matches the standard generic-frame conservative ADM target exactly. What remains symbolic is no longer the main coefficient ledger itself, but absolute normalization, generalized dynamic throat response, finite-size structure, open-system or dissipative physics, and the fully solved moving-throat bulk PDE completion.

Table 2: Status ledger for the main quantities entering the conservative two-body 2PN derivation.

| Quantity                                                                                  | Status                       | Comments                                                                                                     |
|-------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------|
| $\kappa_\rho = 1, n = 5$                                                                  | fixed carry-forward          | Newtonian and optical lower-order backbone; historical gravity-side $q = 1$ is now written $\kappa_\rho = 1$ |
| $\kappa_{\text{add}} = 1/2, C_{\parallel} = -7/2, C_L = -1/2$                             | fixed carry-forward          | frozen topology/wake data from the 1PN program                                                               |
| $\kappa_{\text{PV}} = 3/2, \beta_{\text{1PN}} = 3$                                        | protocol-fixed carry-forward | adiabatic one-DOF breathing closure                                                                          |
| $g_1 = 57/64$                                                                             | protocol-fixed carry-forward | breathing slope used in the one-body 2PN closure                                                             |
| $D_{\text{eff}}(u) = C_s[1 + \frac{32768}{3249}(G-1)^2]$                                  | fixed within hierarchy       | exact one-body denominator repair through 2PN                                                                |
| $\lambda_\rho = 1/2$                                                                      | fixed                        | pure-static isotropic gate                                                                                   |
| $q_i, t_i, s_1$ of equations (85) to (87)                                                 | fixed                        | unique comparable-mass residual solve                                                                        |
| full assembled $L_{\text{cons}}^{\text{derived}}$                                         | fixed                        | exact generic-frame conservative ADM match through 2PN                                                       |
| $R_{1\perp} = 7/2, R_{10} = 4$                                                            | appendix-side fixed          | odd zero-frequency throat residues                                                                           |
| $R_0 = R_{20} = R_{21} = R_{22} = 1$                                                      | appendix-side fixed          | even support residues                                                                                        |
| $J = (4/\sqrt{5}, 5/4, 0, 0, 0, 0)$                                                       | appendix-side fixed          | scalar source vector                                                                                         |
| $\Delta_{\text{geom}} = 281/80$                                                           | appendix-side fixed          | pure- $U$ geometry-closure deficit                                                                           |
| $\Omega_{1\perp}, \Omega_{10}, \Omega_0, \Omega_{20}, \Omega_{21}, \Omega_{22}, \Omega_g$ | symbolic/open                | dynamic DtN/PDE pole scales                                                                                  |
| $K_{\text{EOS}}, W(w), Z(w)$                                                              | symbolic                     | dimensional normalization and profile data                                                                   |
| finite-size moments / higher multipoles                                                   | symbolic                     | beyond the universal monopole 2PN limit                                                                      |
| dissipative, radiative, open-system sectors                                               | symbolic                     | outside the conservative near-zone theorem                                                                   |

That is a stronger statement than was available at 1PN. At 1PN, the main open items still included the comparable-mass 2PN cross content and its throat-side constructive packaging. After the present paper, the remaining problem has narrowed to a much more specific one: to derive from the inner-throat dynamics the pole structure and dynamic completion of a zero-frequency operator whose conservative 2PN coefficient ledger is already known.

## 9 Verification and reproducibility ledger

The derivation in this paper is analytic rather than computational. The accompanying Wolfram Language (WL) files and symbolic derivation notes are therefore not presented as black-box generators of the final result. Their role is narrower and more auditable: they act as a referee archive for the algebraic identities, carried regressions, coefficient solves, and end-to-end assembly statements used in the main text.

At 2PN, the verification archive is split into two layers. The first layer consists of compact executable harnesses, whose job is to check the frozen lower-order ledger, the one-body closure, the self/static seed, the test-mass target, and the finite-size gate in explicit PASS/FAIL form. The second layer consists of symbolic derivation ledgers, whose job is to record the comparable-mass ADM residual solve and the appendix-side throat-response reconstruction in a way that can be

rechecked coefficient by coefficient. This split reflects the actual structure of the project archive: the compact self/static and one-body checks are already bundled as executable scripts, whereas the heavier comparable-mass and throat-response modules are currently documented as exact symbolic solves rather than as a single released WL harness.

This distinction matters for interpretation. The archive verifies that the coefficient ledger derived in the previous sections is internally consistent *within the stated closure hierarchy*. It does not replace any analytic step, and it does not upgrade the paper into an assumption-free theorem of a fully solved moving-throat bulk dynamics. Its purpose is reproducibility and auditability: every load-bearing statement of the conservative 2PN assembly can be checked either by a named executable test or by an explicit symbolic residual recorded in the derivation ledger.

## 9.1 Structure of the verification archive

The verification archive used for the present paper is summarized in [table 3](#). It has three layers: the inherited 1PN freeze, the compact executable 2PN harness, and the symbolic 2PN derivation ledger.

The division of labor among these artifacts mirrors the logic of the paper itself. The inherited 1PN suite freezes the lower-order ledger that the present paper is not allowed to retune. The compact 2PN harness checks the one-body, self, static, and test-mass parts of the proof in explicit executable form. The comparable-mass ADM lift and the appendix-side throat-response modules are then made reproducible by a manifest-driven suite whose component scripts expose their residual bases and solved coefficient sets explicitly, so that the final match can be checked coefficient by coefficient rather than trusted as an opaque notebook output.

## 9.2 What is actually being checked

The archive is designed around the same claim taxonomy used in the main text, and the checks naturally fall into five groups.

**(i) Frozen lower-order regressions.** Before any 2PN step is accepted, the carried lower-order ledger is rechecked. The executable archive verifies, among other items,

$$\kappa_\rho = 1, \quad n = 5, \quad \kappa_{\text{add}} = \frac{1}{2}, \quad \kappa_{\text{PV}} = \frac{3}{2}, \quad \beta_{\text{1PN}} = 3, \quad (103)$$

together with the one-body scalar/optical 1PN self coefficient and the pair-counted static 1PN coefficient. This matters because the present 2PN paper is not permitted to improve its target by silently shifting any part of the already-solved Newtonian+1PN backbone.

**(ii) One-body and self/static 2PN checks.** The compact 2PN harness verifies the exact  $n = 5$  Bernoulli continuation, including the optical-index expansion

$$N(U) = 1 + 2\frac{U}{c^2} + 6\frac{U^2}{c^4} + O(c^{-6}), \quad (104)$$

and then checks the minimal self-sector exactification,

$$\frac{L_{\text{self,min}}}{m} = -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 6U^2v^2 \right) + O(c^{-6}), \quad (105)$$

including the specific coefficients  $1/16$ ,  $7/8$ , and  $6$ . It also verifies the cautionary check that a fully Bernoulli-exactified mass prefactor would generate a forbidden  $+3/2$  static coefficient already at 1PN.

**(iii) Static hierarchy and response-template checks.** The same harness verifies that quadratic local mass scaling yields

$$L_{2,\text{static}} = \frac{\lambda_\rho G^3 m_A m_B (m_A^2 + m_B^2)}{2r^3}, \quad (106)$$

checks selected unit-normalized three-body and four-body cubic-static monomials, and verifies the generic one-fast-mode adiabatic-elimination template through cubic order. These checks do not close the full throat dynamics, but they certify that the exact algebra used to package the self/static seed and the one-degree-of-freedom response correction is internally consistent.

**(iv) Test-mass target and regime checks.** The compact executable harness confirms that the two-body self/static reduction agrees exactly with the direct one-body construction, that the isotropic Schwarzschild target has its required static and mixed coefficients, and that the pure-static isotropic gate fixes

$$\lambda_\rho = \frac{1}{2}. \quad (107)$$

It then checks that the remaining self/static discrepancy is exactly

$$\Delta L_{\text{self+static} \rightarrow \text{iso}} = 4U^2 v^2, \quad (108)$$

which is precisely the missing one-body slot repaired in [section 4](#). Finally, it verifies the universal finite-size gate

$$\frac{F_{\text{finite-size}}^{(2)}}{F_{\text{universal}}^{(2)}} = \left( \frac{ac^2}{GM} \right)^2, \quad (109)$$

so the regime of validity of the universal conservative 2PN force is made explicit.

**(v) Comparable-mass ADM lift and appendix-side constructive checks.** The exact comparable-mass lift is verified in a different style. The symbolic derivation ledger records the perturbative Legendre identity

$$H_1 = -L_1(v_0), \quad H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0, \quad (110)$$

with  $v_0 = p/m$  and  $A_0 = (\partial L_1 / \partial v)_{v=v_0}$ , then writes the residual in an explicit invariant basis and solves it uniquely for  $q_i$ ,  $t_i$ , and  $s_1$ . Because the basis and coefficients are given in closed form, the full Hamiltonian residual can be rechecked term by term. Likewise, the appendix-side constructive program records an explicit final low-frequency throat module whose symbolic residual against the solved added conservative 2PN cross target vanishes identically. These are not presently packaged as one compact released PASS/FAIL script, but they are still reproducible because the residual basis, coefficient ledger, and zero-residual statements are written out explicitly.

### 9.3 Reference run summaries

For the executable layers of the archive, the PASS/FAIL status can be read off directly from the reference transcripts. The inherited 1PN master harness and end-to-end assembly script report

$$\text{PASS count} = 60, \quad \text{FAIL count} = 0, \quad \text{Passes: } 37, \quad \text{Fails: } 0, \quad \text{Skips: } 0, \quad (111)$$

while the constructive wake module reports the real-parameter solution

$$a_H = 0, \quad a_L = \pm \frac{\sqrt{3}}{2}, \quad K_{\text{vec}} = \frac{2}{\pi^2}, \quad C_{\parallel} = -\frac{7}{2}, \quad C_L = -\frac{1}{2}. \quad (112)$$

For the compact executable 2PN layer, the reference transcript reports

$$\text{Passes: } 42, \quad \text{Fails: } 0, \quad \text{Skips: } 0, \quad (113)$$

and terminates with an explicit all-pass summary. In other words, every executable check currently bundled with the 2PN project snapshot closes cleanly.

For the manifest-driven symbolic 2PN suite, the current run reports

$$\text{Files passed: } 22, \quad \text{Files failed: } 0, \quad \text{Detected FAIL lines: } 0, \quad (114)$$

while preserving the exact coefficient-level outputs of the individual derivation scripts.

The symbolic derivation suite is still organized as an exact coefficient solve rather than as a single monolithic transcript. Its reproducibility metadata is the sequence of exact outputs written in the ledger scripts: the unique comparable-mass coefficients  $(q_i, t_i, s_1)$ , the full assembled  $L_{2,\text{full}}$ , the exact generic-frame conservative ADM equality

$$H_2 = H_{2\text{PN}}^{\text{ADM}}, \quad (115)$$

and the final appendix-side statement that the low-frequency throat module has a symbolic residual that vanishes identically against the solved added cross target. These should be read as symbolic reproducibility metadata rather than as substitutes for the derivation itself.

## 9.4 What the verification archive does *not* verify

It is equally important to state the scope limits of the archive.

First, neither the executable harnesses nor the symbolic derivation notes solve the fully dynamical moving-throat bulk PDE for arbitrary motion. The paper remains a conservative near-zone 2PN derivation within a declared closure hierarchy, built on the same controlled worldtube and response reductions used elsewhere in the project.

Second, the compact executable 2PN harness is intentionally narrower than the full paper. It covers the frozen 1PN regressions, the Bernoulli continuation, the self/static seed, the isotropic one-body target, and the finite-size gate. The exact comparable-mass ADM lift and the appendix-side throat-response packaging are executed as a manifest-driven suite of focused symbolic ledger scripts rather than folded into the compact referee harness itself. This is a statement about how the archive is currently organized, not about any missing analytic step in the paper itself.

Third, the archive verifies only the conservative sector addressed in the present paper. It does not claim to cover dissipative leakage, radiation reaction, spin couplings, higher-PN orders, strong-field completion, or non-adiabatic internal dynamics beyond the declared response closures.

In short, the archive verifies the paper *as stated*, not a stronger claim than the paper itself makes.

## 9.5 Reproducibility conventions

Four reproducibility conventions are worth recording explicitly.

**Named executable checks.** Every nontrivial statement in the compact harnesses is given a human-readable PASS/FAIL name. This makes it possible to compare the paper’s prose directly with the verification transcript.

**Local assumptions.** The executable checks do not hide their regime assumptions globally. Quasi-staticity, positivity domains, small-body ordering, and the declared response closures are attached to the relevant symbolic reductions, so the reader can see which equalities are exact and which are conditional.

**Explicit residual bases.** For the comparable-mass lift and the appendix-side throat modules, reproducibility comes from explicit residual bases and explicit solved coefficient ledgers. The point is not to hide a large notebook behind a single “zero” output, but to expose the exact polynomial basis on which the residual is being solved.

**Modular audit trail.** The full verification story is intentionally modular. Lower-order freeze, one-body closure, self/static seed, comparable-mass lift, and constructive throat packaging are kept separate so that failure in one sector cannot be masked by compensating changes in another. This is especially important at 2PN, where it would otherwise be easy to blur together one-body repairs, static bookkeeping, and genuinely new comparable-mass cross content.

For readability, the main text reports only the archive structure, the all-pass summaries, and the zero-residual assembly statements. The compact master harness is now supplemented by a manifest-driven full-suite runner; a future single-file transcript could still be added later as a repository note or supplementary appendix, but the present paper already contains enough metadata for a referee to audit the derivation sector by sector.

**Bottom line.** The purpose of the verification archive is not to ask the reader to trust a computation instead of a derivation. It is to provide an auditable backstop showing that the carried and newly introduced coefficients close exactly under the stated assumptions:

$$D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3), \quad \lambda_\rho = \frac{1}{2}, \quad H_2 = H_{2\text{PN}}^{\text{ADM}}. \quad (116)$$

The appendix-side constructive program then goes further and shows that the solved added cross sector can already be packaged into a minimal low-frequency throat-response module with zero symbolic residual.

**Claim class.** *Documentary / reproducibility note. This section does not add a new dynamical proof. It records how the inherited 1PN suite, the compact executable 2PN harness, and the symbolic 2PN derivation ledger jointly verify the exact identities, controlled reductions, protocol closures, and full-match assembly statements used elsewhere in the paper..*

## 10 Conclusions

### 10.1 What has been proven

The main result of this paper is that the 4D toy model now supports a full *conservative two-body 2PN derivation* within a clearly stated closure hierarchy. The point of the paper is not merely that the model can be tuned to resemble the standard generic-frame ADM Hamiltonian through order  $c^{-4}$ . Rather, once the derivation is organized by sector and the already-proved Newtonian+1PN ledger is frozen, every coefficient in the conservative two-body 2PN block can be traced to a named ingredient of the parent 4D program or to one exact residual solve whose basis is fully explicit.

The proof chain established here is:

1. exact projection identities, the quasi-static Poisson hook, and the controlled coherent-defect / small-body worldtube reduction inherited from the earlier 4D papers;
2. the frozen Newtonian+1PN ledger, including the exact EIH match, which the present paper is not allowed to retune;
3. a minimal one-body DtN-invariant denominator closure,

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \frac{32768}{3249} (G(u) - 1)^2 \right] = 1 - 4u + 8u^2 + O(u^3), \quad (117)$$

which closes the missing test-mass 2PN response slot while preserving the already-frozen 1PN coefficient;

4. a minimal self/static 2PN seed consisting of Bernoulli exactification in the self sector together with quadratic local mass scaling in the static sector, the latter being fixed by the pure-static isotropic gate to

$$\lambda_\rho = \frac{1}{2}; \quad (118)$$

5. an exact comparable-mass ADM lift, which solves the remaining quartic-velocity, mixed  $G^2/r^2$ , and static-cross residual in a compact, symmetry-respecting invariant basis; and
6. final assembly, for which the full derived conservative two-body Lagrangian satisfies

$$L_{\text{cons}}^{\text{derived}} = L_0 + \frac{1}{c^2}L_1 + \frac{1}{c^4}L_{2,\text{full}} + O(c^{-6}), \quad H_2 = H_{2\text{PN}}^{\text{ADM}}. \quad (119)$$

That last equality is the paper’s central **Full assembly** statement. Within the stated closure hierarchy, the conservative two-body 2PN coefficient ledger is no longer open. The assembled answer is not just correct in the strict test-mass limit and not just compatible with a change of gauge; its perturbative Legendre transform is exactly the standard generic-frame conservative ADM target through 2PN.

Conceptually, the paper does three things at once. First, it upgrades the earlier program from a full conservative 1PN derivation to full conservative 2PN order. Second, it keeps that upgrade readable by separating the proof line from the larger constructive throat-response story: the main text proves the 2PN theorem, while the appendices record the tensor-wake, mouth-port, support-minus-closure, PDE-side DtN, and Family-1 developments that explain how the new comparable-mass sector may be understood from the inner-throat side. Third, it sharpens the bookkeeping of what is fixed, what is protocol-fixed, and what should still remain symbolic.

In that sharpened sense, the paper closes more than one algebraic gap. It shows that the missing one-body 2PN response slot can be repaired without spoiling the frozen 1PN ledger; it shows that the pure-static one-body target selects a unique quadratic local-mass-scaling coefficient; and it shows that the remaining comparable-mass discrepancy is not mysterious but a finite linear residual solve with a unique answer. Once these points are separated cleanly, the final conservative 2PN assembly is surprisingly short.

As at 1PN, the status labels matter. Some steps are **Exact** identities or exact algebraic residual solves. Some are **Controlled reduction** reductions, such as the Poisson hook and the worldtube particle limit. Some are **Protocol closure** closures, such as the low-frequency invariant one-body packaging and minimal self/static response hierarchy. The final equality [equation \(119\)](#) is then a **Full assembly** statement once those previously declared ingredients are inserted.

That is the correct strength of the result. This paper does *not* claim an assumption-free theorem derived from a fully solved moving-throat bulk background for arbitrary motion. It establishes a full conservative two-body 2PN derivation *within a stated closure hierarchy*. Within that hierarchy, however, there are no remaining free dimensionless coefficients in the main conservative two-body 2PN ledger.

## 10.2 What remains open

The scope of the result should be kept equally explicit on the negative side.

First, the paper does not solve the fully dynamical moving-throat PDE from first principles and then derive the particle limit as an automatic theorem. The worldtube / coherent-defect reduction remains a controlled effective description. Likewise, the one-body denominator closure and the minimal self/static continuation are not claimed as universal facts of arbitrary throat

dynamics; they are the minimal conservative closures used in the present paper and are justified precisely by the way they preserve the frozen lower-order ledger and close the exact 2PN target.

Second, the paper does not claim that the appendix-side throat-response reconstruction is already dynamically complete. What the appendices show is that the solved conservative 2PN cross block admits a compact zero-frequency packaging in terms of the carried odd dipole wake, a new even  $P_0 \oplus P_2$  support layer, and a pure-potential geometry-closure term. At that level, important static data are already fixed. But the dynamic pole scales remain genuine PDE/DtN observables. A convenient representative list is

$$\Omega_{1\perp}, \quad \Omega_{10}, \quad \Omega_0, \quad \Omega_{20}, \quad \Omega_{21}, \quad \Omega_{22}, \quad \Omega_g. \quad (120)$$

These are the first genuinely new inner-throat dynamic observables after the conservative zero-frequency 2PN coefficient ledger has been fixed.

Third, the present paper is restricted to the conservative near-zone 2PN sector. It does not address dissipative leakage, radiation reaction, spin couplings, strong-field nonperturbative completion, or the general finite-size / higher-multipole problem beyond the universal monopole regime. The finite-size gate derived in the main text and appendices should therefore be read as part of the honest scope statement rather than as a hidden footnote: the universal conservative 2PN ledger applies in the declared small-body regime, not as a claim about arbitrary extended objects.

These non-claims do not weaken the main theorem. They tell us exactly where the genuine remaining physics now sits. Before the present paper, it was still unclear whether the conservative two-body 2PN ledger could be closed cleanly at all. After the present paper, that algebraic question is settled. What remains open is the dynamical throat explanation of already-solved conservative data, not the existence of the data themselves.

### 10.3 Interface to the next paper

This gives the project a much cleaner handoff than before. The next true job is no longer to continue coefficient matching at conservative zero-frequency 2PN order. That work is finished. The next job is to derive the appendix-side throat data directly from the inner-throat PDE / DtN problem and to turn the present static and low-frequency packaging into a fully dynamical response module.

Put differently, the conservative zero-frequency target has now been promoted from a heuristic guide to a precise construction target. The isotropic cavity has become a useful no-go unit test rather than a vague model. The minimal axisymmetric channel structure is known. The zero-frequency support/source ledger is known. And even the pure-geometry closure deficit is known. So the next paper can focus on a narrower and more physical question: can the Family-1 throat dynamics, with its wall and endcap structure treated honestly, derive the open pole data and dynamic response functions whose static limits are already fixed here?

That is a qualitatively better situation than at the end of the 1PN paper. At 1PN, one still had to discover the new comparable-mass 2PN cross block and determine whether it admitted any constructive throat-response interpretation. After the present paper, the conservative 2PN algebra is complete and the appendix program already shows that the answer is yes: the new cross content does admit a compact throat-response packaging. The remaining task is therefore to derive its dynamics, not to guess its static form.

The most natural immediate follow-ups are then clear. One is a dedicated dynamical throat-response paper, centered on the pole structure, soft-wall / surface inertia completion, and the low-frequency DtN operator. Another is a more systematic finite-size and higher-multipole extension of the worldtube reduction. A third is the move beyond conservative near-zone physics toward dissipation, radiation, and higher post-Newtonian order. But those are now properly sequenced follow-ups rather than unfinished business inside the present paper.

What has been shown here is that the 4D toy model can reproduce the full conservative two-body 2PN ledger in a way that is sector-by-sector transparent, algebraically exact at final assembly, and explicit about its remaining scope conditions. That is the sense in which the project has moved from a successful 1PN derivation to a genuine conservative 2PN proof.

**Claim class.** *Full assembly* as a summary statement of the paper’s central result, with the important scope condition that the derivation is complete within the stated closure hierarchy rather than as an assumption-free theorem of the fully solved parent 4D dynamics..

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## A Notation and conventions

This appendix records the notation and bookkeeping conventions used throughout the paper. It is intentionally shorter than the corresponding appendix in the 1PN paper. The goal here is only to collect the symbols and conventions that actually enter the conservative two-body 2PN derivation and the appendices that support it.

The main text is organized by derivation sector rather than by symbol class, so the same quantities reappear in several different roles: as exact carry-forward data, as low-frequency response invariants, as coefficients in the generic-frame two-body Lagrangian, and as appendix-side throat-response parameters. The purpose of the present appendix is to keep those uses unambiguous and to state explicitly which equalities are literal definitions, which are reduction conventions, and which are claim-class labels.

### A.1 Coordinates, pair variables, and PN bookkeeping

We work in a parent spacetime with coordinates

$$X^M = (t, x, y, z, w) \equiv (t, \mathbf{x}, w), \quad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \quad \mathbf{x} = (x, y, z) \in \mathbb{R}^3. \quad (121)$$

Here  $\mathbf{X}$  denotes the full spatial position in the parent 4-space, whereas  $\mathbf{x}$  denotes the ordinary brane spatial position. Except where appendix-side carry-forward formulas explicitly refer back to the parent construction, the main paper is written entirely in terms of brane-observed effective particle variables.

Body labels  $A, B, C, \dots$  are never summed implicitly. By contrast, ordinary spatial Cartesian components are summed in the usual Euclidean way when dot products are written in index-free form. An overdot always denotes differentiation with respect to coordinate time  $t$ :

$$\mathbf{v}_A \equiv \dot{\mathbf{x}}_A, \quad \mathbf{v}_B \equiv \dot{\mathbf{x}}_B. \quad (122)$$

All dot products such as  $\mathbf{v}_A \cdot \mathbf{v}_B$  and  $\mathbf{n} \cdot \mathbf{v}_A$  are ordinary Euclidean 3-dimensional brane dot products.

For the two-body problem we define

$$\mathbf{r} \equiv \mathbf{x}_A - \mathbf{x}_B, \quad r \equiv |\mathbf{r}|, \quad \mathbf{n} \equiv \frac{\mathbf{r}}{r}, \quad (123)$$

and use the invariant velocity combinations

$$v_A^2 \equiv \mathbf{v}_A \cdot \mathbf{v}_A, \quad v_B^2 \equiv \mathbf{v}_B \cdot \mathbf{v}_B, \quad v_{AB} \equiv \mathbf{v}_A \cdot \mathbf{v}_B, \quad (124)$$

$$v_{An} \equiv \mathbf{v}_A \cdot \mathbf{n}, \quad v_{Bn} \equiv \mathbf{v}_B \cdot \mathbf{n}. \quad (125)$$

These are the only kinematic invariants needed for the generic-frame conservative 2PN Lagrangian basis used in the main text.

The local Newtonian pair potentials are written

$$U_A \equiv \frac{Gm_B}{r}, \quad U_B \equiv \frac{Gm_A}{r}. \quad (126)$$

In the strict one-body reduction, where body  $B$  is promoted to a fixed heavy source, we write

$$m_A \rightarrow m, \quad m_B \rightarrow M, \quad \mathbf{v}_B = 0, \quad U \equiv \frac{GM}{r}, \quad u \equiv \frac{U}{c^2}. \quad (127)$$

The dimensionless variable  $u$  is the natural weak-field bookkeeping variable in the one-body DtN closure and isotropic Schwarzschild comparisons.

We keep both  $G$  and  $c$  explicit throughout the paper; no geometric-unit convention ( $G = c = 1$ ) is adopted. The post-Newtonian counting is standard:

$$\frac{v^2}{c^2} \sim \frac{U}{c^2} \sim O(c^{-2}). \quad (128)$$

Accordingly, the conservative two-body Lagrangian is expanded as

$$L_{\text{cons}} = L_0 + \frac{1}{c^2} L_1 + \frac{1}{c^4} L_2 + O(c^{-6}), \quad (129)$$

where  $L_0$  is the Newtonian block,  $L_1$  the full frozen conservative 1PN block, and  $L_2$  the full conservative 2PN block derived in this paper. Likewise,

$$H_{\text{cons}} = H_0 + \frac{1}{c^2} H_1 + \frac{1}{c^4} H_2 + O(c^{-6}) \quad (130)$$

denotes the perturbatively Legendre-transformed Hamiltonian.

When canonical variables are needed in the ADM comparison, we write  $\mathbf{p}_A$  and  $\mathbf{p}_B$  for the momenta conjugate to  $\mathbf{x}_A$  and  $\mathbf{x}_B$ . As in the main text, the perturbative Legendre transform is performed in the generic frame rather than in a center-of-mass-specialized basis.

The paper uses the standard Lagrangian sign convention

$$L = T - V. \quad (131)$$

Because the Newtonian pair potential energy is

$$V_{AB} = -\frac{Gm_A m_B}{r}, \quad (132)$$

the Newtonian interaction term appears in the Lagrangian as

$$L_{\text{int}}^{(AB)} = +\frac{Gm_A m_B}{r}. \quad (133)$$

This positive Newtonian pair term is used uniformly throughout the paper and its appendices.

Two further bookkeeping conventions are worth making explicit. First, unless stated otherwise, equalities between displayed Lagrangians in the main text are literal equalities in the chosen generic-frame representative, not merely equivalences modulo an unspecified contact transformation. Second, when the paper states an “exact ADM match,” the precise claim is that the perturbative Legendre transform of the assembled Lagrangian satisfies

$$H_2 = H_{2\text{PN}}^{\text{ADM}} \quad (134)$$

in the standard generic-frame conservative target basis.

Finally, the appendix-side throat-response sections use the labels

$$R_{1\perp}, R_{10}, R_0, R_{20}, R_{21}, R_{22}, J, \Delta_{\text{geom}}, \quad (135)$$

for zero-frequency odd and even mouth-response data, source weights, and the pure-potential geometry-closure deficit, and

$$\Omega_{1\perp}, \Omega_{10}, \Omega_0, \Omega_{20}, \Omega_{21}, \Omega_{22}, \Omega_g \quad (136)$$

for the corresponding dynamical pole scales. These quantities are used only in the appendix-side constructive program and are not needed to read the main 2PN proof. In particular, the symbol  $J$  in this appendix notation is a throat-response source-weight vector rather than a Maxwell current.

## A.2 Claim-class labels

The paper uses four claim-class labels to distinguish exact statements from controlled reductions, declared closures, and final end-to-end matches. These labels are part of the scientific bookkeeping of the paper, not merely an expository convenience.

### Exact

Statements that follow directly from definitions, exact algebra, exact projection identities already established in the parent theory, or exact coefficient solves once a residual basis has been declared. In the present paper, the perturbative Legendre algebra and the final residual solve for the added comparable-mass block are typical examples.

### Controlled reductionreduction

Statements that require an explicit regime assumption or reduction hierarchy, such as the quasi-static Poisson hook, the coherent-defect / small-body worldtube reduction, weak-field truncation, or the universal small-body regime used in the self/static 2PN sector. These are not arbitrary guesses, but they are not exact theorems of the fully dynamical parent bulk problem either.

### Protocol closureclosure

Statements that depend on an explicitly declared internal-response or operator closure. In the present paper, this includes the adiabatic breathing-response input inherited from the 1PN program and the minimal low-frequency invariant packaging used to close the one-body missing 2PN denominator slot.

### Full assembly

End-to-end equalities that become exact once all previously declared exact ingredients, controlled reductions, and protocol closures are inserted. The primary example in the present paper is the final statement that the assembled conservative two-body 2PN Lagrangian Legendre-transforms exactly to the standard generic-frame conservative ADM Hamiltonian target.

These labels should be read cumulatively rather than competitively. A final **Full assembly** statement may rely on earlier **Exact** algebra, on explicitly named **Controlled reduction** reductions, and on one or more declared **Protocol closure** closures. The point of the labels is to keep that dependence visible instead of hiding it.

**Practical reading guide.** Three conventions are especially useful to keep in mind while reading the paper:

1. the generic-frame two-body calculation is written entirely in terms of the invariants  $v_A^2$ ,  $v_B^2$ ,  $v_{AB}$ ,  $v_{An}$ , and  $v_{Bn}$ ;
2. the one-body variable  $u = U/c^2$  is reserved for the strict test-mass reduction; and
3. “exact ADM match” always refers to equality of the Legendre-transformed Hamiltonian block, not merely to similarity of the Lagrangian coefficient pattern.

These three points account for most of the otherwise avoidable notational ambiguity in the main derivation.

## B Carry-forward formulas from the earlier 4D and 1PN papers

This appendix collects the specific formulas imported from the earlier 4D and full conservative 1PN papers and used without rederivation in the present 2PN work. It is not a replacement for those earlier derivations. Its job is narrower: to place in one location the exact projection identities, controlled near-zone reductions, and frozen 1PN coefficient formulas that the main text assumes as already established.

Only the load-bearing formulas are repeated here. In particular, we do *not* reprint the full parent action, the full projected momentum appendix, or the longer intermediate derivations that led to the wake and breathing-response closures. The present paper uses those earlier results only through the formulas recorded below.

### B.1 Exact projection formulas carried from the parent 4D paper

The parent 4D paper defines brane observables by a normalized projection kernel  $W(w)$ :

$$\mathcal{P}_W[Q](t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w) Q(t, \mathbf{x}, w) dw, \quad \int_{-\infty}^{\infty} W(w) dw = 1. \quad (137)$$

This is a measurement prescription, not an approximation. Projection and reduction are kept distinct throughout the program.

For the bulk density  $\rho$  and bulk spatial current  $j^i$ , the brane-facing density, current, and velocity are

$$\rho_{\text{br}}(t, \mathbf{x}) \equiv \mathcal{P}_W[\rho], \quad \mathbf{J}_{\text{br}}(t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w) \mathbf{j}_{xyz}(t, \mathbf{x}, w) dw, \quad \mathbf{v}_{\text{br}} \equiv \frac{\mathbf{J}_{\text{br}}}{\rho_{\text{br}}}, \quad (138)$$

where  $\mathbf{j}_{xyz} = (j^x, j^y, j^z)$ . The point to remember is that  $\mathbf{v}_{\text{br}}$  is a ratio of projected quantities, not the projection of the bulk velocity.

Starting from exact bulk continuity,

$$\partial_t \rho + \partial_i j^i = 0, \quad i \in \{x, y, z, w\}, \quad (139)$$

projection gives the exact open-system brane continuity law

$$\partial_t \rho_{\text{br}} + \nabla_3 \cdot \mathbf{J}_{\text{br}} = S_{\text{leak}}, \quad (140)$$

with leakage source

$$S_{\text{leak}}(\mathbf{x}, t) = -[W(w)j^w(\mathbf{x}, w, t)]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w)j^w(\mathbf{x}, w, t)dw. \quad (141)$$

Under fast decay of  $Wj^w$  at infinity this simplifies to

$$S_{\text{leak}}(\mathbf{x}, t) = \int_{-\infty}^{\infty} W'(w)j^w(\mathbf{x}, w, t)dw. \quad (142)$$

Next decompose the brane velocity by the standard Helmholtz split,

$$\mathbf{v}_{\text{br}} = \nabla_3 \phi_{\text{br}} + \mathbf{v}_T, \quad \nabla_3 \cdot \mathbf{v}_T = 0. \quad (143)$$

Then [equation \(140\)](#) becomes the exact longitudinal identity

$$\rho_{\text{br}} \nabla_3^2 \phi_{\text{br}} = S_{\text{leak}} - \partial_t \rho_{\text{br}} - (\nabla_3 \rho_{\text{br}}) \cdot (\nabla_3 \phi_{\text{br}} + \mathbf{v}_T). \quad (144)$$

This identity is one of the core exact carry-forward formulas from the parent 4D paper. It is not yet a Poisson equation; the Poisson equation appears only after an explicit near-zone reduction.

## B.2 Controlled Poisson hook and Newtonian point-particle limit

In the quasi-static near-zone regime, with approximately constant projected density, subleading advection, and slow leakage/time variation, [equation \(144\)](#) reduces to the Poisson hook

$$\nabla_3^2 \phi_{\text{br}} \simeq \frac{S_{\text{eff}}}{\rho_0}. \quad (145)$$

For a localized effective source,

$$S_{\text{eff}}(\mathbf{x}) \sim \mathcal{S} \delta^{(3)}(\mathbf{x}), \quad (146)$$

this gives the standard inverse-square longitudinal profile

$$\phi_{\text{br}}(\mathbf{x}) \sim -\frac{\mathcal{S}}{4\pi\rho_0} \frac{1}{r}, \quad \nabla_3 \phi_{\text{br}}(\mathbf{x}) \sim \frac{\mathcal{S}}{4\pi\rho_0} \frac{\hat{\mathbf{r}}}{r^2}. \quad (147)$$

The measured Newtonian potential is then defined by a normalization of this scalar channel,

$$\Phi_N(\mathbf{x}, t) = \lambda_\Phi \phi_{\text{br}}(\mathbf{x}, t). \quad (148)$$

For an isolated compact source  $B$ , the near-zone exterior field is carried forward as

$$\Phi_B(\mathbf{x}, t) = -\frac{Gm_B}{|\mathbf{x} - \mathbf{X}_B(t)|} + O\left(\frac{Gm_B a_B^2}{|\mathbf{x} - \mathbf{X}_B|^3}\right). \quad (149)$$

The coherent-defect / small-body reduction then turns a compact worldtube into an effective particle source. For a defect with center-of-mass position  $\mathbf{X}_{\text{cm}}(t)$ , characteristic size  $a$ , and external field smooth across the support, the leading equation of motion is Newtonian,

$$M\ddot{\mathbf{X}}_{\text{cm}} = -M\nabla_3 \Phi_N(\mathbf{X}_{\text{cm}}, t) + O(Q \nabla^3 \Phi_N), \quad (150)$$

with the first finite-size correction quadrupolar and suppressed by

$$\frac{|\mathbf{F}_{\text{quad}}|}{|\mathbf{F}_{\text{mono}}|} = O\left(\frac{a^2}{r^2}\right). \quad (151)$$

The corresponding one-body Newtonian point-particle Lagrangian is

$$L_N = \frac{1}{2}M\dot{\mathbf{X}}_{\text{cm}}^2 - M\Phi_N(\mathbf{X}_{\text{cm}}, t), \quad (152)$$

and pair counting for two compact defects gives the frozen Newtonian two-body block

$$L_0 = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 + \frac{Gm_A m_B}{r_{AB}}. \quad (153)$$

This is the only Newtonian particle mechanics imported by the present 2PN paper.

### B.3 Frozen scalar/optical 1PN worldline formulas

The 1PN paper fixed the self-sector and universal free-particle coefficients by the reduced scalar/optical worldline ansatz

$$L_{\text{sc}} = -M_0 \left( 1 + \kappa_\rho \frac{\Phi_N}{c^2} \right) c^2 \sqrt{1 - \frac{v^2/c^2}{1 + (n-1)\Phi_N/c^2}}. \quad (154)$$

Its weak-field expansion is

$$\begin{aligned} L_{\text{sc}} = & -M_0 c^2 - \kappa_\rho M_0 \Phi_N + \frac{1}{2} M_0 v^2 + \frac{1}{8} M_0 \frac{v^4}{c^2} \\ & + M_0 \left( \frac{\kappa_\rho}{2} - \frac{n-1}{2} \right) \frac{\Phi_N v^2}{c^2} + O(c^{-4}). \end{aligned} \quad (155)$$

Newtonian matching fixes

$$\kappa_\rho = 1, \quad (156)$$

where  $\kappa_\rho$  is the historical gravity-side scalar mass-dressing coefficient rather than an electric charge. Weak-field optical consistency for the polytropic equation of state

$$P(\rho) = K_{\text{EOS}} \rho^n \quad (157)$$

fixes

$$n = 5. \quad (158)$$

With these values, the free-particle part of [equation \(155\)](#) becomes

$$L_{\text{free}} = -M_0 c^2 + \frac{1}{2} M_0 v^2 + \frac{1}{8} M_0 \frac{v^4}{c^2} + O(c^{-4}), \quad (159)$$

so the frozen 1PN Lagrangian coefficient of  $v^4$  is  $1/8$ .

The mixed coefficient in [equation \(155\)](#) is

$$\mathcal{C}_{\text{self}} = \frac{\kappa_\rho}{2} - \frac{n-1}{2}. \quad (160)$$

Using [equations \(156\)](#) and [\(158\)](#) gives

$$\mathcal{C}_{\text{self}} = \frac{1}{2} - 2 = -\frac{3}{2}. \quad (161)$$

Since  $\Phi_B(\mathbf{X}_A) = -Gm_B/r_{AB}$ , the symmetric two-body self term is therefore

$$L_{\text{self}}^{(AB)} = \frac{Gm_A m_B}{c^2 r_{AB}} \frac{3}{2} (v_A^2 + v_B^2). \quad (162)$$

This  $+3/2$  coefficient is one of the central frozen inputs to the present 2PN paper.

### B.4 Frozen static and response ledgers at 1PN

The static nonlinear 1PN term comes from local mass scaling and exact pair counting. The effective mass of body  $A$  is written

$$m_A^{\text{eff}} = m_A \left( 1 + \kappa_\rho \frac{\Phi_A^{\text{loc}}}{c^2} \right) + O(c^{-4}), \quad \Phi_A^{\text{loc}} = - \sum_{C \neq A} \frac{Gm_C}{r_{AC}}, \quad (163)$$

with the pair-counted potential

$$L_{\text{pot}} = -\frac{1}{2} \sum_A m_A^{\text{eff}} \Phi_A^{\text{loc}}. \quad (164)$$

Expanding [equation \(164\)](#) gives the compact many-body static formula

$$L_{\text{stat}} = -\frac{\kappa_\rho G^2}{2c^2} \sum_A m_A \left( \sum_{C \neq A} \frac{m_C}{r_{AC}} \right)^2. \quad (165)$$

For two bodies, with  $\kappa_\rho = 1$ , this reduces to

$$L_{\text{stat}}^{(AB)} = -\frac{G^2 m_A m_B (m_A + m_B)}{2c^2 r_{AB}^2}. \quad (166)$$

Thus the frozen static 1PN coefficient is exactly  $-1/2$ .

The 1PN paper also fixed the self-sector ledger in the familiar split

$$\beta_{\text{1PN}} = \kappa_\rho + \kappa_{\text{add}} + \kappa_{\text{PV}}. \quad (167)$$

The topology calculation for a  $w$ -uniform translating throat gives

$$\kappa_{\text{add}} = \frac{1}{2}, \quad (168)$$

while the counterfactual compact 4D bubble would give

$$\kappa_{\text{add}}^{(B^4)} = \frac{1}{3}. \quad (169)$$

This distinction is carried forward as a geometry/topology discriminator rather than as a tunable inertial parameter.

Within the declared adiabatic one-degree-of-freedom breathing closure, the response side is fixed to

$$\kappa_{\text{PV}} = \frac{3}{2}, \quad E_w : E_f : E_{\text{PV}} = 11 : 2 : 5, \quad \frac{d \ln a_*}{d \ln \rho} = -\frac{57}{64}. \quad (170)$$

Therefore the full frozen 1PN self ledger is

$$\beta_{\text{1PN}} = \kappa_\rho + \kappa_{\text{add}} + \kappa_{\text{PV}} = 1 + \frac{1}{2} + \frac{3}{2} = 3. \quad (171)$$

The present 2PN paper imports all of [equations \(168\)](#), [\(170\)](#) and [\(171\)](#) without reopening their derivation.

## B.5 Frozen wake formulas and the parity-even cross sector

The parity-even velocity cross sector is imported from the constructive isotropic wake calculation of the 1PN paper. For two bodies,

$$L_{\text{vec}}^{(AB)} = \frac{G m_A m_B}{c^2 r_{AB}} \left[ C_{\parallel} \mathbf{v}_A \cdot \mathbf{v}_B + C_L (\mathbf{v}_A \cdot \hat{\mathbf{n}}_{AB}) (\mathbf{v}_B \cdot \hat{\mathbf{n}}_{AB}) \right]. \quad (172)$$

The isotropic Fourier-space wake basis gives

$$C_{\parallel} = K_{\text{vec}} \pi^2 (-1 + a_H^2 - \alpha^2), \quad C_L = K_{\text{vec}} \pi^2 (-1 + a_H^2 + \alpha^2), \quad (173)$$

where  $\alpha^2 \equiv a_L^2$  is the longitudinal wake fraction.

The thermodynamic wake closure imported from the 1PN paper is

$$\alpha^2 = 1 - \frac{U}{P} = 1 - \frac{1}{n-1}. \quad (174)$$

Using the already-frozen value  $n = 5$  gives

$$\alpha^2 = \frac{3}{4}. \quad (175)$$

The parity-even real solution then collapses to

$$a_H = 0, \quad K_{\text{vec}} = \frac{2}{\pi^2}, \quad (176)$$

which yields the exact EIH cross coefficients

$$C_{\parallel} = -\frac{7}{2}, \quad C_L = -\frac{1}{2}. \quad (177)$$

Thus the frozen two-body velocity cross term carried into the present paper is

$$L_{\text{cross}}^{(AB)} = \frac{Gm_A m_B}{c^2 r_{AB}} \left[ -\frac{7}{2} \mathbf{v}_A \cdot \mathbf{v}_B - \frac{1}{2} (\mathbf{v}_A \cdot \hat{\mathbf{n}}_{AB})(\mathbf{v}_B \cdot \hat{\mathbf{n}}_{AB}) \right]. \quad (178)$$

No part of the present 2PN paper is allowed to retune these 1PN cross coefficients.

## B.6 Full carried conservative 1PN assembly

Combining the Newtonian sector, free quartic term, self term, cross term, and static term recorded above gives the full conservative two-body 1PN Lagrangian imported from the earlier paper:

$$\begin{aligned} L_{\text{1PN}}^{\text{carried}} = & \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{m_A v_A^4 + m_B v_B^4}{8c^2} + \frac{Gm_A m_B}{r_{AB}} \\ & + \frac{Gm_A m_B}{c^2 r_{AB}} \left[ \frac{3}{2} (v_A^2 + v_B^2) - \frac{7}{2} \mathbf{v}_A \cdot \mathbf{v}_B - \frac{1}{2} (\mathbf{v}_A \cdot \hat{\mathbf{n}}_{AB})(\mathbf{v}_B \cdot \hat{\mathbf{n}}_{AB}) \right] \\ & - \frac{G^2 m_A m_B (m_A + m_B)}{2c^2 r_{AB}^2}. \end{aligned} \quad (179)$$

The exact equality carried forward from the 1PN paper is

$$L_{\text{1PN}}^{\text{carried}} = L_{\text{1PN}}^{\text{EIH}}. \quad (180)$$

This is one of the main frozen inputs to the present 2PN assembly: the current paper adds only the order- $c^{-4}$  block and is not allowed to revise any coefficient in [equation \(179\)](#).

For later reference, the one-body reduction of [equation \(179\)](#) is

$$\frac{L_{\text{test}}}{m} = \frac{1}{2} v^2 + \frac{\mu}{r} + \frac{1}{c^2} \left[ \frac{1}{8} v^4 + \frac{3\mu}{2r} v^2 - \frac{1}{2} \frac{\mu^2}{r^2} \right], \quad \mu \equiv GM, \quad (181)$$

which can be repackaged in the standard orbit-level form

$$\mathcal{L}_{\text{orb}} = \frac{1}{2} (1 + \sigma(r)) (\dot{r}^2 + r^2 \dot{\phi}^2) + \frac{\mu}{r}, \quad \sigma(r) = \frac{\beta_{\text{1PN}} \mu}{c^2 r}. \quad (182)$$

Using  $\beta_{\text{1PN}} = 3$  then gives the carried perihelion result

$$\Delta\phi_{\text{model}} = \frac{2\pi\beta_{\text{1PN}}\mu}{c^2 a(1-e^2)} = \frac{6\pi\mu}{c^2 a(1-e^2)} = \Delta\phi_{\text{GR}}. \quad (183)$$

This is not used directly in the 2PN coefficient solve, but it remains part of the frozen lower-order status ledger inherited by the present paper.

**Claim class.** *Exact* for the projection formulas [equations \(137\), \(140\) and \(144\)](#), for pair counting in [equation \(165\)](#), for the algebraic assembly [equation \(179\)](#), and for the exact equality [equation \(180\)](#) once the imported coefficients are inserted. **Controlled reduction** for the Poisson hook, the small-body worldtube reduction, the local mass-scaling ansatz, the topology result  $\kappa_{\text{add}} = 1/2$ , and the wake closure leading to [equations \(175\) to \(177\)](#). **Protocol closure** for the adiabatic breathing-response input [equation \(170\)](#) and therefore for the carried value  $\beta_{\text{1PN}} = 3$ . **Full assembly** for the exact EIH equality [equation \(180\)](#) and the carried perihelion formula [equation \(183\)](#)..

## C One-body DtN closure details

This appendix records the detailed algebra behind the one-body 2PN closure used in [section 4](#). The main text gave only the minimal proof line: the low-frequency monopole mouth operator supplies two normalized invariants, the exact isotropic Schwarzschild test-mass target fixes the required effective velocity denominator, and preservation of the frozen 1PN slot then leaves a unique minimal quadratic geometry correction.

The purpose of the present appendix is to unpack that chain in a more constructive way. We first write the simplest unit-test mouth operator explicitly, then show why its normalized invariant combinations isolate sound-speed and geometry information, then derive the target denominator directly from the isotropic Schwarzschild one-body expansion, and finally show how the earlier raw resonance-proxy fit factorizes into a clean *raw DtN piece* times a *port correction*. Nothing in this appendix is needed to state the main theorem, but it makes the one-body closure much less opaque.

### C.1 Low-frequency monopole mouth operator on the cylinder unit-test branch

The simplest conservative mouth model retained in the 2PN notes is the monopole branch of a cylinder-like throat with one axial coordinate  $z \in [0, L]$ , a mouth at  $z = 0$ , and a Neumann bottom at  $z = L$ . This is a unit-test branch, not the final moving-throat PDE. Its value is that the low-frequency Dirichlet-to- Neumann map can be written in closed form and already exposes the two geometric quantities that matter for the one-body closure: the effective sound speed  $c_s$  and the throat depth  $L$ .

Let  $\psi(z, \omega)$  satisfy the one-dimensional Helmholtz equation

$$\psi'' + k^2 \psi = 0, \quad k \equiv \frac{\omega}{c_s}, \quad (184)$$

with bottom condition

$$\psi'(L, \omega) = 0 \quad (185)$$

and mouth datum

$$\psi(0, \omega) = \psi_m(\omega). \quad (186)$$

The unique regular solution is

$$\psi(z, \omega) = \psi_m(\omega) \frac{\cos(k(L - z))}{\cos(kL)}. \quad (187)$$

Taking the outward normal derivative at the mouth gives

$$-\partial_z \psi(0, \omega) = -k \tan(kL) \psi_m(\omega). \quad (188)$$

Thus the monopole mouth operator is

$$Z_{00}(\omega; u) = -k \tan(kL) = -\frac{\omega}{c_s} \tan\left(\frac{\omega L}{c_s}\right). \quad (189)$$

Expanding at low frequency yields

$$Z_{00}(\omega; u) = Z_2(u) \omega^2 + Z_4(u) \omega^4 + O(\omega^6), \quad (190)$$

with

$$Z_2(u) = -\frac{L(u)}{c_s^2(u)}, \quad Z_4(u) = -\frac{L^3(u)}{3c_s^4(u)}. \quad (191)$$

The absence of an  $\omega^0$  term is automatic in this conservative unit-test branch: once the static monopole potential is accounted for separately, the retained mouth operator begins at even time-derivative order.

The main text uses the normalized invariant combinations

$$C_s(u) \equiv \frac{Z_4/Z_2^3}{(Z_4/Z_2^3)_0}, \quad G(u) \equiv \frac{Z_4/Z_2^2}{(Z_4/Z_2^2)_0}, \quad (192)$$

where the subscript 0 means evaluation at  $u = 0$ . In the cylinder branch, these become exactly

$$C_s(u) = \frac{c_s^2(u)}{c_{s0}^2}, \quad G(u) = \frac{L(u)}{L_0}. \quad (193)$$

This is why the choice [equation \(192\)](#) is so useful: all irrelevant normalizations and numerical factors cancel, leaving one invariant that tracks the sound-speed sector and one invariant that tracks the geometry / depth sector.

Equivalently, the low-frequency coefficients themselves can be reconstructed from the invariants as

$$\frac{Z_2(u)}{Z_2(0)} = \frac{G(u)}{C_s(u)}, \quad \frac{Z_4(u)}{Z_4(0)} = \frac{G^3(u)}{C_s^2(u)}. \quad (194)$$

This inversion is often the most convenient route for checking explicit series.

## C.2 Current conservative freeze and explicit series data

The one-body 2PN closure is not performed in a vacuum; it inherits specific lower-order data from the frozen 1PN program.

First, the exact Bernoulli continuation of the already-fixed  $n = 5$  barotropic sector gives the sound-speed factor

$$\frac{c_s^2(u)}{c^2} = 1 - 4u, \quad u \equiv \frac{U}{c^2}, \quad U = \frac{GM}{r}. \quad (195)$$

In the present normalized language this is simply

$$C_s(u) = 1 - 4u. \quad (196)$$

Second, the carried low-frequency throat closure writes the geometric depth factor as

$$G(u) = \frac{L(u)}{L_0} = a(u), \quad (197)$$

with the already-fixed 1PN breathing slope

$$g_1 \equiv \left. \frac{dG}{du} \right|_{u=0} = \frac{57}{64}. \quad (198)$$

At the current closure level, the explicit series used in the notes is

$$G(u) = 1 + \frac{57}{64}u + \frac{298821}{131072}u^2 + O(u^3). \quad (199)$$

Substituting [equations \(196\) and \(199\)](#) into [equation \(194\)](#) yields

$$\frac{Z_2(u)}{Z_2(0)} = 1 + \frac{313}{64}u + \frac{2862917}{131072}u^2 + O(u^3), \quad (200)$$

$$\frac{Z_4(u)}{Z_4(0)} = 1 + \frac{683}{64}u + \frac{10301487}{131072}u^2 + O(u^3). \quad (201)$$

Conversely, inserting these series back into [equation \(192\)](#) reproduces

$$C_s(u) = 1 - 4u, \quad G(u) = 1 + \frac{57}{64}u + \frac{298821}{131072}u^2 + O(u^3), \quad (202)$$

so the normalized invariant extraction is self-consistent.

Two features of [equation \(202\)](#) matter for the rest of the appendix. First, the sound-speed invariant is already completely fixed through the current order. Second, only the *linear* geometry slope  $g_1 = 57/64$  will enter into the actual 2PN denominator match; the coefficient of  $u^2$  in  $G(u)$  does not affect the conservative one-body target through  $c^{-4}$ .

### C.3 The exact isotropic Schwarzschild target as a denominator problem

The exact isotropic Schwarzschild test-mass Lagrangian is

$$\frac{L_{\text{Schw}}}{m} = -c^2 \sqrt{\left(\frac{1-u/2}{1+u/2}\right)^2 - (1+u/2)^4 \frac{v^2}{c^2}}. \quad (203)$$

Expanding through total post-Newtonian order  $c^{-4}$  gives

$$\begin{aligned} \frac{L_{\text{Schw}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3 \right) + O(c^{-6}). \end{aligned} \quad (204)$$

In the carried toy-model assembly, the  $v^6$  and  $Uv^4$  terms are already produced by the minimal Bernoulli exactification of the self sector, while the pure-static  $U^3$  term is handled separately by quadratic local mass scaling. So the genuinely missing one-body response datum is the coefficient of  $U^2v^2$ , which must be

$$2. \quad (205)$$

To isolate that datum, write the reduced one-body worldline in the denominator form

$$L_{\text{red}} = -mc^2(1-u)\sqrt{1 - \frac{v^2/c^2}{D(u)}}, \quad D(u) = 1 - 4u + d_2u^2 + O(u^3). \quad (206)$$

Expanding [equation \(206\)](#) gives

$$\begin{aligned} \frac{L_{\text{red}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + \left( 6 - \frac{d_2}{2} \right) U^2v^2 \right) + O(c^{-6}). \end{aligned} \quad (207)$$

Comparing [equation \(207\)](#) with the exact target [equation \(204\)](#) shows that the effective denominator must satisfy

$$6 - \frac{d_2}{2} = 2, \quad \text{hence} \quad d_2 = 8. \quad (208)$$

So the exact isotropic one-body target can be restated as the denominator condition

$$D_{\text{target}}(u) = 1 - 4u + 8u^2 + O(u^3). \quad (209)$$

This repackaging is the reason the low-frequency mouth invariants are useful at all: the entire one-body closure problem reduces to producing the correct quadratic coefficient in [equation \(209\)](#) without changing the already-frozen linear term.

## C.4 Minimal invariant ansatz and uniqueness of the quadratic geometry correction

Within the current conservative hierarchy, the sound-speed invariant  $C_s(u)$  is already frozen by the Bernoulli continuation, so any genuinely new one-body response information must enter through the geometry invariant  $G(u)$ . The smallest analytic invariant ansatz is therefore

$$D_{\text{eff}}(u) = C_s(u) F(G(u)), \quad F(1) = 1, \quad (210)$$

with  $F$  expanded near  $G = 1$ . Through 2PN, this is equivalent to

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \alpha (G - 1) + \beta (G - 1)^2 \right] + O(u^3). \quad (211)$$

Write

$$G(u) = 1 + g_1 u + g_2 u^2 + O(u^3), \quad C_s(u) = 1 - 4u + O(u^2). \quad (212)$$

Then

$$D_{\text{eff}}(u) = 1 + (-4 + \alpha g_1)u + (\beta g_1^2 + \dots)u^2 + O(u^3). \quad (213)$$

Because the  $u$ -coefficient is already frozen by the exact 1PN ledger, preservation of the lower-order slot requires

$$\alpha g_1 = 0. \quad (214)$$

Since the carried breathing slope is nonzero,

$$g_1 = \frac{57}{64} \neq 0, \quad (215)$$

we must have

$$\alpha = 0. \quad (216)$$

So the first admissible correction is quadratic:

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \mu (G(u) - 1)^2 \right]. \quad (217)$$

Expanding [equation \(217\)](#) gives

$$D_{\text{eff}}(u) = 1 - 4u + \mu g_1^2 u^2 + O(u^3). \quad (218)$$

Matching the exact denominator target [equation \(209\)](#) therefore forces

$$\mu g_1^2 = 8. \quad (219)$$

Using [equation \(198\)](#) gives

$$\mu = \frac{8}{g_1^2} = \frac{32768}{3249} \approx 10.0855647892. \quad (220)$$

Hence the final one-body closure used in the main text is

$$D_{\text{eff}}(u) = C_s(u) \left[ 1 + \frac{32768}{3249} (G(u) - 1)^2 \right], \quad (221)$$

which indeed expands to

$$D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3). \quad (222)$$

Three remarks are worth making explicit. First, the uniqueness here is a *through*-2PN uniqueness within the stated analytic invariant hierarchy. Nothing stronger is claimed. Second, only the linear breathing slope  $g_1$  matters for the coefficient match; the higher geometric series coefficients of  $G$  do not enter until higher PN order. Third, the closure is automatically compatible with the frozen 1PN ledger because that compatibility is exactly what forced  $\alpha = 0$ .

**Claim 6** (One-body invariant closure through 2PN). *Within the conservative analytic ansatz [equation \(211\)](#), the frozen 1PN denominator coefficient forces the geometry correction to begin quadratically in  $G - 1$ , and the exact isotropic Schwarzschild test-mass target then fixes the quadratic coefficient uniquely to*

$$\mu = \frac{32768}{3249}. \quad (223)$$

*Equivalently, the completed one-body denominator is exactly*

$$D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3) \quad (224)$$

*through conservative 2PN order.*

### C.5 Relation to the earlier raw resonance-proxy fit

Before the invariant packaging [equation \(221\)](#) was written in its final form, the same low-frequency DtN data had been summarized by a more primitive “raw resonance proxy,” namely

$$D_{\text{raw}}(u) \equiv \frac{[Z_2/Z_4](u)}{[Z_2/Z_4](0)}. \quad (225)$$

Using [equations \(191\)](#) and [\(193\)](#), this can be written exactly as

$$D_{\text{raw}}(u) = \frac{C_s(u)}{G(u)^2}. \quad (226)$$

With the current series data,

$$D_{\text{raw}}(u) = 1 - \frac{185}{32}u + \frac{324075}{65536}u^2 + O(u^3). \quad (227)$$

The problem with [equation \(227\)](#) is immediate: it does *not* preserve the already-frozen 1PN coefficient, because the linear term is  $-185/32$  instead of  $-4$ .

The final closure shows exactly how that raw proxy must be corrected. Define the multiplicative port factor

$$P_{\text{port}}(u) \equiv G(u)^2 [1 + \mu (G(u) - 1)^2]. \quad (228)$$

Then

$$D_{\text{eff}}(u) = D_{\text{raw}}(u) P_{\text{port}}(u). \quad (229)$$

With  $\mu = 32768/3249$ , the current series is

$$P_{\text{port}}(u) = 1 + \frac{57}{32}u + \frac{875093}{65536}u^2 + O(u^3). \quad (230)$$

Multiplying [equations \(227\)](#) and [\(230\)](#) gives back the exact target series

$$D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3). \quad (231)$$

So the earlier port fit was not an arbitrary extra correction. It can be read more cleanly as a factorization statement: the ratio  $Z_2/Z_4$  supplies a raw DtN proxy, while the missing geometry-port dressing is exactly the factor required to preserve the frozen 1PN slot and hit the 2PN one-body target.

This factorization is also useful conceptually. It shows that the one-body closure is not inventing a new denominator from scratch; it is reorganizing the same low-frequency mouth data into a product of a raw response ratio and a compensating geometry-port factor. The raw piece by itself is too naive, but it already contains the right low-frequency ingredients.

## C.6 Scope of the one-body closure

The one-body closure derived above is deliberately narrow. It closes the missing conservative test-mass 2PN response slot and nothing more. Several things are fixed, and several things remain open.

What is fixed is the following. The sound-speed invariant is carried as  $C_s(u) = 1 - 4u$ . The geometry slope is carried as  $g_1 = 57/64$ . Within the declared conservative analytic invariant hierarchy, these data uniquely fix  $\mu = 32768/3249$  and therefore fix the denominator target  $D_{\text{eff}}(u) = 1 - 4u + 8u^2 + O(u^3)$ . As a result, the one-body  $U^2v^2$  slot is no longer open in the main 2PN assembly.

What remains open is the deeper derivation of this closure from the fully dynamical inner-throat PDE and its frequency-dependent DtN operator. In particular, the present appendix does not derive the analytic structure of  $F(G)$  beyond the minimal quadratic term needed at 2PN, nor does it determine the dynamic pole scales of the appendix-side mouth-response operator. Those are genuine inner-throat observables and belong to the later PDE / DtN program rather than to the conservative algebraic proof completed in the main text.

**Claim class.** *Controlled reduction* for the cylinder / Neumann-bottom unit-test branch and the import of the Bernoulli and breathing series data. **Protocol closure** for the analytic invariant ansatz  $D_{\text{eff}} = C_s[1 + \alpha(G - 1) + \beta(G - 1)^2]$ . **Exact** for the algebraic elimination of  $\alpha$ , for the determination of  $\mu = 32768/3249$ , and for the factorization  $D_{\text{eff}} = D_{\text{raw}}P_{\text{port}}$  once the closure hierarchy is declared. **Full assembly** for the statement that the resulting denominator reproduces the exact isotropic Schwarzschild one-body target through conservative 2PN order..

## D Bernoulli exactification, quadratic local mass scaling, and the static hierarchy

This appendix collects the detailed self/static formulas used in [section 5](#). The main text kept only the proof line: exact  $n = 5$  Bernoulli continuation fixes the sound-speed sector, minimal optical exactification supplies the raw self contribution through 2PN, quadratic local mass scaling fixes the static hierarchy, and the pure-static isotropic test-mass gate selects  $\lambda_\rho = 1/2$ . The purpose of the present appendix is to write those statements in a form that can be checked term by term.

There are four points to keep separate. First, the barotropic  $n = 5$  Bernoulli continuation is exact within the reduced matter model and already determines the optical index through 2PN. Second, only the optical denominator is Bernoulli-exactified in the self sector; a naïve full exactification of the mass prefactor would spoil the already-frozen 1PN static ledger. Third, quadratic local mass scaling generates a compact many-body cubic-static hierarchy, of which the two-body term is selected by the isotropic one-body target and the higher-body terms are predictions of the same bookkeeping rule. Fourth, any interpretation of these terms as universal  $1/r^4$ -order structure must respect the worldtube finite-size gate derived at the end of the appendix.

### D.1 Exact $n = 5$ continuation of the barotropic sector

The barotropic sector carried from the earlier papers is

$$P(\rho) = K_{\text{EOS}}\rho^5, \quad U_{\text{int}}(\rho) = \frac{K_{\text{EOS}}}{4}\rho^5, \quad (232)$$

so the enthalpy variable is

$$h(\rho) \equiv \frac{dU_{\text{int}}}{d\rho} = \frac{5K_{\text{EOS}}}{4}\rho^4, \quad (233)$$

and the sound speed satisfies

$$c_s^2(\rho) = \frac{1}{m_{\text{mat}}} \frac{dP}{d\rho} = \frac{5K_{\text{EOS}}}{m_{\text{mat}}} \rho^4. \quad (234)$$

In the reduced matter units used in the harness, one sets  $m_{\text{mat}} = 1$  and fixes the background normalization by demanding

$$c_s^2(\rho_0) = c^2. \quad (235)$$

This gives

$$K_{\text{EOS}} = \frac{c^2}{5\rho_0^4}, \quad h_0 \equiv h(\rho_0) = \frac{c^2}{4}. \quad (236)$$

The Bernoulli continuation is then exact:

$$h(\rho) = h_0 + \Phi, \quad (237)$$

with  $\Phi$  the local Newtonian potential. Using [equations \(233\)](#) and [\(236\)](#), one solves [equation \(237\)](#) directly for the density profile,

$$\rho_{\text{Bern}}(\Phi) = \rho_0 \left(1 + \frac{4\Phi}{c^2}\right)^{1/4}. \quad (238)$$

Substituting [equation \(238\)](#) into [equation \(234\)](#) gives

$$\frac{c_s^2(\Phi)}{c^2} = 1 + \frac{4\Phi}{c^2}. \quad (239)$$

Equivalently, in the attractive one-body notation  $\Phi = -U$  with  $u \equiv U/c^2$ ,

$$\frac{c_s^2}{c^2} = 1 - 4u. \quad (240)$$

This is the exact Bernoulli input used throughout the 2PN one-body and self-sector analysis.

The corresponding optical index is

$$N(\Phi) \equiv \frac{c}{c_s(\Phi)} = \left(1 + \frac{4\Phi}{c^2}\right)^{-1/2}, \quad (241)$$

so in  $U = -\Phi$  variables,

$$N(U) = \left(1 - \frac{4U}{c^2}\right)^{-1/2} = 1 + 2\frac{U}{c^2} + 6\frac{U^2}{c^4} + O(c^{-6}). \quad (242)$$

The coefficients in [equation \(242\)](#) are exactly the ones checked by the compact 2PN harness and are the source of the carried  $\frac{7}{8}Uv^4$  and raw  $6U^2v^2$  terms in the self sector.

With this optical continuation in hand, the minimal self exactification used in the main text is

$$L_{\text{self},\text{min}} = -mc^2 \left(1 + \frac{\Phi}{c^2}\right) \sqrt{1 - \frac{v^2}{c^2(1 + 4\Phi/c^2)}}. \quad (243)$$

Expanding [equation \(243\)](#) through total 2PN order gives

$$\begin{aligned} \frac{L_{\text{self},\text{min}}}{m} = & -c^2 + \frac{1}{2}v^2 - \Phi + \frac{1}{c^2} \left( \frac{1}{8}v^4 - \frac{3}{2}\Phi v^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 - \frac{7}{8}\Phi v^4 + 6\Phi^2 v^2 \right) + O(c^{-6}). \end{aligned} \quad (244)$$

Writing  $\Phi = -U$  converts this to the more familiar attractive-field form

$$\begin{aligned} \frac{L_{\text{self,min}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 6U^2v^2 \right) + O(c^{-6}). \end{aligned} \quad (245)$$

Thus the exact  $n = 5$  Bernoulli continuation plus minimal optical exactification already fixes the raw self-sector coefficients

$$\frac{1}{16}, \quad \frac{7}{8}, \quad 6. \quad (246)$$

The important obstruction appears if one tries to Bernoulli-exactify the mass prefactor as well. The naïve aggressive continuation would be

$$L_{\text{self,aggr}} = -mc^2 \left( 1 + \frac{4\Phi}{c^2} \right)^{1/4} \sqrt{1 - \frac{v^2}{c^2(1 + 4\Phi/c^2)}}. \quad (247)$$

Its 1PN expansion is

$$\frac{L_{\text{self,aggr}}}{m} = -c^2 + \frac{1}{2}v^2 - \Phi + \frac{1}{c^2} \left( \frac{1}{8}v^4 - \frac{3}{2}\Phi v^2 + \frac{3}{2}\Phi^2 \right) + O(c^{-4}). \quad (248)$$

Since  $\Phi^2 = U^2$ , this would generate a  $+\frac{3}{2}U^2/c^2$  static term at 1PN, which is incompatible with the already-frozen pair-counted static ledger  $-\frac{1}{2}U^2/c^2$ . That is why the present paper exactifies only the optical denominator and leaves the mass prefactor at its frozen linear form.

**Claim class.** *Exact for the Bernoulli continuation [equations \(238\), \(239\) and \(241\)](#). **Controlled reduction** for the minimal self-sector continuation [equation \(243\)](#), whose role is to preserve the frozen 1PN self ledger while generating the raw universal 2PN self coefficients. **Protocol closure** for the decision not to Bernoulli-exactify the mass prefactor beyond linear order because doing so would violate the already-frozen 1PN static regression..*

## D.2 Two-body static sector and $\lambda_\rho = 1/2$

The static sector is extended independently of the optical/self continuation by quadratic local mass scaling. For each body,

$$m_A^{\text{eff}} = m_A \left( 1 + \frac{\Phi_A^{\text{loc}}}{c^2} + \lambda_\rho \frac{(\Phi_A^{\text{loc}})^2}{c^4} \right), \quad m_B^{\text{eff}} = m_B \left( 1 + \frac{\Phi_B^{\text{loc}}}{c^2} + \lambda_\rho \frac{(\Phi_B^{\text{loc}})^2}{c^4} \right), \quad (249)$$

where

$$\Phi_A^{\text{loc}} = -\frac{Gm_B}{r}, \quad \Phi_B^{\text{loc}} = -\frac{Gm_A}{r} \quad (250)$$

for the two-body system. The pair-counted potential is

$$L_{\text{pot}} = -\frac{1}{2} \left( m_A^{\text{eff}} \Phi_A^{\text{loc}} + m_B^{\text{eff}} \Phi_B^{\text{loc}} \right). \quad (251)$$

Expanding [equation \(251\)](#) gives the complete two-body static sector through 2PN:

$$L_{\text{static}}^{(AB)} = \frac{Gm_A m_B}{r} - \frac{G^2 m_A m_B (m_A + m_B)}{2c^2 r^2} + \frac{\lambda_\rho G^3 m_A m_B (m_A^2 + m_B^2)}{2c^4 r^3} + O(c^{-6}). \quad (252)$$

Thus the quadratic local-mass rule fixes the two-body 2PN static coefficient to

$$\frac{\lambda_\rho}{2}. \quad (253)$$

In the strict test-mass reduction, [equation \(252\)](#) becomes

$$\frac{L_{\text{static,1body}}}{m} = U - \frac{1}{2} \frac{U^2}{c^2} + \frac{\lambda_\rho}{2} \frac{U^3}{c^4} + O(c^{-6}). \quad (254)$$

The exact isotropic Schwarzschild target requires the pure-static coefficient of  $U^3/c^4$  to be  $+\frac{1}{4}$ , so the isotropic gate immediately fixes

$$\lambda_\rho = \frac{1}{2}. \quad (255)$$

Once [equation \(255\)](#) is imposed, the two-body static block becomes

$$L_{2,\text{static}}^{(AB)} = \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}, \quad (256)$$

where, as throughout the paper,  $L_2$  denotes the coefficient of  $c^{-4}$  in the post-Newtonian expansion.

Combining the static contribution [equation \(254\)](#) with the minimal self exactification [equation \(245\)](#) gives the raw one-body self/static candidate

$$\begin{aligned} \frac{L_{\text{self+static}}^{\text{raw}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left( \frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left( \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 6U^2v^2 + \frac{\lambda_\rho}{2}U^3 \right) + O(c^{-6}). \end{aligned} \quad (257)$$

Fixing  $\lambda_\rho = 1/2$  matches the pure-static isotropic coefficient exactly and leaves

$$\Delta L_{\text{raw} \rightarrow \text{iso}} = \frac{4U^2v^2}{c^4} \quad (258)$$

as the sole remaining one-body discrepancy. This is precisely the missing test-mass slot closed by the DtN-invariant denominator repair in [section 4](#) and [appendix C](#).

**Claim class.** ***Exact** for the pair-counted two-body static expansion [equation \(252\)](#). **Controlled reduction** for the quadratic local mass-scaling rule. **Full assembly** for the pure-static isotropic selection [equation \(255\)](#) and for the statement that, after this selection, the unrepaired one-body discrepancy sits entirely in the  $U^2v^2$  sector..*

### D.3 Selected three-body and four-body static terms

The same quadratic local mass-scaling rule generates a compact many-body static hierarchy. For an arbitrary collection of bodies with local potentials

$$\Phi_A^{\text{loc}} = - \sum_{C \neq A} \frac{Gm_C}{r_{AC}}, \quad (259)$$

the 2PN static sector is simply

$$L_{2,\text{static}}^{(N)} = -\frac{\lambda_\rho}{2} \sum_A m_A (\Phi_A^{\text{loc}})^3. \quad (260)$$

[Equation \(260\)](#) is the entire higher-body static hierarchy in compact form. The point of the present subsection is only to record the first representative monomials that were checked explicitly in the harness.

For three bodies  $A, B, C$ , one has

$$\Phi_A^{\text{loc}} = -G \left( \frac{m_B}{r_{AB}} + \frac{m_C}{r_{AC}} \right), \quad \Phi_B^{\text{loc}} = -G \left( \frac{m_A}{r_{AB}} + \frac{m_C}{r_{BC}} \right), \quad \Phi_C^{\text{loc}} = -G \left( \frac{m_A}{r_{AC}} + \frac{m_B}{r_{BC}} \right). \quad (261)$$

Expanding [equation \(260\)](#) gives selected cubic-static terms of the form

$$L_{2,\text{static}}^{(ABC)} \supset \frac{3\lambda_\rho G^3}{2} \left[ \frac{m_A m_B^2 m_C}{r_{AB}^2 r_{AC}} + \frac{m_A m_B m_C^2}{r_{AB} r_{AC}^2} + \text{cyclic relabelings} \right], \quad (262)$$

where the overall  $c^{-4}$  factor is understood as part of  $L_2$ . The corresponding unit-normalized coefficients are exactly unity, in agreement with the compact harness checks.

Just as important is what does *not* appear. The coefficient of the pure triangle triple product

$$\frac{1}{r_{AB} r_{AC} r_{BC}} \quad (263)$$

vanishes identically in this hierarchy. Equivalently,

$$\left[ \frac{1}{r_{AB} r_{AC} r_{BC}} \right] L_{2,\text{static}}^{(ABC)} = 0. \quad (264)$$

This is not accidental. Every term in [equation \(260\)](#) is anchored on one body, so each cubic monomial is a product of three pair potentials that all emanate from the same anchor body. A pure triangle product involving three different edges and no common anchor is therefore absent.

The first genuine four-body monomial appears when one body couples simultaneously to three others. For four bodies  $A, B, C, D$ , the harness explicitly checks the coefficient of

$$\frac{1}{r_{AB} r_{AC} r_{AD}}, \quad (265)$$

which is

$$L_{2,\text{static}}^{(ABCD)} \supset \frac{3\lambda_\rho G^3 m_A m_B m_C m_D}{r_{AB} r_{AC} r_{AD}} + \text{relabelings of the anchor body}. \quad (266)$$

Again, the unit-normalized coefficient is exactly one. Thus the first higher-body static prediction of the quadratic local mass-scaling rule is not a pure triangle term, but a common-anchor cubic product.

After imposing the isotropic gate  $\lambda_\rho = 1/2$ , the selected coefficients in [equations \(262\)](#) and [\(266\)](#) are simply halved relative to the formal  $\lambda_\rho$ -normalized hierarchy. The pattern itself, however, is fixed already by [equation \(260\)](#) and does not depend on the test-mass matching step.

**Claim class.** ***Exact** for the compact many-body formula [equation \(260\)](#) once quadratic local mass scaling is declared. **Controlled reduction** for the decision to interpret this hierarchy as the static extension of the same reduced local-potential bookkeeping used at lower order..*

#### D.4 Worldtube finite-size gate

The static hierarchy above is algebraically clean, but any claim that it represents a universal  $1/r^4$ -order force law must be interpreted with care. The same small-body worldtube logic used throughout the project implies an explicit finite-size gate.

Consider a compact body of characteristic radius  $a$  moving in the field of a central mass  $M$  at separation  $r$ . The Newtonian force scale is

$$F_N \sim \frac{GMm}{r^2}. \quad (267)$$

The leading finite-size correction from a quadrupolar worldtube distortion is then of order

$$F_{\text{quad}} \sim F_N \left( \frac{a}{r} \right)^2. \quad (268)$$

By contrast, a universal conservative 2PN correction scales as

$$F_{\text{universal}}^{(2)} \sim F_N \left( \frac{GM}{rc^2} \right)^2. \quad (269)$$

Therefore the ratio of finite-size to universal-2PN effects is

$$\frac{F_{\text{quad}}}{F_{\text{universal}}^{(2)}} \sim \frac{(a/r)^2}{(GM/(rc^2))^2} = \left( \frac{ac^2}{GM} \right)^2. \quad (270)$$

The striking feature of [equation \(270\)](#) is that the separation  $r$  cancels: once one compares against a universal 2PN claim, the relevant gate is no longer merely geometric smallness relative to the orbit.

The required regime is therefore

$$a \ll \frac{GM}{c^2}, \quad (271)$$

not just  $a \ll r$ . This is the honest interpretation bound for any statement that the static hierarchy of the present appendix yields a universal conservative  $1/r^4$  sector. The algebraic coefficient ledger remains correct once the reduction is declared, but the physical scope of universality is narrower than a casual “small body” phrase might suggest.

In practice this means the static hierarchy should be read in two layers. As an algebraic output of quadratic local mass scaling, it is exact within the declared reduced bookkeeping. As a literal universal force law for finite extended bodies, it requires the stronger gate [equation \(271\)](#). This distinction is why the main text states the finite-size caveat explicitly even though it plays no role in the final comparable-mass ADM match.

**Claim class.** *Controlled reduction for the worldtube estimate [equations \(270\) and \(271\)](#). This is not a new coefficient solve; it is the regime statement needed to interpret the static hierarchy honestly..*

## E Perturbative Legendre transform and ADM residual-solve details

This appendix records the algebra behind the main-text 2PN lift of [section 6](#). There are two separate points to keep visible. First, once the Newtonian+1PN ledger is frozen, the perturbative Legendre transform simplifies enough that any *new* 2PN Lagrangian block enters the Hamiltonian linearly. Second, after the repaired self/static seed is inserted, the remaining mismatch with the standard generic-frame conservative ADM target lives in a small, symmetry-respecting comparable-mass basis. The exact 2PN lift is therefore an ordinary linear coefficient solve rather than an open-ended fit.

### E.1 Legendre-transform bookkeeping through 2PN

Write the conservative two-body Lagrangian in the bookkeeping form

$$L = L_0 + \epsilon L_1 + \epsilon^2 L_2, \quad \epsilon \equiv c^{-2}, \quad (272)$$

with quadratic Newtonian block

$$L_0 = \frac{1}{2} v^T M v + V_0, \quad V_0 \equiv \frac{G m_A m_B}{r}, \quad M \equiv \text{diag}(m_A I_3, m_B I_3). \quad (273)$$

Here

$$v \equiv (\mathbf{v}_A, \mathbf{v}_B), \quad p \equiv (\mathbf{p}_A, \mathbf{p}_B), \quad (274)$$

with six-vector dot products understood in the obvious block-Euclidean sense. Because  $V_0$  is velocity-independent, the canonical momenta are

$$p = \frac{\partial L}{\partial v} = Mv + \epsilon \frac{\partial L_1}{\partial v} + \epsilon^2 \frac{\partial L_2}{\partial v}. \quad (275)$$

We seek the inverse relation in the form

$$v = v_0 + \epsilon v_1 + \epsilon^2 v_2 + O(\epsilon^3). \quad (276)$$

At leading order,

$$p = Mv_0, \quad \text{so that} \quad v_0 = M^{-1}p = \left( \frac{\mathbf{p}_A}{m_A}, \frac{\mathbf{p}_B}{m_B} \right). \quad (277)$$

Now define the frozen 1PN gradient, Hessian, and the raw 2PN gradient at  $v = v_0$ :

$$A_0 \equiv \frac{\partial L_1}{\partial v} \Big|_{v=v_0}, \quad B_0 \equiv \frac{\partial^2 L_1}{\partial v \partial v} \Big|_{v=v_0}, \quad C_0 \equiv \frac{\partial L_2}{\partial v} \Big|_{v=v_0}. \quad (278)$$

Expanding [equation \(275\)](#) order by order gives

$$Mv_1 + A_0 = 0, \quad Mv_2 + B_0 v_1 + C_0 = 0. \quad (279)$$

Hence

$$v_1 = -M^{-1}A_0, \quad v_2 = -M^{-1}(B_0 v_1 + C_0). \quad (280)$$

The Hamiltonian is

$$H = p \cdot v - L. \quad (281)$$

Expanding the quadratic Newtonian part gives

$$p \cdot v - \frac{1}{2} v^T M v = \frac{1}{2} v_0^T M v_0 - \frac{\epsilon^2}{2} v_1^T M v_1 + O(\epsilon^3), \quad (282)$$

while Taylor expansion of the higher-order blocks around  $v_0$  gives

$$L_1(v) = L_1(v_0) + \epsilon A_0 \cdot v_1 + O(\epsilon^2), \quad L_2(v) = L_2(v_0) + O(\epsilon). \quad (283)$$

Substituting into [equation \(281\)](#) yields

$$\begin{aligned} H &= \left( \frac{1}{2} v_0^T M v_0 - V_0 \right) + \epsilon [-L_1(v_0)] \\ &\quad + \epsilon^2 \left[ -L_2(v_0) - A_0 \cdot v_1 - \frac{1}{2} v_1^T M v_1 \right] + O(\epsilon^3). \end{aligned} \quad (284)$$

Using [equation \(280\)](#), one has

$$A_0 \cdot v_1 = -A_0^T M^{-1} A_0, \quad v_1^T M v_1 = A_0^T M^{-1} A_0, \quad (285)$$

so the  $\epsilon^2$  block collapses to

$$H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0. \quad (286)$$

Thus the full perturbative Legendre transform through 2PN is

$$H_0 = \frac{1}{2} v_0^T M v_0 - V_0, \quad H_1 = -L_1(v_0), \quad H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0. \quad (287)$$

This is the appendix derivation of the compact main-text identity [equation \(56\)](#).

The key structural point is that  $A_0$  depends only on the already-frozen 1PN block. Therefore the quadratic correction  $\frac{1}{2}A_0^T M^{-1}A_0$  is fixed once and for all before any new 2PN comparable-mass information is introduced. In particular, for any *new* 2PN Lagrangian block  $L_{2,\text{new}}$ ,

$$\Delta H_{2,\text{new}} = -L_{2,\text{new}}(v_0). \quad (288)$$

This is the algebraic reason the lift in [section 6](#) is linear.

For completeness, the explicit frozen 1PN gradient entering [equation \(286\)](#) is

$$A_{0,A} = \frac{p_A^2}{2m_A^2} \mathbf{p}_A + \frac{Gm_A m_B}{r} \left[ \frac{3}{m_A} \mathbf{p}_A - \frac{7}{2m_B} \mathbf{p}_B - \frac{\mathbf{p}_B \cdot \mathbf{n}}{2m_B} \mathbf{n} \right], \quad (289)$$

$$A_{0,B} = \frac{p_B^2}{2m_B^2} \mathbf{p}_B + \frac{Gm_A m_B}{r} \left[ \frac{3}{m_B} \mathbf{p}_B - \frac{7}{2m_A} \mathbf{p}_A - \frac{\mathbf{p}_A \cdot \mathbf{n}}{2m_A} \mathbf{n} \right], \quad (290)$$

where  $A_0 = (A_{0,A}, A_{0,B})$ . We never need to expand  $\frac{1}{2}A_0^T M^{-1}A_0$  by hand in the paper, but it is useful to record [equations \(289\)](#) and [\(290\)](#) explicitly because they show that the entire quadratic correction is fixed by the already-carried 1PN coefficients.

## E.2 Residual basis and linear solve

The repaired self/static 2PN seed carried into the lift is

$$L_{2,\text{self+static}} = \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{7Gm_A m_B}{8r} (v_A^4 + v_B^4) + \frac{2G^2 m_A m_B}{r^2} (m_B v_A^2 + m_A v_B^2) + \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}, \quad (291)$$

with the overall factor  $c^{-4}$  understood. As explained in the main text, this seed is already exact in the strict one-body sector, so any remaining mismatch with the standard generic-frame conservative ADM target must vanish in the test-mass limit. That observation fixes the shape of the admissible comparable-mass ansatz.

For convenience, we restate the compact generic-frame basis used in [section 6](#). In velocity variables, the quartic  $G/r$  sector is spanned by

$$\begin{aligned} Q_1 &= v_{AB}(v_A^2 + v_B^2), & Q_2 &= v_{An}v_{Bn}(v_A^2 + v_B^2), & Q_3 &= v_A^2 v_B^2, \\ Q_4 &= v_{AB}^2, & Q_5 &= v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2, & Q_6 &= v_{AB}v_{An}v_{Bn}, & Q_7 &= v_{An}^2 v_{Bn}^2, \end{aligned} \quad (292)$$

while the quadratic  $G^2/r^2$  sector is spanned by

$$\begin{aligned} T_1 &= m_B v_A^2 + m_A v_B^2, & T_2 &= m_A v_A^2 + m_B v_B^2, & T_3 &= (m_A + m_B)v_{AB}, \\ T_4 &= (m_A + m_B)v_{An}v_{Bn}, & T_5 &= m_B v_{An}^2 + m_A v_{Bn}^2, & T_6 &= m_A v_{An}^2 + m_B v_{Bn}^2, \end{aligned} \quad (293)$$

and the static comparable-mass slot is

$$S_1 \equiv \frac{G^3 m_A^2 m_B^2}{r^3}. \quad (294)$$

The trial added block is therefore

$$L_{2,\text{cross}}^{\text{ans}} = \frac{Gm_A m_B}{r} \sum_{i=1}^7 q_i Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{i=1}^6 t_i T_i + s_1 S_1. \quad (295)$$

On the Hamiltonian side one simply substitutes  $v \rightarrow v_0 = p/m$ . It is therefore useful to define the induced momentum-basis monomials

$$\hat{Q}_i \equiv Q_i(v_0), \quad \hat{T}_i \equiv T_i(v_0), \quad \hat{S}_1 \equiv S_1. \quad (296)$$

Explicitly,

$$\begin{aligned}
\hat{Q}_1 &= \frac{\mathbf{p}_A \cdot \mathbf{p}_B}{m_A m_B} \left( \frac{p_A^2}{m_A^2} + \frac{p_B^2}{m_B^2} \right), & \hat{Q}_2 &= \frac{(\mathbf{p}_A \cdot \mathbf{n})(\mathbf{p}_B \cdot \mathbf{n})}{m_A m_B} \left( \frac{p_A^2}{m_A^2} + \frac{p_B^2}{m_B^2} \right), \\
\hat{Q}_3 &= \frac{p_A^2 p_B^2}{m_A^2 m_B^2}, & \hat{Q}_4 &= \frac{(\mathbf{p}_A \cdot \mathbf{p}_B)^2}{m_A^2 m_B^2}, \\
\hat{Q}_5 &= \frac{p_A^2 (\mathbf{p}_B \cdot \mathbf{n})^2 + p_B^2 (\mathbf{p}_A \cdot \mathbf{n})^2}{m_A^2 m_B^2}, & \hat{Q}_6 &= \frac{(\mathbf{p}_A \cdot \mathbf{p}_B)(\mathbf{p}_A \cdot \mathbf{n})(\mathbf{p}_B \cdot \mathbf{n})}{m_A^2 m_B^2}, \\
\hat{Q}_7 &= \frac{(\mathbf{p}_A \cdot \mathbf{n})^2 (\mathbf{p}_B \cdot \mathbf{n})^2}{m_A^2 m_B^2}, & & 
\end{aligned} \tag{297}$$

$$\begin{aligned}
\hat{T}_1 &= \frac{m_B}{m_A^2} p_A^2 + \frac{m_A}{m_B^2} p_B^2, & \hat{T}_2 &= \frac{p_A^2}{m_A} + \frac{p_B^2}{m_B}, & \hat{T}_3 &= \frac{m_A + m_B}{m_A m_B} \mathbf{p}_A \cdot \mathbf{p}_B, \\
\hat{T}_4 &= \frac{m_A + m_B}{m_A m_B} (\mathbf{p}_A \cdot \mathbf{n})(\mathbf{p}_B \cdot \mathbf{n}), & \hat{T}_5 &= \frac{m_B}{m_A^2} (\mathbf{p}_A \cdot \mathbf{n})^2 + \frac{m_A}{m_B^2} (\mathbf{p}_B \cdot \mathbf{n})^2, \\
\hat{T}_6 &= \frac{(\mathbf{p}_A \cdot \mathbf{n})^2}{m_A} + \frac{(\mathbf{p}_B \cdot \mathbf{n})^2}{m_B}. & & & 
\end{aligned} \tag{298}$$

By [equation \(288\)](#), the added Hamiltonian contribution is simply

$$H_{2,\text{cross}}^{\text{ans}} = -\frac{G m_A m_B}{r} \sum_{i=1}^7 q_i \hat{Q}_i - \frac{G^2 m_A m_B}{r^2} \sum_{i=1}^6 t_i \hat{T}_i - s_1 \hat{S}_1. \tag{299}$$

Now let  $H_{2,\text{seed}}$  denote the 2PN Hamiltonian obtained by Legendre-transforming the frozen lower-order ledger plus [equation \(291\)](#). The exact generic-frame ADM residual is

$$\Delta H_2 \equiv (H_{2,\text{seed}} + H_{2,\text{cross}}^{\text{ans}}) - H_{2\text{PN}}^{\text{ADM}}. \tag{300}$$

Because the basis

$$\mathfrak{B} = \{\hat{Q}_1, \dots, \hat{Q}_7, \hat{T}_1, \dots, \hat{T}_6, \hat{S}_1\} \tag{301}$$

is linearly independent in a generic frame,  $\Delta H_2 = 0$  is equivalent to the vanishing of each coefficient separately. In other words, the lift reduces to a  $14 \times 14$  linear system for

$$c = (q_1, \dots, q_7, t_1, \dots, t_6, s_1)^T. \tag{302}$$

The symbolic derivation ledger solves this system exactly and uniquely, giving

$$\begin{aligned}
q_1 &= -\frac{7}{4}, & q_2 &= -\frac{1}{4}, & q_3 &= \frac{11}{8}, & q_4 &= \frac{1}{4}, \\
q_5 &= -\frac{5}{8}, & q_6 &= \frac{3}{2}, & q_7 &= \frac{3}{8}, & & 
\end{aligned} \tag{303}$$

$$t_1 = 0, \quad t_2 = \frac{11}{8}, \quad t_3 = -\frac{15}{4}, \quad t_4 = 0, \quad t_5 = 0, \quad t_6 = \frac{15}{8}, \tag{304}$$

$$s_1 = \frac{5}{4}. \tag{305}$$

So three quadratic basis coefficients vanish automatically,

$$t_1 = t_4 = t_5 = 0. \tag{306}$$

That is a useful structural output: the exact lift does not need every symmetry-allowed quadratic monomial.

Substituting [equations \(303\) to \(305\)](#) into [equation \(295\)](#) reproduces exactly the main-text added block [equation \(66\)](#). The solution is therefore unique not merely in the space of all possible 2PN corrections, but already in the smallest compact basis that one would naturally try first.

### E.3 Exact equality to the standard generic-frame ADM target

With the solved coefficients inserted, the trial 2PN Hamiltonian is

$$H_{2,\text{trial}} \equiv H_{2,\text{seed}} + H_{2,\text{cross}}^{\text{ans}} \Big|_{\text{solved } q,t,s}. \quad (307)$$

By construction,

$$H_{2,\text{trial}} - H_{2\text{PN}}^{\text{ADM}} = 0. \quad (308)$$

This is the appendix form of the main theorem used in [section 7](#): once the repaired one-body/self/static seed is frozen, the exact comparable-mass lift closes the remaining generic-frame conservative 2PN residual identically.

It is worth recording explicitly what is being cancelled. The seed [equation \(291\)](#) already reproduces the full strict test-mass sector, so [equation \(308\)](#) does *not* rely on any retuning of the one-body closure. All of the new information lives in comparable-mass structures that vanish when one body is held fixed and the other is taken test-mass light. This is why the lift can be interpreted as a genuine comparable-mass completion rather than as a disguised one-body repair.

The static cross upgrade is a particularly transparent check. The seed contributes

$$\frac{G^3 m_A m_B}{4r^3} (m_A^2 + m_B^2), \quad (309)$$

while the added cross block contributes

$$\frac{5G^3 m_A^2 m_B^2}{4r^3} = \frac{G^3 m_A m_B}{4r^3} (5m_A m_B). \quad (310)$$

Therefore the full assembled static mass polynomial is

$$\frac{G^3 m_A m_B}{4r^3} (m_A^2 + 5m_A m_B + m_B^2), \quad (311)$$

which is exactly the mass structure required by the standard conservative generic-frame ADM target.

A second useful check is the test-mass limit. Setting body  $B$  heavy and at rest, all quartic cross monomials vanish immediately, the  $G^2/r^2$  comparable-mass terms vanish after division by the light test mass, and the static cross term is likewise suppressed by one additional power of the light mass. Thus the whole added block drops out in the strict one-body limit, as it must.

Finally, note that no contact transformation is being used here to hide a mismatch. The statement [equation \(308\)](#) is literal equality of the perturbatively Legendre-transformed Hamiltonian to the standard generic-frame conservative ADM target in the chosen representative. The main-text full assembled Lagrangian therefore satisfies

$$L_{2,\text{full}} \xrightarrow{\text{Legendre}} H_{2\text{PN}}^{\text{ADM}} \quad (312)$$

with no leftover residual and no unassigned comparable-mass coefficients.

**Claim class.** *Exact* for the derivation of [equation \(287\)](#) and for the coefficient-by-coefficient linear solve in the independent basis [equation \(301\)](#). **Full assembly** for the exact equality [equation \(308\)](#) once the carried lower-order ledger, the one-body closure, and the self/static seed are inserted..

## F Tensor-wake decomposition of the solved added cross block

This appendix gives the first constructive rewrite of the uniquely solved added comparable-mass conservative 2PN cross block. The exact ADM residual solve fixed that block algebraically. The point here is different: to show that, once solved, the new coefficients already organize into a small and intelligible wake/tensor module rather than an opaque polynomial list.

The decomposition has four parts. First, two quartic coefficients are forced by a universal kinetic dressing of the frozen 1PN vector wake. Second, the remaining quartic residual closes exactly in a minimal five-channel rank-2 tensor basis. Third, the solved  $G^2/r^2$  velocity block is reproduced by a compact local-potential dressing of the vector/tensor channels. Fourth, the static cross term is a single scalar overlap. The later appendices sharpen this further into a body-local  $P_0 \oplus P_2$  mouth-port language and then into the support-minus-closure EFT. The tensor-wake level is the intermediate constructive layer that makes those later rewritings possible.

As in the rest of the paper,  $L_2$  denotes the coefficient of  $c^{-4}$  in the conservative Lagrangian expansion  $L_{\text{cons}} = L_0 + c^{-2}L_1 + c^{-4}L_2 + O(c^{-6})$ . Accordingly, the explicit factor  $c^{-4}$  is omitted throughout this appendix.

### F.1 Solved added cross block to be decomposed

The solved added conservative two-body 2PN cross block is

$$\begin{aligned} L_{2,\text{cross}} = & \frac{Gm_A m_B}{r} \left[ -\frac{7}{4} v_{AB}(v_A^2 + v_B^2) - \frac{1}{4} v_{An} v_{Bn}(v_A^2 + v_B^2) + \frac{11}{8} v_A^2 v_B^2 + \frac{1}{4} v_{AB}^2 \right. \\ & \left. - \frac{5}{8} (v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2} v_{AB} v_{An} v_{Bn} + \frac{3}{8} v_{An}^2 v_{Bn}^2 \right] \\ & + \frac{G^2 m_A m_B}{r^2} \left[ \frac{11}{8} (m_A v_A^2 + m_B v_B^2) - \frac{15}{4} (m_A + m_B) v_{AB} + \frac{15}{8} (m_A v_{An}^2 + m_B v_{Bn}^2) \right] \\ & + \frac{5G^3 m_A^2 m_B^2}{4r^3}. \end{aligned} \quad (313)$$

Here

$$\mathbf{n} \equiv \frac{\mathbf{r}}{r}, \quad v_{AB} \equiv \mathbf{v}_A \cdot \mathbf{v}_B, \quad v_{An} \equiv \mathbf{v}_A \cdot \mathbf{n}, \quad v_{Bn} \equiv \mathbf{v}_B \cdot \mathbf{n}, \quad (314)$$

and the local pair potentials are

$$U_A \equiv \frac{Gm_B}{r}, \quad U_B \equiv \frac{Gm_A}{r}. \quad (315)$$

The rest of the appendix reconstructs [equation \(313\)](#) from a minimal tensor-wake channel system.

### F.2 Universal kinetic dressing of the frozen 1PN vector wake

The carried 1PN vector-wake kernel is

$$\mathcal{W}_{AB}^{(1)} \equiv C_{\parallel} v_{AB} + C_L v_{An} v_{Bn}, \quad C_{\parallel} = -\frac{7}{2}, \quad C_L = -\frac{1}{2}. \quad (316)$$

The first two quartic coefficients in [equation \(313\)](#) are then exactly

$$-\frac{7}{4} v_{AB}(v_A^2 + v_B^2) - \frac{1}{4} v_{An} v_{Bn}(v_A^2 + v_B^2) = \frac{1}{2} (v_A^2 + v_B^2) \mathcal{W}_{AB}^{(1)}. \quad (317)$$

So these terms are not genuinely new 2PN fits at all; they are a universal leg-dressing of the frozen 1PN wake with coefficient

$$\sigma = \frac{1}{2}. \quad (318)$$

It is therefore natural to define

$$L_{2,\text{vec},\text{kin}} \equiv \frac{Gm_A m_B}{r} \frac{1}{2} (v_A^2 + v_B^2) \mathcal{W}_{AB}^{(1)}. \quad (319)$$

Subtracting [equation \(319\)](#) from the quartic  $G/r$  part of [equation \(313\)](#) isolates the genuinely new quartic residual.

### F.3 Minimal rank-2 tensor basis for the quartic residual

Define, relative to the pair direction  $\mathbf{n}$ , the transverse and longitudinal scalar channels

$$T_A \equiv v_A^2 - v_{An}^2, \quad L_A \equiv v_{An}^2, \quad T_B \equiv v_B^2 - v_{Bn}^2, \quad L_B \equiv v_{Bn}^2, \quad (320)$$

and the two genuinely rank-2 cross channels

$$S_{AB} \equiv (v_{AB} - v_{An}v_{Bn})^2 - \frac{1}{2}T_A T_B, \quad (321)$$

$$M_{AB} \equiv 2(v_{AB} - v_{An}v_{Bn})v_{An}v_{Bn}. \quad (322)$$

The interpretations are

$$T : \text{transverse-trace scalar}, \quad L : \text{longitudinal scalar}, \quad S : \text{transverse shear}, \quad M : \text{mixed transverse-longitudinal shear}, \quad (323)$$

With these definitions, the quartic residual after removing the universal vector-wake leg dressing is exactly

$$L_{2,\text{tensor},\text{quartic}} = \frac{Gm_A m_B}{r} \left[ \frac{3}{2} T_A T_B + \frac{1}{4} S_{AB} + M_{AB} + \frac{3}{4} (T_A L_B + L_A T_B) + \frac{9}{4} L_A L_B \right]. \quad (324)$$

Expanding [equation \(324\)](#) back into the velocity invariants gives

$$L_{2,\text{tensor},\text{quartic}} = \frac{Gm_A m_B}{r} \left[ \frac{11}{8} v_A^2 v_B^2 + \frac{1}{4} v_{AB}^2 - \frac{5}{8} (v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2} v_{AB} v_{An} v_{Bn} + \frac{3}{8} v_{An}^2 v_{Bn}^2 \right], \quad (325)$$

which is precisely the remaining quartic part of [equation \(313\)](#). Thus the genuinely new quartic 2PN content is an exact minimal five-channel rank-2 tensor overlap with coefficients

$$k_{TT} = \frac{3}{2}, \quad k_S = \frac{1}{4}, \quad k_M = 1, \quad k_{TL} = \frac{3}{4}, \quad k_{LL} = \frac{9}{4}. \quad (326)$$

### F.4 Positive scalar subsector

Inside the tensor decomposition, the scalar  $(T, L)$  subsector is governed by

$$K_{\text{scalar}} = \begin{pmatrix} \frac{3}{2} & \frac{3}{4} \\ \frac{3}{4} & \frac{9}{4} \end{pmatrix}. \quad (327)$$

This matrix is positive:

$$\det K_{\text{scalar}} = \frac{45}{16} > 0, \quad \text{tr } K_{\text{scalar}} = \frac{15}{4} > 0, \quad (328)$$

with eigenvalues

$$\lambda_{\pm} = \frac{15 \pm 3\sqrt{5}}{8}. \quad (329)$$

So the scalar part of the new quartic tensor sector is constructive rather than sign-indefinite. This is why the later diagonalization into the body-local  $P_0 \oplus P_2$  mouth-port basis is so clean.

## F.5 Local-potential dressing of the $G^2/r^2$ block

The solved quadratic-velocity sector of [equation \(313\)](#) is

$$L_{2,\text{quad,vel}} = \frac{G^2 m_A m_B}{r^2} \left[ \frac{11}{8} (m_A v_A^2 + m_B v_B^2) - \frac{15}{4} (m_A + m_B) v_{AB} + \frac{15}{8} (m_A v_{An}^2 + m_B v_{Bn}^2) \right]. \quad (330)$$

This block is reproduced exactly by the local-potential dressing

$$L_{2,\text{quad,pot}} = \frac{G m_A m_B}{r} \left[ \tau_{\parallel} (U_A + U_B) C_{\parallel} v_{AB} + \beta_T (U_A T_B + U_B T_A) + \beta_L (U_A L_B + U_B L_A) \right]. \quad (331)$$

Matching to [equation \(330\)](#) gives the unique coefficients

$$\tau_{\parallel} = \frac{15}{14}, \quad \beta_T = \frac{11}{8}, \quad \beta_L = \frac{13}{4}. \quad (332)$$

Inserting [equations \(315\), \(316\), \(320\)](#) and [\(332\)](#) into [equation \(331\)](#) gives

$$\begin{aligned} L_{2,\text{quad,pot}} &= \frac{G^2 m_A m_B}{r^2} \left[ -\frac{15}{4} (m_A + m_B) v_{AB} + \frac{11}{8} (m_A v_A^2 + m_B v_B^2) \right. \\ &\quad \left. + \left( \frac{13}{4} - \frac{11}{8} \right) (m_A v_{An}^2 + m_B v_{Bn}^2) \right] \\ &= \frac{G^2 m_A m_B}{r^2} \left[ \frac{11}{8} (m_A v_A^2 + m_B v_B^2) - \frac{15}{4} (m_A + m_B) v_{AB} + \frac{15}{8} (m_A v_{An}^2 + m_B v_{Bn}^2) \right], \end{aligned} \quad (333)$$

which is exactly [equation \(330\)](#).

Structurally, the solved answer is smaller than a generic ansatz would suggest: only the parallel vector dressing is needed, and the tensor-potential dressing is diagonal in the transverse and longitudinal projector channels.

## F.6 Static scalar overlap and full constructive reconstruction

The remaining purely static comparable-mass piece is

$$L_{2,\text{static,cross}} = \frac{5G^3 m_A^2 m_B^2}{4r^3} = \frac{G m_A m_B}{r} \frac{5}{4} U_A U_B. \quad (334)$$

Therefore the full solved added cross block can be reconstructed exactly as

$$L_{2,\text{cross}} = L_{2,\text{vec,kin}} + L_{2,\text{tensor,quartic}} + L_{2,\text{quad,pot}} + L_{2,\text{static,cross}}. \quad (335)$$

By direct expansion, the right-hand side of [equation \(335\)](#) reproduces [equation \(313\)](#) term by term.

This gives the tensor-wake interpretation of the solved added cross block:

1. a carried-forward 1PN vector wake with universal kinetic dressing,
2. a genuinely new five-channel rank-2 tensor residual,
3. a compact local-potential dressing of the vector/tensor projector channels, and
4. a single scalar static overlap.

In this language the algebraically solved ADM residual is already telling us what a constructive inner-throat derivation should reproduce.

## F.7 Bridge to the mouth-port language

The tensor-wake basis is not yet the final throat-response basis. It is the most economical intermediate basis for seeing how the solved quartic residual organizes before one commits to explicit mouth ports. The next appendix shows that the same scalar tensor block diagonalizes into  $P_0 \oplus P_2$  channels. At the scalar level one writes

$$P_0 = T + L = v^2, \quad P_{20} = -T + 2L = 3(v \cdot n)^2 - v^2, \quad (336)$$

and the scalar kernel [equation \(327\)](#) diagonalizes exactly in that basis. So the tensor-wake decomposition should be read as the constructive bridge between “the coefficients solved by the ADM lift” and “the small  $P_0 \oplus P_2$  mouth-port hierarchy” used in the next appendices.

**Claim class.** *Exact for the algebraic decomposition of the solved added cross block into [equation \(335\)](#) once the target coefficients are inserted. Controlled reduction for the interpretation of these exact overlaps as a constructive wake/tensor response module that the inner-throat derivation should reproduce. This appendix is explanatory rather than independent: it does not alter the solved coefficients, but reorganizes them into a minimal carried-vector plus rank-2 tensor channel structure..*

## G $P_0 \oplus P_2$ mouth-port rebuild and support-minus-closure EFT

This appendix continues the constructive program started in [appendix F](#). There the solved added conservative two-body 2PN comparable-mass cross block was rewritten as a carried 1PN vector wake plus a minimal rank-2 tensor residual and a compact local-potential dressing. The purpose of the present appendix is to sharpen that constructive rewrite into a body-local mouth-port language and then into a small signed auxiliary-channel EFT.

Two statements are proved here. First, after the universal leg-dressing piece inherited from the frozen 1PN wake is removed, the genuinely new quartic residual diagonalizes exactly into one real monopole-like port and the full real quadrupole multiplet,  $P_0 \oplus P_2$ . Second, once the solved  $U \times \Pi$  terms and the static cross term are restored, the new even block factorizes as a positive support bundle minus a single pure-potential closure channel. That is the sense in which the solved added 2PN cross sector becomes a genuine *support-minus-closure throat response theory* rather than an opaque coefficient list.

As in the preceding appendix,  $L_2$  denotes the coefficient of  $c^{-4}$  in the conservative expansion  $L_{\text{cons}} = L_0 + c^{-2}L_1 + c^{-4}L_2 + O(c^{-6})$ , so the explicit overall factor  $c^{-4}$  is suppressed throughout.

### G.1 Body-local $P_0 \oplus P_2$ mouth-port basis

To display the real mouth-port channels it is convenient to work in the pair frame with

$$\mathbf{n} \equiv \frac{\mathbf{r}}{r} = \hat{\mathbf{z}}. \quad (337)$$

Write each velocity as

$$\mathbf{v}_A = \mathbf{u}_A + d_A \mathbf{n}, \quad \mathbf{v}_B = \mathbf{u}_B + d_B \mathbf{n}, \quad \mathbf{u}_A \cdot \mathbf{n} = \mathbf{u}_B \cdot \mathbf{n} = 0, \quad (338)$$

so that  $d_A = v_{An}$  and  $d_B = v_{Bn}$ . The real transverse components are denoted

$$\mathbf{u}_A = (u_{Ax}, u_{Ay}), \quad \mathbf{u}_B = (u_{Bx}, u_{By}). \quad (339)$$

Rotational covariance guarantees that the final reconstruction is independent of the choice of pair frame; the  $\hat{\mathbf{z}}$  presentation is only a convenient way to write the real channels explicitly.

The frozen 1PN wake is already a dipole kernel,

$$L_{1\text{PN}}^{\text{wake}} = -\frac{7}{2} \mathbf{v}_A \cdot \mathbf{v}_B - \frac{1}{2} d_A d_B = -\left[ \Pi_A^{(1\pm)} \cdot \Pi_B^{(1\pm)} + \Pi_A^{(10)} \Pi_B^{(10)} \right], \quad (340)$$

with the body-local odd dipole ports

$$\Pi_A^{(1\pm)} = \sqrt{\frac{7}{2}} \mathbf{u}_A, \quad \Pi_A^{(10)} = 2d_A, \quad (341)$$

(and similarly for body  $B$ ). So the 1PN sector already carries a local  $\ell = 1$  mouth-response language. The new question is how the genuinely new quartic residual organizes after the universal kinetic leg dressing

$$\frac{1}{2}(v_A^2 + v_B^2) L_{1\text{PN}}^{\text{wake}} \quad (342)$$

is removed.

The scalar tensor block derived in [appendix F](#) used the variables

$$T = u^2, \quad L = d^2, \quad (343)$$

with scalar kernel

$$K_{TL} = \begin{pmatrix} \frac{3}{2} & \frac{3}{4} \\ \frac{3}{4} & \frac{9}{4} \end{pmatrix}. \quad (344)$$

Introducing the scalar combinations

$$P_0 = T + L = v^2, \quad P_{20} = -T + 2L = 3(v \cdot n)^2 - v^2, \quad (345)$$

this kernel diagonalizes exactly to

$$K_{P_0 P_{20}} = \begin{pmatrix} \frac{5}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}. \quad (346)$$

This immediately suggests the canonically normalized scalar ports

$$\Pi^{(0)} = \sqrt{\frac{5}{4}} P_0 = \frac{\sqrt{5}}{2} v^2, \quad \Pi^{(20)} = \sqrt{\frac{1}{4}} P_{20} = \frac{1}{2} (3(v \cdot n)^2 - v^2). \quad (347)$$

The full real quadrupole multiplet is then completed by

$$\Pi^{(21c)} = \sqrt{2} (v \cdot n) u_x, \quad \Pi^{(21s)} = \sqrt{2} (v \cdot n) u_y, \quad (348)$$

$$\Pi^{(22c)} = \frac{u_x^2 - u_y^2}{2\sqrt{2}}, \quad \Pi^{(22s)} = \frac{2u_x u_y}{2\sqrt{2}}. \quad (349)$$

Using these definitions for each body, the quartic residual left after subtracting the universal carried-forward piece [equation \(342\)](#) is exactly

$$\begin{aligned} Q_{2\text{PN}}^{\text{new}} &= \Pi_A^{(0)} \Pi_B^{(0)} + \Pi_A^{(20)} \Pi_B^{(20)} + \Pi_A^{(21c)} \Pi_B^{(21c)} + \Pi_A^{(21s)} \Pi_B^{(21s)} \\ &\quad + \Pi_A^{(22c)} \Pi_B^{(22c)} + \Pi_A^{(22s)} \Pi_B^{(22s)}. \end{aligned} \quad (350)$$

So the genuinely new quartic 2PN tensor sector is a positive Gram overlap of one real  $P_0$  channel and the full real  $P_2$  multiplet. This is the exact body-local mouth-port version of the tensor-wake residual from the previous appendix.

The solved  $G^2/r^2$  velocity block also becomes simpler in this basis. Using the local pair potentials

$$U_A \equiv \frac{Gm_B}{r}, \quad U_B \equiv \frac{Gm_A}{r}, \quad (351)$$

it can be written as

$$-\frac{15}{4}(U_A + U_B)(\mathbf{v}_A \cdot \mathbf{v}_B) + U_A \left( \frac{4}{\sqrt{5}}\Pi_B^{(0)} + \frac{5}{4}\Pi_B^{(20)} \right) + U_B \left( \frac{4}{\sqrt{5}}\Pi_A^{(0)} + \frac{5}{4}\Pi_A^{(20)} \right). \quad (352)$$

So only the scalar axisymmetric ports  $P_0$  and  $P_{20}$  are driven directly by the local-potential source terms.

Finally, the static added cross piece is simply

$$\frac{5}{4}U_A U_B. \quad (353)$$

Equations [equations \(342\), \(350\), \(352\)](#) and [\(353\)](#) therefore reconstruct the full added conservative 2PN cross block in an explicit body-local mouth-port basis.

## G.2 Support-minus-closure factorization

The new even sector becomes especially transparent once the solved  $U \times \Pi$  coefficients are absorbed into shifted body-local support channels. Define

$$\mathcal{S}_A = \left( \Pi_A^{(0)} + \frac{4}{\sqrt{5}}U_A, \Pi_A^{(20)} + \frac{5}{4}U_A, \Pi_A^{(21c)}, \Pi_A^{(21s)}, \Pi_A^{(22c)}, \Pi_A^{(22s)} \right), \quad (354)$$

with the obvious analogous definition for  $\mathcal{S}_B$ . Then the positive support channels alone would generate a pure static coefficient

$$\left( \frac{4}{\sqrt{5}} \right)^2 + \left( \frac{5}{4} \right)^2 = \frac{381}{80}. \quad (355)$$

But the solved static coefficient in [equation \(353\)](#) is only

$$\frac{5}{4} = \frac{100}{80}. \quad (356)$$

So the exact closure deficit is

$$\Delta_{\text{geom}} = \frac{381}{80} - \frac{100}{80} = \frac{281}{80}. \quad (357)$$

This deficit is removed by one negative closure channel,

$$\mathcal{C}_A = \sqrt{\frac{281}{80}} U_A, \quad \mathcal{C}_B = \sqrt{\frac{281}{80}} U_B. \quad (358)$$

The entire even block is therefore exactly

$$L_{2\text{PN}}^{\text{even}} = \mathcal{S}_A \cdot \mathcal{S}_B - \mathcal{C}_A \mathcal{C}_B. \quad (359)$$

The important structural fact is that the negative direction lives entirely in the pure potential  $U$ -direction. It does *not* mix with the genuine mouth ports. That is why it is natural to interpret the indefinite part as a geometry-closure term rather than as a pathology of the mouth-response sector itself.

In the ordered source basis

$$X = (\Pi^{(0)}, \Pi^{(20)}, \Pi^{(21c)}, \Pi^{(21s)}, \Pi^{(22c)}, \Pi^{(22s)}, U), \quad (360)$$

the exact even response matrix is

$$M_{\text{even}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 4/\sqrt{5} \\ 0 & 1 & 0 & 0 & 0 & 0 & 5/4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 4/\sqrt{5} & 5/4 & 0 & 0 & 0 & 0 & 5/4 \end{pmatrix}. \quad (361)$$

It factorizes exactly as

$$M_{\text{even}} = R_{\text{support}}^T R_{\text{support}} - R_{\text{geom}}^T R_{\text{geom}}, \quad (362)$$

with

$$R_{\text{support}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 4/\sqrt{5} \\ 0 & 1 & 0 & 0 & 0 & 0 & 5/4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad R_{\text{geom}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \sqrt{281/80} \end{pmatrix}. \quad (363)$$

So the even block is literally a positive-semidefinite mouth-response block minus a rank-1 geometry-closure block.

### G.3 Local auxiliary-channel EFT and exact on-shell reconstruction

The factorization above has an immediate local auxiliary-field interpretation. For a positive support channel  $q$  with body sources  $S_A, S_B$ , use

$$L_q = -\frac{1}{2}q^2 + q(S_A + S_B). \quad (364)$$

Here the auxiliary variable  $q$  is a local EFT support channel and is unrelated to electric charge or to the gravity-side coefficient  $\kappa_\rho$ . Eliminating  $q$  gives the cross term  $+S_A S_B$ . For the negative closure channel  $g$  with sources  $C_A, C_B$ , use

$$L_g = +\frac{1}{2}g^2 + g(C_A + C_B). \quad (365)$$

Eliminating  $g$  gives the cross term  $-C_A C_B$ . Finally, for the odd dipole wake channel  $d$  with body sources  $D_A, D_B$ , use

$$L_d = -\frac{1}{2}d^2 + d(D_A - D_B), \quad (366)$$

which yields the odd cross term  $-D_A D_B$  on shell. Thus the solved 2PN sector is exactly the on-shell result of a small signed auxiliary-channel EFT.

It is useful to keep the odd channels visible explicitly. Define the odd dipole bundle

$$\mathcal{D}_A = \left( \sqrt{\frac{7}{2}}v_{Ax}, \sqrt{\frac{7}{2}}v_{Ay}, 2v_{Az} \right), \quad \mathcal{D}_B = \left( \sqrt{\frac{7}{2}}v_{Bx}, \sqrt{\frac{7}{2}}v_{By}, 2v_{Bz} \right), \quad (367)$$

so that the frozen 1PN wake is

$$L_{\text{1PN}}^{\text{wake}} = -\mathcal{D}_A \cdot \mathcal{D}_B. \quad (368)$$

The added odd 2PN block is then simply the universal kinetic dressing plus the solved local-potential dressing from [appendix F](#), namely

$$L_{\text{odd,dressed}}^{\text{add}} = \frac{1}{2}(v_A^2 + v_B^2)L_{\text{1PN}}^{\text{wake}} - \frac{15}{4}(U_A + U_B)(\mathbf{v}_A \cdot \mathbf{v}_B). \quad (369)$$

Putting everything together, the entire solved added conservative 2PN cross block is reconstructed exactly as

$$L_{2\text{PN}, \text{cross}}^{\text{add}} = \frac{Gm_A m_B}{r} \left[ L_{\text{odd}, \text{dressed}}^{\text{add}} + \mathcal{S}_A \cdot \mathcal{S}_B - \mathcal{C}_A \mathcal{C}_B \right]. \quad (370)$$

By direct expansion, [equation \(370\)](#) reproduces the full solved added ADM-lifted cross target term by term.

## G.4 Why this is the right constructive split

This appendix matters because it is the first point where the new 2PN cross sector stops looking like “a polynomial found by the ADM lift” and starts looking like a real throat-response theory. The constructive split is now very sharp:

1. the frozen 1PN wake survives as an odd dipole backbone,
2. the genuinely new even 2PN sector is a positive  $P_0 \oplus P_2$  mouth-response bundle, and
3. the only negative piece is a single pure-potential geometry-closure channel.

That last point is especially important. Because the negative direction lives entirely in the  $U$ -direction, the present factorization does *not* say that the mouth operator itself is indefinite. It says something more controlled: the support sector is positive and minimal, while a separate geometry-energy closure term subtracts the excess static overlap needed to reach the exact solved coefficient. This is why the next appendices can cleanly separate “mouth response” from “geometry closure” and then package the result into the minimal irreducible-channel throat operator.

In this language, the remaining constructive task is no longer vague. The inner-throat program must explain two specific pieces:

1. why the positive support bundle closes exactly on the  $P_0 \oplus P_2$  channels with the solved source shifts  $4/\sqrt{5}$  and  $5/4$ , and
2. why geometry contributes the single rank-1 pure- $U$  closure deficit  $\Delta_{\text{geom}} = 281/80$ .

So the support-minus-closure EFT is not merely another algebraic repackaging. It is the first form in which the solved added 2PN cross block becomes a small, body-local throat-response theory with a clear division between mouth support and geometry closure.

**Claim class.** *Exact for the algebraic equivalence of [equation \(370\)](#) with the solved added conservative two-body 2PN cross block once the coefficients are inserted. Controlled reduction for the interpretation of the positive block as a minimal mouth-response sector and of the rank-1 pure- $U$  subtraction as a geometry-closure channel. This appendix is constructive and explanatory: it does not introduce new fitted coefficients, but reveals that the solved ADM cross block is already the on-shell output of a small signed auxiliary-channel EFT..*

## H Minimal irreducible-channel throat operator and dynamic DtN scaffolding

This appendix turns the constructive rewrites of [appendices F](#) and [G](#) into the smallest actual low-frequency throat operator that reproduces the solved added conservative two-body 2PN cross target. The point is no longer only that the solved coefficients admit a useful decomposition. The stronger statement is that they already determine a minimal channel content for the throat-response problem.

At zero frequency, the full added conservative 2PN cross block is reproduced by three ingredients:

1. a carried odd  $\ell = 1$  dipole sector with distinct transverse and longitudinal residues,
2. a new even  $\ell = 0 \oplus 2$  support bundle with canonical unit support residues, and
3. one direct pure-potential geometry-closure channel.

Once that static operator is identified, the remaining dynamic problem becomes much sharper. One must explain why the zero-frequency residues take the values already solved by the conservative 2PN algebra and then derive the corresponding low-frequency pole structure from the inner-throat PDE / DtN problem.

The appendix is organized accordingly. First, we record the minimal irreducible-channel operator fixed by the solved static target. Next, we show that the same operator already predicts the first local-potential three-body lift. Finally, we build the minimal frequency-dependent DtN scaffolding: an isotropic 4D-ball unit test, which fails in exactly the ways expected from the algebra, and a one-pole-per-channel axisymmetric completion, which reduces to the solved static operator as  $\omega \rightarrow 0$ .

As elsewhere in the paper,

$$L_{\text{cons}} = L_0 + \frac{1}{c^2} L_1 + \frac{1}{c^4} L_2 + O(c^{-6}), \quad (371)$$

so the explicit overall factor  $c^{-4}$  is suppressed throughout this appendix.

## H.1 Minimal irreducible channel content at zero frequency

Work in the pair frame

$$\mathbf{n} \equiv \frac{\mathbf{r}}{r} = \hat{\mathbf{z}}, \quad \mathbf{v}_A = \mathbf{u}_A + d_A \mathbf{n}, \quad \mathbf{v}_B = \mathbf{u}_B + d_B \mathbf{n}, \quad (372)$$

with  $\mathbf{u}_A \cdot \mathbf{n} = \mathbf{u}_B \cdot \mathbf{n} = 0$  and

$$U_A \equiv \frac{Gm_B}{r}, \quad U_B \equiv \frac{Gm_A}{r}. \quad (373)$$

The frozen odd dipole wake may be written as

$$L_{\text{1PN}}^{\text{wake}} = -\frac{7}{2} \mathbf{u}_A \cdot \mathbf{u}_B - 4d_A d_B, \quad (374)$$

which already displays two inequivalent odd channels:

$$1 \perp: \mathbf{u}, \quad 10: d. \quad (375)$$

The solved added odd conservative 2PN block is then

$$L_{\text{odd}}^{\text{add}} = \frac{1}{2} (v_A^2 + v_B^2) L_{\text{1PN}}^{\text{wake}} - (U_A + U_B) \left[ \frac{15}{4} \mathbf{u}_A \cdot \mathbf{u}_B + \frac{15}{4} d_A d_B \right]. \quad (376)$$

Dividing the solved potential dressings by the frozen static odd residues gives the body source dressings

$$\eta_{\perp} = \frac{15}{14}, \quad \eta_{\parallel} = \frac{15}{16}, \quad (377)$$

so the odd sector is fully fixed at conservative zero frequency.

The even sector comes from the body-local  $P_0 \oplus P_2$  mouth-port basis already constructed in [appendix G](#). Using the six real ports

$$\Pi \equiv (\Pi^{(0)}, \Pi^{(20)}, \Pi^{(21c)}, \Pi^{(21s)}, \Pi^{(22c)}, \Pi^{(22s)}), \quad (378)$$

the solved even block can be written as

$$L_{2\text{PN}}^{\text{even}} = (\Pi_A + JU_A) \cdot (\Pi_B + JU_B) - \Delta_{\text{geom}} U_A U_B, \quad (379)$$

with source vector

$$J = \left( \frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right) \quad (380)$$

and geometry-closure deficit

$$\Delta_{\text{geom}} = \frac{281}{80}. \quad (381)$$

The support part alone contributes the static coefficient

$$J \cdot J = \left( \frac{4}{\sqrt{5}} \right)^2 + \left( \frac{5}{4} \right)^2 = \frac{381}{80}, \quad (382)$$

so the direct geometry subtraction in [equation \(381\)](#) is exactly what lowers the pure-static coefficient to the solved target value

$$\frac{381}{80} - \frac{281}{80} = \frac{5}{4}. \quad (383)$$

This yields the canonical zero-frequency channel data:

$$R_{1\perp} = \frac{7}{2}, \quad R_{10} = 4, \quad (384)$$

$$R_0 = R_{20} = R_{21} = R_{22} = 1, \quad (385)$$

together with the scalar source vector [equation \(380\)](#) and the closure deficit [equation \(381\)](#). In other words, the smallest irreducible channel set required by the solved conservative added 2PN cross block is

$$\{1 \perp, 10, 0, 20, 21c, 21s, 22c, 22s, g\}, \quad (386)$$

where  $g$  denotes the single pure-potential geometry-closure channel.

A compact way to summarize the full added conservative cross block is therefore

$$L_{2\text{PN}, \text{cross}}^{\text{add}} = \frac{Gm_A m_B}{r} \left[ L_{\text{odd}}^{\text{add}} + (\Pi_A + JU_A) \cdot (\Pi_B + JU_B) - \Delta_{\text{geom}} U_A U_B \right]. \quad (387)$$

Equation [\(387\)](#) is the minimal irreducible-channel throat operator at zero frequency.

## H.2 Immediate local-potential 3-body lift

The operator above already predicts the first local-potential three-body lift. Promote body-local pair potentials to

$$U_A^{\text{loc}} = G \left( \frac{m_B}{r_{AB}} + \frac{m_C}{r_{AC}} \right), \quad U_B^{\text{loc}} = G \left( \frac{m_A}{r_{AB}} + \frac{m_C}{r_{BC}} \right). \quad (388)$$

Then the pair- $AB$  operator immediately induces a background- $C$  three-body lift. The velocity-dependent piece is

$$L_{AB|C}^{(3\text{-body, vel})} = \frac{G^2 m_A m_B m_C}{8r_{AB}} \left[ \frac{11v_B^2 + 15d_B^2 - 30(\mathbf{v}_A \cdot \mathbf{v}_B)}{r_{AC}} + \frac{11v_A^2 + 15d_A^2 - 30(\mathbf{v}_A \cdot \mathbf{v}_B)}{r_{BC}} \right], \quad (389)$$

with

$$d_A \equiv \mathbf{v}_A \cdot \mathbf{n}_{AB}, \quad d_B \equiv \mathbf{v}_B \cdot \mathbf{n}_{AB}, \quad (390)$$

and the static piece is

$$L_{AB|C}^{(3\text{-body, stat})} = \frac{5G^3 m_A m_B m_C}{4} \left[ \frac{m_A}{r_{AB}^2 r_{AC}} + \frac{m_B}{r_{AB}^2 r_{BC}} + \frac{m_C}{r_{AB} r_{AC} r_{BC}} \right]. \quad (391)$$

These terms are not used in the main two-body 2PN theorem, but they matter constructively: once the minimal throat operator is fixed at zero frequency, it already carries nontrivial  $N$ -body content.

### H.3 Isotropic 4D-ball DtN unit test

The cleanest PDE-side unit test is the regular interior Helmholtz problem in a 4D ball. For harmonic degree  $\ell$ , the regular radial mode is

$$u_\ell(r) \propto r^{-1} J_{\ell+1}(kr), \quad z \equiv ka, \quad (392)$$

so the boundary Dirichlet-to-Neumann eigenvalue at radius  $a$  is

$$\Lambda_\ell(z) = -\frac{1}{a} + \frac{z}{a} \frac{J'_{\ell+1}(z)}{J_{\ell+1}(z)}. \quad (393)$$

Its small- $z$  expansions for the first three channels are

$$\Lambda_0(z) = -\frac{z^2}{4a} - \frac{z^4}{96a} + O(z^6), \quad (394)$$

$$\Lambda_1(z) = \frac{1}{a} - \frac{z^2}{6a} - \frac{z^4}{288a} + O(z^6), \quad (395)$$

$$\Lambda_2(z) = \frac{2}{a} - \frac{z^2}{8a} - \frac{z^4}{640a} + O(z^6). \quad (396)$$

Accordingly, the first admittances are

$$Y_1(z) = \frac{1}{\Lambda_1(z)} = a + \frac{az^2}{6} + \frac{az^4}{32} + O(z^6), \quad (397)$$

$$Y_2(z) = \frac{1}{\Lambda_2(z)} = \frac{a}{2} + \frac{az^2}{32} + \frac{3az^4}{1280} + O(z^6). \quad (398)$$

This isotropic branch is useful precisely because it fails in the right way. It gets several structural features right: it is diagonal in harmonic channels, it supports positive  $\ell = 1$  and  $\ell = 2$  towers, and after canonical normalization it is compatible with an identity-like support metric within a fixed  $\ell$ -sector. But it also fails two requirements of the solved conservative operator immediately.

First, there is no finite static monopole stiffness:

$$\Lambda_0(0) = 0. \quad (399)$$

So a pure isotropic cavity cannot generate the solved finite static monopole support or source structure. Second, a spherical branch cannot split the odd dipole residues. It contains one degenerate  $\ell = 1$  multiplet rather than the two distinct static values [equation \(384\)](#). Therefore the isotropic 4D-ball branch is a useful no-go unit test, but not the full throat.

#### H.4 Minimal axisymmetric pole-completed DtN scaffolding

The static operator above therefore forces the minimal PDE-side symmetry reduction

$$O(3) \longrightarrow O(2), \quad (400)$$

which splits the dipole and quadrupole multiplets exactly as required by the solved conservative channel structure. The smallest low-frequency completion consistent with the zero-frequency data is then a one-pole-per-channel model with admittances

$$Y_{1\perp}(\omega) = \frac{7/2}{1 - \omega^2/\Omega_{1\perp}^2}, \quad Y_{10}(\omega) = \frac{4}{1 - \omega^2/\Omega_{10}^2}, \quad (401)$$

$$Y_0(\omega) = \frac{1}{1 - \omega^2/\Omega_0^2}, \quad Y_{20}(\omega) = \frac{1}{1 - \omega^2/\Omega_{20}^2}, \quad (402)$$

$$Y_{21}(\omega) = \frac{1}{1 - \omega^2/\Omega_{21}^2}, \quad Y_{22}(\omega) = \frac{1}{1 - \omega^2/\Omega_{22}^2}, \quad Y_g(\omega) = \frac{1}{1 - \omega^2/\Omega_g^2}. \quad (403)$$

Here  $Y_{21}$  stands for both real  $|m| = 1$  quadrupole components and  $Y_{22}$  for both real  $|m| = 2$  components. The corresponding bare DtN kernels are simply the inverses,

$$Z_\alpha(\omega) = Y_\alpha(\omega)^{-1}, \quad (404)$$

for  $\alpha \in \{1\perp, 10, 0, 20, 21, 22, g\}$ . Explicitly,

$$Z_{1\perp}(\omega) = \frac{1 - \omega^2/\Omega_{1\perp}^2}{7/2}, \quad Z_{10}(\omega) = \frac{1 - \omega^2/\Omega_{10}^2}{4}, \quad (405)$$

with analogous formulas for the even and geometry channels.

In the ordered source basis

$$X = (P_0, P_{20}, P_{21c}, P_{21s}, P_{22c}, P_{22s}, U), \quad (406)$$

the frequency-dependent even response matrix becomes

$$M_{\text{even}}(\omega) = \begin{pmatrix} Y_0 & 0 & 0 & 0 & 0 & 0 & J_0 Y_0 \\ 0 & Y_{20} & 0 & 0 & 0 & 0 & J_{20} Y_{20} \\ 0 & 0 & Y_{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Y_{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & Y_{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & Y_{22} & 0 \\ J_0 Y_0 & J_{20} Y_{20} & 0 & 0 & 0 & 0 & J_0^2 Y_0 + J_{20}^2 Y_{20} - \Delta_{\text{geom}} Y_g \end{pmatrix}, \quad (407)$$

with

$$J_0 = \frac{4}{\sqrt{5}}, \quad J_{20} = \frac{5}{4}. \quad (408)$$

This is the dynamic continuation of the static support-minus-closure matrix from [appendix G](#). The same signed factorization survives at all frequencies:

$$M_{\text{even}}(\omega) = R_{\text{support}}(\omega)^T R_{\text{support}}(\omega) - R_{\text{geom}}(\omega)^T R_{\text{geom}}(\omega), \quad (409)$$

with the support block dressed by the channel admittances and the geometry block dressed by  $\sqrt{\Delta_{\text{geom}}} Y_g(\omega)$ .

The scalar drive vector is correspondingly promoted to

$$J_{\text{eff}}(\omega) = (J_0 Y_0(\omega), J_{20} Y_{20}(\omega), 0, 0, 0, 0), \quad (410)$$

and the effective pure-potential coefficient becomes

$$S_{\text{eff}}(\omega) = J_0^2 Y_0(\omega) + J_{20}^2 Y_{20}(\omega) - \Delta_{\text{geom}} Y_g(\omega). \quad (411)$$

At zero frequency,

$$S_{\text{eff}}(0) = \frac{5}{4}, \quad (412)$$

exactly as required by the solved static cross term. Its low-frequency expansion is

$$S_{\text{eff}}(\omega) = \frac{5}{4} + \omega^2 \left( \frac{16}{5\Omega_0^2} + \frac{25}{16\Omega_{20}^2} - \frac{281}{80\Omega_g^2} \right) + O(\omega^4). \quad (413)$$

So the first nontrivial dynamic scalar data are now isolated sharply.

The static check is immediate:

$$\lim_{\omega \rightarrow 0} Y_{1\perp} = \frac{7}{2}, \quad \lim_{\omega \rightarrow 0} Y_{10} = 4, \quad \lim_{\omega \rightarrow 0} Y_0 = \lim_{\omega \rightarrow 0} Y_{20} = \lim_{\omega \rightarrow 0} Y_{21} = \lim_{\omega \rightarrow 0} Y_{22} = 1, \quad \lim_{\omega \rightarrow 0} Y_g = 1, \quad (414)$$

so the axisymmetric pole-completed model reduces exactly to the minimal zero-frequency operator data already solved by the conservative 2PN algebra.

## H.5 What is fixed and what remains open

The constructive status after this step is unusually clean. The conservative two-body 2PN program has already fixed the entire zero-frequency operator ledger:

$$R_{1\perp} = \frac{7}{2}, \quad R_{10} = 4, \quad R_0 = R_{20} = R_{21} = R_{22} = 1, \quad (415)$$

$$J = \left( \frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right), \quad \Delta_{\text{geom}} = \frac{281}{80}. \quad (416)$$

What remains open is not the static conservative operator, but its dynamic PDE-side completion. In the one-pole-per-channel model, the still-open observables are exactly the seven pole scales

$$\Omega_{1\perp}, \quad \Omega_{10}, \quad \Omega_0, \quad \Omega_{20}, \quad \Omega_{21}, \quad \Omega_{22}, \quad \Omega_g. \quad (417)$$

An optional near-spherical support simplification is

$$\Omega_{20} = \Omega_{21} = \Omega_{22}, \quad (418)$$

but it is not required anywhere in the main proof.

So the remaining PDE task is now very specific. It is no longer to guess the conservative zero-frequency throat operator from scratch. It is to derive dynamically, from the inner-throat wall and bulk physics, the pole structure of a channel system whose static residues, source vector, and geometry-closure deficit are already known.

**Claim class.** ***Exact** for the zero-frequency operator data [equations \(380\), \(381\), \(384\) and \(385\)](#) and for the statement that the axisymmetric pole-completed model reduces to those data in its  $\omega \rightarrow 0$  limit. **Controlled reduction** for the isotropic 4D-ball unit test as a PDE-side no-go benchmark and for the one-pole-per-channel axisymmetric DtN completion as the minimal dynamic scaffolding consistent with the solved conservative 2PN cross target. This appendix does not claim a full moving-throat PDE solution; it identifies the exact zero-frequency target and the first open dynamic observables that the inner-throat program must still derive..*

## I Family-1 soft-wall and wall-stress program

This appendix records the constructive Family-1 chain that turns the solved static even support sector of [appendix G.2](#) into an explicit local wall law on a flared soft wall. The main text does not need this material for the conservative 2PN proof. Its role is explanatory: it shows that the passed  $P_0 \oplus P_2$  support/source data are not merely a solved operator table, but can be organized as the shadow of a definite Family-1 throat program.

The chain has five steps. First, the solved support and source profiles already live in the same low-order Family-1 flare basis  $\{1, \mu^2, \mu^4\}$ . Second, the passed static support data admit an exact reduced variational wall-Hessian. Third, the strict isotropic steep-wall pullback fails, so a genuinely new wall degree of freedom is required. Fourth, one axisymmetric tangential wall-stress profile repairs the support sector exactly and predicts the size of the remaining monopole add-on. Finally, that monopole add-on can itself be reinterpreted as a reduced geometry-Hessian compressibility rather than as a disconnected auxiliary insertion.

The result is a compact constructive reading of the static Family-1 wall sector:

$$\boxed{\text{local support/source wall law} + \text{one reduced global monopole breathing mode.}} \quad (419)$$

This is the static wall-side companion of the minimal dynamic channel scaffolding assembled later in [appendix H](#).

### I.1 The solved support/source sector already sits in the Family-1 flare basis

As in [appendix G.2](#), the static even support targets are

$$K_{00} = \frac{4}{45}, \quad K_{1\perp} = \frac{2}{7}, \quad K_{10} = \frac{1}{4}, \quad K_{20} = \frac{4}{9}, \quad K_{21} = \frac{2}{3}, \quad K_{22} = \frac{8}{3}. \quad (420)$$

The first constructive simplification is that the full passed  $\ell = 1, 2$  support data can be written as one dipole-base profile plus one curvature correction:

$$Z_{\ell m} = \langle z_{\text{base}}(\mu) \rangle_{\ell m} + (\lambda_\ell - 2) \langle z_{\text{curv}}(\mu) \rangle_{\ell m}, \quad \lambda_\ell \equiv \ell(\ell + 1), \quad \mu \equiv \cos \theta, \quad (421)$$

with

$$z_{\text{base}}(\mu) = \frac{17}{56} - \frac{5}{56}\mu^2, \quad z_{\text{curv}}(\mu) = \frac{593}{672} - \frac{1553}{672}\mu^2 + \frac{7}{8}\mu^4. \quad (422)$$

The entire quadrupole-only correction is therefore carried by the factor  $\lambda_\ell - 2$ , so the dipole is special because  $\lambda_1 - 2 = 0$ .

This is exactly the basis generated by the carried Family-1 flare profile

$$a(z) = a_0 \left( 1 + \beta e^{-z^2/z_m^2} \right). \quad (423)$$

Near the mouth center, with  $a_m = a(0)$  and suitable dimensionless  $q, r$  that are purely flare-shape coefficients rather than charges, one has

$$\frac{a(z)}{a_m} = 1 - q\mu^2 + r\mu^4 + O(\mu^6). \quad (424)$$

So the solved support operator already sits in the same low-order polynomial basis produced by the actual Family-1 flare geometry:

$$\{1, \mu^2, \mu^4\}. \quad (425)$$

The source sector collapses into the same basis:

$$S(\mu) = \frac{11}{8} + \frac{15}{8}P_2(\mu) = \frac{7}{16} + \frac{45}{16}\mu^2. \quad (426)$$

So both the support operator and the scalar source sector already live in the same simple Family-1 flare basis before any dynamical completion is attempted.

A useful equivalent operator notation is

$$\mathcal{B}_{\text{eff}} = \partial_n + z_{\text{base}}(\mu) + \frac{1}{2} \left\{ -\Delta_S - 2, z_{\text{curv}}(\mu) \right\}, \quad (427)$$

restricted to the support sector. On a spherical harmonic  $Y_{\ell m}$ , the anticommutator produces precisely the factor  $\lambda_\ell - 2$ .

## I.2 Exact reduced variational wall-Hessian

The next constructive step is stronger: the passed static support data admit an exact reduced wall-Hessian on the truncated  $\ell = 0, 1, 2$  mouth basis. Write

$$K_{00} = K_{\text{mono}}, \quad (428)$$

and, for  $\ell = 1, 2$ ,

$$K_{\ell m} = K_{\text{mono}} + T_0 \ell(\ell + 1) + (A_0 + A_1 \ell(\ell + 1)) \langle P_2 \rangle_{\ell m} + (B_0 + B_1 \ell(\ell + 1)) \langle P_2^2 \rangle_{\ell m}. \quad (429)$$

The required exact moments are

$$\langle P_2 \rangle_{1\perp} = -\frac{1}{5}, \quad \langle P_2 \rangle_{10} = \frac{2}{5}, \quad \langle P_2 \rangle_{20} = \frac{2}{7}, \quad \langle P_2 \rangle_{21} = \frac{1}{7}, \quad \langle P_2 \rangle_{22} = -\frac{2}{7}, \quad (430)$$

$$\langle P_2^2 \rangle_{1\perp} = \frac{1}{7}, \quad \langle P_2^2 \rangle_{10} = \frac{11}{35}, \quad \langle P_2^2 \rangle_{20} = \frac{3}{7}, \quad \langle P_2^2 \rangle_{21} = \frac{1}{7}, \quad \langle P_2^2 \rangle_{22} = \frac{1}{7}. \quad (431)$$

Solving [equation \(429\)](#) against the passed support data gives

$$K_{\text{mono}} = \frac{4}{45}, \quad T_0 = \frac{23}{135}, \quad (432)$$

$$A_0 = \frac{9095}{3528}, \quad A_1 = -\frac{25559}{21168}, \quad B_0 = -\frac{109}{56}, \quad B_1 = \frac{1765}{3024}. \quad (433)$$

Substituting these values back reproduces all six channel stiffnesses with zero residual. Thus the static axisymmetric support operator is not just a fitted Robin table; it is the Hessian of a concrete reduced variational wall law.

This reduced wall-Hessian has a direct flare interpretation. Define

$$\xi \equiv \frac{a_0^2}{z_m^2}. \quad (434)$$

Expanding the Family-1 flare profile through second order gives

$$\frac{\delta a(\theta)}{a_0} = \beta \left[ 1 - \frac{\xi}{3} + \frac{\xi^2}{18} \right] + \beta \left[ -\frac{2\xi}{3} + \frac{2\xi^2}{9} \right] P_2 + \beta \left[ \frac{2\xi^2}{9} \right] P_2^2 + O(\xi^3). \quad (435)$$

So the actual frozen Family-1 flare automatically generates the exact axisymmetric basis  $\{1, P_2, P_2^2\}$  required by the reduced wall-Hessian.

There is also a useful sharp sub-result. If the anisotropic gradient block of the reduced wall law is assumed linear in the Family-1 flare profile, then

$$\frac{B_1}{A_1} = \frac{\xi}{\xi - 3}. \quad (436)$$

Using [equation \(433\)](#) gives

$$\xi = \frac{12355}{12638} \approx 0.9776072163, \quad \frac{z_m}{a_0} \approx 1.0113880097, \quad (437)$$

so the minimal linear-gradient reading points to a flare width of order the throat radius.

### I.3 Strict isotropic steep-wall no-go

The next question is whether the solved support operator could already come from the strictest possible steep-wall interpretation of the Family-1 geometry: one isotropic normal penetration moment, one isotropic tangential-gradient moment, and the full geometric pullback of the flare. Write

$$\sigma(\mu) = 1 - q\mu^2 + r\mu^4, \quad (438)$$

and let  $J(\mu)$ ,  $F_\theta(\mu)$ , and  $F_\phi(\mu)$  denote the resulting quadratic-flare pullback factors. The strict boundary-layer surface energy is

$$E_{\text{BL}}^{\text{strict}} = \frac{1}{2} \int d\Omega \left[ A J(\mu) \Psi^2 + B \left( F_\theta(\mu) (\partial_\theta \Psi)^2 + F_\phi(\mu) \frac{(\partial_\phi \Psi)^2}{\sin^2 \theta} \right) \right]. \quad (439)$$

Projecting [equation \(439\)](#) onto the  $\ell = 0, 1, 2$  mouth harmonics gives exact channel formulas polynomial in  $(A, B, q, r)$ . But a multi-seed least-squares scan shows that the best-fit residual is not small:

$$\sum_i \Delta_i^2 \approx 0.5536733194, \quad \max_i |\Delta_i| \approx 0.4390744081. \quad (440)$$

So the passed static support sector is *not* just the strict isotropic steep-wall layer pulled back through Family-1 flare geometry.

This is a useful negative result. It means that pure isotropic penetration plus pure geometry pullback is too small a model, and that at least one extra local wall-stress, traction, or profile degree of freedom is genuinely required.

### I.4 Exact traction-balance completion and local wall-stress law

The smallest exact repair is to keep the monopole wall channel separate and solve the  $\ell = 1, 2$  support sector with one base pressure profile and one tangential wall-stress / curvature-compliance profile. On the support sector, write

$$K_{\ell m} = \langle z_{\text{base}} \rangle_{\ell m} + (\ell(\ell + 1) - 2) \langle t \rangle_{\ell m}, \quad (441)$$

with low-order Family-1 profiles

$$z_{\text{base}}(\mu) = b_0 + b_2 \mu^2, \quad t(\mu) = t_0 + t_2 \mu^2 + t_4 \mu^4. \quad (442)$$

Matching the five carried support targets uniquely fixes

$$b_0 = \frac{17}{56}, \quad b_2 = -\frac{5}{56}, \quad t_0 = \frac{593}{672}, \quad t_2 = -\frac{1553}{672}, \quad t_4 = \frac{7}{8}, \quad (443)$$

so that

$$z_{\text{base}}(\mu) = \frac{17}{56} - \frac{5}{56} \mu^2, \quad t(\mu) = \frac{593}{672} - \frac{1553}{672} \mu^2 + \frac{7}{8} \mu^4. \quad (444)$$

In Legendre form,

$$z_{\text{base}}(\mu) = \frac{23}{84} - \frac{5}{84} P_2(\mu), \quad t(\mu) = \frac{211}{1008} - \frac{129}{112} P_2(\mu) + \frac{7}{18} P_2(\mu)^2. \quad (445)$$

This is exactly the same Family-1 low-order basis that kept appearing in the flare and reduced-Hessian steps.

The equivalent local quadratic form is

$$E_{\text{supp}}[\Psi] = \frac{1}{2} \int d\Omega \left[ p(\mu) \Psi^2 + t(\mu) |\nabla_S \Psi|^2 \right], \quad (446)$$

with

$$p(\mu) = z_{\text{base}}(\mu) - 2t(\mu) - \frac{1}{2}\Delta_{st}(\mu). \quad (447)$$

Using

$$\Delta_{st}(\mu) = -\frac{35}{2}\mu^4 + \frac{2729}{112}\mu^2 - \frac{1553}{336}, \quad (448)$$

one finds the exact pressure profile

$$p(\mu) = \frac{571}{672} - \frac{5141}{672}\mu^2 + 7\mu^4. \quad (449)$$

Thus the solved support sector is an explicit local wall model: one base pressure profile, one tangential wall-stress / curvature-compliance profile, and the induced pressure profile that follows from the anticommutator structure. All five  $\ell = 1, 2$  support residuals then vanish identically.

## I.5 Minimal anisotropic PDE-side completion and universal monopole prediction

The traction-balance rewrite above still leaves open whether a genuine PDE-side boundary-layer completion could generate the needed tangential wall law. The minimal new test is to keep the normal penetration moment isotropic, keep the same strict flare pullback, and promote only the tangential wall moment from a constant to the smallest axisymmetric profile

$$B_{\text{tan}}(\mu) = B_0 + B_2\mu^2. \quad (450)$$

The boundary-layer energy becomes

$$E_{\text{BL}}^{\text{aniso}} = \frac{1}{2} \int d\Omega \left[ A J(\mu) \Psi^2 + B_{\text{tan}}(\mu) \left( F_\theta(\mu) (\partial_\theta \Psi)^2 + F_\phi(\mu) \frac{(\partial_\phi \Psi)^2}{\sin^2 \theta} \right) \right]. \quad (451)$$

Projecting onto the  $\ell = 1, 2$  channels gives exact formulas linear in  $(A, B_0, B_2)$  and polynomial in  $(q, r)$ . The resulting system has exact real solutions, so the strict isotropic no-go is repaired by a single new local degree of freedom: an axisymmetric tangential wall-stress profile.

A natural Family-1-like branch is the one with  $q > 0$ ,  $r > 0$ , and  $\sigma(\mu) > 0$  on  $[-1, 1]$ . On that branch,

$$A \approx -0.281923219302567, \quad B_0 \approx 0.648914709228733, \quad B_2 \approx -1.085434603876673, \quad (452)$$

$$q \approx 2.370915717168574, \quad r \approx 2.758343474052255, \quad (453)$$

and the support residuals vanish at machine precision. In Legendre form,

$$B_{\text{tan}}(\mu) = \beta_0 + \beta_2 P_2(\mu), \quad \beta_0 = B_0 + \frac{1}{3}B_2 \approx 0.287103174603175, \quad \beta_2 = \frac{2}{3}B_2 \approx -0.723623069251115. \quad (454)$$

So the minimal successful PDE-side completion is an axisymmetric  $P_2$ -modulated tangential traction profile.

The most useful structural consequence is a universal monopole identity:

$$K_{00} = K_{1\perp} + \frac{1}{2}K_{10} - \frac{1}{10}K_{20} - \frac{1}{5}K_{21} - \frac{1}{5}K_{22}. \quad (455)$$

Once the  $\ell = 1, 2$  support targets are matched exactly, the raw boundary-layer monopole is fixed automatically:

$$K_{00}^{\text{BL}} = \frac{2}{7} + \frac{1}{2} \cdot \frac{1}{4} - \frac{1}{10} \cdot \frac{4}{9} - \frac{1}{5} \cdot \frac{2}{3} - \frac{1}{5} \cdot \frac{8}{3} = -\frac{757}{2520}. \quad (456)$$

Since the carried monopole target is  $K_{00}^{\text{target}} = 4/45$ , the separate monopole wall add-on is fixed to

$$\Delta K_{00}^{\text{mono}} = \frac{4}{45} - \left( -\frac{757}{2520} \right) = \frac{109}{280}. \quad (457)$$

So the support-sector repair does not eliminate the separate monopole channel; it predicts its exact required size.

## I.6 Support/source locking and the canonical scalar source vector

The Family-1 wall program becomes even tighter once the scalar source sector is included. With the exact carried base profile from [equation \(444\)](#), the exact source profile obeys

$$S(\mu) = 10 - \frac{63}{2} z_{\text{base}}(\mu). \quad (458)$$

So the source sector is not an independent fit. Once the base support profile is fixed, the source profile is fixed automatically.

Equivalently, write the axisymmetric quadratic base profile as

$$z_{\text{base}}(\mu) = a_0 + a_2 P_2(\mu). \quad (459)$$

Then the dipole support residues reconstruct it exactly:

$$a_0 = \frac{K_{10} + 2K_{1\perp}}{3}, \quad a_2 = \frac{5}{3}(K_{10} - K_{1\perp}). \quad (460)$$

If the scalar source is written as

$$S(\mu) = p_{\text{iso}} + q_{\text{ax}} P_2(\mu), \quad (461)$$

then the same constitutive law forces

$$p_{\text{iso}} = 10 - \frac{21}{2}(K_{10} + 2K_{1\perp}), \quad q_{\text{ax}} = -\frac{105}{2}(K_{10} - K_{1\perp}). \quad (462)$$

At the passed dipole support values

$$K_{1\perp} = \frac{2}{7}, \quad K_{10} = \frac{1}{4}, \quad (463)$$

this gives

$$p_{\text{iso}} = \frac{11}{8}, \quad q_{\text{ax}} = \frac{15}{8}, \quad (464)$$

exactly.

Using the canonical normalization convention of [appendices G](#) and [H](#), the scalar source vector is

$$J_0 = \frac{2}{\sqrt{5}} \left( p_{\text{iso}} + \frac{q_{\text{ax}}}{3} \right), \quad J_{20} = \frac{2}{3} q_{\text{ax}}. \quad (465)$$

Substituting [equation \(462\)](#) gives

$$J_0 = \frac{\sqrt{5}}{5} (20 - 56K_{10} - 7K_{1\perp}), \quad J_{20} = 35(K_{1\perp} - K_{10}). \quad (466)$$

so at the passed dipole support values,

$$J = \left( \frac{4}{\sqrt{5}}, \frac{5}{4} \right), \quad (467)$$

exactly. This is the same canonical scalar source vector that appears in [appendices G](#) and [H.1](#). The source sector used in the solved conservative 2PN cross block is therefore already fixed once the dipole support sector is fixed.

The full static local wall operator is correspondingly

$$\mathcal{O}_{\text{loc}} = z_{\text{base}}(\mu) + \frac{1}{2} \{-\Delta_S - 2, t(\mu)\}, \quad (468)$$

with exact channel values

$$K_{00}^{\text{raw}} = -\frac{757}{2520}, \quad K_{1\perp} = \frac{2}{7}, \quad K_{10} = \frac{1}{4}, \quad K_{20} = \frac{4}{9}, \quad K_{21} = \frac{2}{3}, \quad K_{22} = \frac{8}{3}. \quad (469)$$

So the minimal full static even-wall completion is simply

$$\mathcal{O}_{\text{full}} = \mathcal{O}_{\text{loc}} + \frac{109}{280} \mathbb{P}_{00}, \quad (470)$$

with  $\mathbb{P}_{00}$  the monopole projector.

## I.7 Monopole closure from the reduced geometry Hessian

The remaining static question is whether the monopole add-on 109/280 can itself be given a geometry-side origin. The answer is yes: it can be reinterpreted as a low-frequency geometry compressibility of the throat variables  $(a, L)$ .

Use the cylinder-like throat volume and area

$$V(a, L) = \frac{4\pi}{3}a^3L, \quad A(a, L) = 4\pi a^2L + \frac{8\pi}{3}a^3, \quad (471)$$

and the minimal curvature-completed geometry energy

$$E_{\text{geom}}(a, L) = P_{\text{vac}}V(a, L) + \sigma A(a, L) + \kappa_b \frac{a^2}{L}. \quad (472)$$

Treat the monopole port as a pressure-like source coupling through volume work,

$$\delta W = -p \delta V, \quad g = \nabla_{(a,L)} V. \quad (473)$$

If  $H_0$  denotes the Hessian of  $E_{\text{geom}}$  at the reference point  $(a_0, L_0) = (a_0, \Lambda a_0)$ , then integrating out  $(\delta a, \delta L)$  gives the exact reduced monopole coefficient

$$\Delta K_{00}^{\text{geom}} = \frac{g^T H_0^{-1} g}{V_0^2}. \quad (474)$$

So the old global monopole projector is reinterpreted as a genuine low-frequency geometry compressibility.

Write

$$\sigma = \frac{\Sigma}{a_0^3}, \quad P_{\text{vac}} = \rho \frac{\Sigma}{a_0^4}, \quad \kappa_b = \beta \frac{\Sigma}{a_0}. \quad (475)$$

For the baseline model with no curvature completion ( $\beta = 0$ ), the reduced Hessian is indefinite, so the literal  $PV + \sigma A$  geometry energy cannot by itself close the monopole channel. With the curvature completion, positivity requires

$$\beta > \beta_{\text{stab}}(\Lambda, \rho) \equiv \frac{\pi \Lambda^3 (\rho + 2)^2}{2\Lambda \rho + 3\Lambda + 2}, \quad (476)$$

and positivity of the monopole coefficient also requires

$$\beta > \beta_{\Delta}(\Lambda, \rho) \equiv \frac{\pi \Lambda (2\Lambda \rho + 5\Lambda - 2)}{4}. \quad (477)$$

Thus the geometry-side closure exists whenever

$$\beta > \max(\beta_{\text{stab}}, \beta_{\Delta}). \quad (478)$$

On the EM branch,

$$\Lambda_{\text{EM}} = \frac{\sqrt{2}\pi}{x_{01}} \approx 1.847486577120128, \quad x_{01} = 2.40482555769577 \dots, \quad (479)$$

a simple positive worked point is

$$\rho = \frac{1}{10}, \quad \beta = 12, \quad (480)$$

which lies safely inside the passive/positive-support region. Matching the exact target  $\Delta K_{00}^{\text{geom}} = 109/280$  then fixes the overall geometry stiffness scale to

$$\Sigma_* \approx 0.2076143291835488854. \quad (481)$$

At this point,

$$\Delta K_{00}^{\text{geom}} = \frac{109}{280} \quad (482)$$

at machine precision.

The resulting two-degree-of-freedom geometry sector is strongly dominated by one mode. At the worked point above, the mode-resolved contributions to  $\Delta K_{00}^{\text{geom}}$  are approximately

$$(0.00878310, 0.38050262), \quad (483)$$

which sum to  $109/280$ , with the larger mode carrying

$$97.743791\% \quad (484)$$

of the total static monopole response. This explains why the earlier single global monopole auxiliary was already the right effective description: it is a controlled reduction of the full  $(a, L)$  geometry sector, not a disconnected ad hoc insertion.

The cleanest static closure statement is therefore

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| local Family-1 support/source wall law + dominant-coupling reduction of the reduced geometry Hessian. |
|-------------------------------------------------------------------------------------------------------|

(485)

This closes the static Family-1 wall-stress program. The subsequent dynamic tasks – bulk inertia, sidewall and endcap soft-wall corrections, and the final coupled low-frequency response module – belong to the dynamic DtN scaffolding discussed in [appendix H](#).

## J Geometry-breathing and dynamic monopole closures

This appendix records the scalar channel that remained after the static support/source program of [appendix I](#). The logic is now fairly sharp. The local Family-1 wall law closes the full static  $\ell = 1, 2$  support/source sector, but it leaves one scalar remainder in the monopole channel. That remainder is not another arbitrary support coefficient. It is the low-frequency shadow of a genuine reduced geometry response in the throat variables  $(a, L)$ , and once a minimal dynamic reduction is written down it becomes an exact positive two-pole breathing susceptibility.

A notational warning is worth stating first. Two different scalar objects appear in the appendix program. One is the pure-potential geometry-closure deficit  $\Delta_{\text{geom}} = 281/80$  in the support-minus-closure EFT of [appendices G](#) and [H](#); that quantity belongs to the comparable-mass even cross operator. The other is the monopole wall add-on

$$\Delta K_{00} = \frac{109}{280}, \quad (486)$$

which is the missing scalar support needed to lift the local static wall law to the full passed monopole coefficient. The present appendix concerns *this second object*. It is the breathing/compressibility closure of the scalar monopole wall channel, not the pure- $U$  closure subtraction of the even cross EFT.

The appendix has four layers. First, we isolate the scalar static remainder left by the local wall law. Second, we show that the missing coefficient is generated exactly by a reduced geometry Hessian in  $(a, L)$ . Third, we derive the exact dynamic monopole susceptibility of that reduced system and its controlled one-pole low-frequency reduction. Fourth, we fold in the Family-1 bulk, sidewall, and endcap inertia completions to obtain an explicit wall-completed breathing branch.

### J.1 The scalar remainder left by the local static wall law

The static support/source program of [appendix I](#) compresses the full even wall sector into

$$[\text{exact local support/source constitutive law}] + \Delta K_{00} \mathbb{P}_{00}, \quad (487)$$

where  $\mathbb{P}_{00}$  is the monopole projector. The local part reproduces the non-monopole support and source data exactly, and the only remaining scalar gap is

$$\Delta K_{00} = \frac{109}{280}. \quad (488)$$

Equivalently, the carried local wall baseline in the monopole channel is

$$K_{00}^{\text{loc}} = -\frac{757}{2520}, \quad (489)$$

so the full passed monopole coefficient

$$K_{00} = \frac{4}{45} \quad (490)$$

is recovered as

$$K_{00} = K_{00}^{\text{loc}} + \Delta K_{00} = -\frac{757}{2520} + \frac{109}{280} = \frac{4}{45}. \quad (491)$$

Thus the static Family-1 wall program leaves exactly one unresolved scalar breathing coefficient and no unresolved  $\ell = 1, 2$  support coefficient.

This scalar remainder is what the older notes had treated as a “global monopole auxiliary.” The point of the present appendix is that the auxiliary is no longer merely a bookkeeping insertion. It can be derived as a reduced compressibility of the throat geometry itself.

### J.2 Static geometry-Hessian closure in $(a, L)$

Use the cylinder-like throat geometry

$$V(a, L) = \frac{4\pi}{3}a^3L, \quad A(a, L) = 4\pi a^2L + \frac{8\pi}{3}a^3, \quad (492)$$

and the minimal curvature-completed geometry energy

$$E_{\text{geom}}(a, L) = P_{\text{vac}}V(a, L) + \sigma A(a, L) + \kappa_b \frac{a^2}{L}. \quad (493)$$

The curvature term is the smallest explicit completion that repairs the known failure of literal  $P_{\text{vac}}V + \sigma A$  geometry in the monopole channel. Write the reference configuration as

$$(a_0, L_0) = (a_0, \Lambda a_0), \quad V_0 = V(a_0, L_0) = \frac{4\pi}{3}\Lambda a_0^4, \quad (494)$$

and let  $H_0$  be the Hessian of  $E_{\text{geom}}$  at that point. The uniform monopole port couples through volume work,

$$\delta W = -p\delta V, \quad g \equiv \nabla_{(a,L)}V|_{(a_0,L_0)}. \quad (495)$$

Integrating out the reduced geometry variables then gives the exact static compressibility coefficient

$$\boxed{\Delta K_{00}^{\text{geom}} = \frac{g^T H_0^{-1} g}{V_0^2}}. \quad (496)$$

So the old monopole auxiliary is reinterpreted as a genuine reduced geometry-side compressibility.

It is convenient to introduce the scaled parameters

$$\sigma = \frac{\Sigma}{a_0^3}, \quad P_{\text{vac}} = \rho \frac{\Sigma}{a_0^4}, \quad \kappa_b = \beta \frac{\Sigma}{a_0}, \quad (497)$$

for which the exact reduced coefficient becomes

$$\Delta K_{00}^{\text{geom}} = \frac{2\pi\Lambda^2\rho + 5\pi\Lambda^2 - 2\pi\Lambda - 4\beta}{2\pi\Sigma(\pi\Lambda^3\rho^2 + 4\pi\Lambda^3\rho + 4\pi\Lambda^3 - 2\Lambda\beta\rho - 3\Lambda\beta - 2\beta)}. \quad (498)$$

If the curvature completion is removed,  $\beta = 0$ , then the baseline Hessian is

$$H_{\text{base}} = \begin{pmatrix} 8\pi(\Lambda\rho + \Lambda + 2) & 4\pi(\rho + 2) \\ 4\pi(\rho + 2) & 0 \end{pmatrix}, \quad (499)$$

with

$$\det H_{\text{base}} = -16\pi^2(\rho + 2)^2 < 0. \quad (500)$$

So literal  $P_{\text{vac}}V + \sigma A$  geometry is not a passive 2-DOF Hessian and cannot by itself close the monopole channel. This no-go is the static reason a curvature or soft-wall completion is necessary.

For the curvature-completed model, positivity and positive monopole support require

$$\beta > 0, \quad \beta > \beta_{\text{stab}}(\Lambda, \rho) \equiv \frac{\pi\Lambda^3(\rho + 2)^2}{2\Lambda\rho + 3\Lambda + 2}, \quad (501)$$

and

$$\beta > \beta_{\Delta}(\Lambda, \rho) \equiv \frac{\pi\Lambda(2\Lambda\rho + 5\Lambda - 2)}{4}. \quad (502)$$

A representative EM-branch worked point is

$$\Lambda_{\text{EM}} = \frac{\sqrt{2}\pi}{x_{01}} \approx 1.847486577120128, \quad \rho = \frac{1}{10}, \quad \beta = 12, \quad (503)$$

with

$$\beta_{\text{stab}}(\Lambda_{\text{EM}}, \rho) \approx 11.0420171, \quad \beta_{\Delta}(\Lambda_{\text{EM}}, \rho) \approx 11.0377513, \quad (504)$$

so the point lies inside the passive positive-support region. Matching the exact target

$$\Delta K_{00}^{\text{geom}} = \frac{109}{280} \quad (505)$$

then fixes the overall geometry stiffness scale to

$$\Sigma_* \approx 0.2076143291835488854. \quad (506)$$

At this point the normalized static coupling vector

$$\bar{g} \equiv \frac{\nabla V_0}{V_0} = \left(3, \frac{1}{\Lambda_{\text{EM}}}\right) \quad (507)$$

feeds dominantly into one Hessian eigenmode, which is why the historical single breathing auxiliary was already a good static reduction.

### J.3 Exact dynamic monopole susceptibility and the two-pole breathing branch

To dynamize the static compressibility, use the dimensionless geometry variables

$$\mathbf{Q}_{\text{geom}} = \left( \frac{\delta a}{a_0}, \frac{\delta L}{a_0} \right) \quad (508)$$

and the minimal affine displacement field inside the cylinder-like throat,

$$\boldsymbol{\xi}_{\perp} = \frac{\delta a}{a_0} \mathbf{r}, \quad \xi_w = \frac{\delta L}{L_0} w. \quad (509)$$

For a uniform bulk inertia density  $\rho_{\text{eff}}$ , the reduced kinetic energy takes exactly the form

$$T^{(2)} = \frac{1}{2} \rho_{\text{eff}} V_0 a_0^2 \mathbf{Q}_{\text{geom}}^T \widehat{M} \mathbf{Q}_{\text{geom}}, \quad \widehat{M} = \begin{pmatrix} \frac{3}{5} & 0 \\ 0 & \frac{1}{12} \end{pmatrix}. \quad (510)$$

The corresponding quadratic energy is

$$E^{(2)} = \frac{1}{2} \Sigma \mathbf{Q}_{\text{geom}}^T \widehat{H} \mathbf{Q}_{\text{geom}}, \quad \widehat{H} \equiv \frac{a_0^2}{\Sigma} H_0. \quad (511)$$

Define the scaled frequency variable

$$s \equiv \omega^2 \frac{\rho_{\text{eff}} V_0 a_0^2}{\Sigma}. \quad (512)$$

Then the exact reduced geometry response is

$$Y_{\text{geom}}(s) = \frac{1}{\Sigma} \bar{g}^T (\widehat{H} - s \widehat{M})^{-1} \bar{g}. \quad (513)$$

The raw monopole wall channel is therefore

$$K_{00}^{\text{raw}}(s) = -\frac{757}{2520} + Y_{\text{geom}}(s). \quad (514)$$

At zero frequency, [equation \(514\)](#) reduces to the static closure already discussed above.

Because  $\widehat{H}$  and  $\widehat{M}$  are positive on the worked branch,  $Y_{\text{geom}}(s)$  is a two-pole Stieltjes function. Equivalently, if

$$\widehat{H} v_i = \lambda_i \widehat{M} v_i, \quad i \in \{-, +\}, \quad (515)$$

then

$$Y_{\text{geom}}(s) = \frac{R_-}{1 - s/\lambda_-} + \frac{R_+}{1 - s/\lambda_+}, \quad R_{\pm} > 0, \quad R_- + R_+ = \frac{109}{280}. \quad (516)$$

So the monopole wall completion is not a generic ad hoc pole fit. It is the exact dynamic response of a reduced  $(a, L)$  geometry sector.

At the EM worked point of [equations \(503\)](#) and [\(506\)](#), one finds

$$\widehat{H} \approx \begin{pmatrix} 114.33174685 & 19.35786733 \\ 19.35786733 & 3.80598758 \end{pmatrix}, \quad (517)$$

with generalized pole parameters

$$\lambda_- \approx 5.23115613, \quad \lambda_+ \approx 230.99360624, \quad (518)$$

and positive residues

$$R_- \approx 0.00327376153, \quad R_+ \approx 0.38601195275, \quad R_- + R_+ = \frac{109}{280}. \quad (519)$$

The physical pole frequencies are

$$\Omega_{\pm}^2 = \frac{\Sigma_*}{\rho_{\text{eff}} V_0 a_0^2} \lambda_{\pm}. \quad (520)$$

Thus the exact geometry breathing response is an explicit positive two-pole dynamic closure whose static limit reproduces the required monopole coefficient.

#### J.4 Controlled one-pole reduction and the old breathing auxiliary

The historical single breathing auxiliary is recovered by a low-frequency Padé reduction of the exact two-pole function. Expanding [equation \(513\)](#) at small  $s$  gives

$$Y_{\text{geom}}(s) = \Delta_0 + \Delta_2 s + O(s^2), \quad (521)$$

with

$$\Delta_0 = \frac{1}{\Sigma} \bar{g}^T \hat{H}^{-1} \bar{g}, \quad \Delta_2 = \frac{1}{\Sigma} \bar{g}^T \hat{H}^{-1} \hat{M} \hat{H}^{-1} \bar{g}. \quad (522)$$

The associated  $[1/1]$  Padé reduction is

$$\boxed{Y_{\text{geom}}(s) \approx \frac{\Delta_0}{1 - s/\lambda_{\text{eff}}}, \quad \lambda_{\text{eff}} = \frac{\Delta_0}{\Delta_2}.} \quad (523)$$

Since  $\Delta_0 = 109/280$ , this may be written equivalently as

$$Y_{\text{mono}}^{\text{eff}}(s) = \frac{109/280}{1 - s/\lambda_{\text{eff}}}. \quad (524)$$

At the EM affine worked point,

$$\lambda_{\text{eff}} \approx 169.48205088. \quad (525)$$

On the natural low-frequency band

$$0 \leq s \leq 0.1\lambda_{-}, \quad (526)$$

the maximum relative error of the one-pole reduction against the exact two-pole response is only

$$\max |\delta Y/Y| \approx 8.87 \times 10^{-5}. \quad (527)$$

So the old global breathing auxiliary is a controlled low-frequency reduction of the exact reduced geometry dynamics, not a separate phenomenological invention.

#### J.5 Family-1 bulk inertia and the $n = 5$ Thomas–Fermi branch

The affine model above leaves the overall inertia scale  $\rho_{\text{eff}}$  open. The next step is to derive that scale from the Family-1 parent PDE on the filled-to-endcap Thomas–Fermi branch. Use the harmonic interior trap

$$V_{\text{in}}(w) = \frac{1}{2} m_{\psi} \Omega_{\text{in}}^2 w^2, \quad (528)$$

with frozen barotrope

$$P(\rho) = K_{\text{EOS}} \rho^n, \quad h(\rho) = \frac{n K_{\text{EOS}}}{n-1} \rho^{n-1}. \quad (529)$$

In the Thomas–Fermi/hydrostatic limit,

$$h(\rho) + \frac{1}{2} m_{\psi} \Omega_{\text{in}}^2 w^2 = \mu_{\text{TF}}. \quad (530)$$

On the filled-to-endcap branch,

$$\mu_{\text{TF}} = \frac{1}{2} m_{\psi} \Omega_{\text{in}}^2 \left( \frac{L_0}{2} \right)^2, \quad (531)$$

so the exact axial density profile is

$$\rho_0(w) = \rho_c \left( 1 - \frac{4w^2}{L_0^2} \right)^{1/(n-1)}, \quad |w| \leq \frac{L_0}{2}, \quad (532)$$

with central density

$$\rho_c = \left[ \frac{(n-1)m_\psi \Omega_{\text{in}}^2 L_0^2}{8nK_{\text{EOS}}} \right]^{1/(n-1)}. \quad (533)$$

Define

$$c_0(n) = \frac{1}{2} \int_{-1}^1 (1-u^2)^{1/(n-1)} du. \quad (534)$$

Then the effective bulk inertia density entering the monopole breathing channel is

$$\boxed{\rho_{\text{eff}}^{\text{TF}}(n) = c_0(n)\rho_c}, \quad (535)$$

with exact closed form

$$\boxed{c_0(n) = \frac{\sqrt{\pi} \Gamma\left(1 + \frac{1}{n-1}\right)}{2 \Gamma\left(\frac{3}{2} + \frac{1}{n-1}\right)}}. \quad (536)$$

For the frozen  $n = 5$  branch,

$$\rho_{\text{eff}}^{\text{TF}}(5) = \frac{\sqrt{\pi} \Gamma(1/4)}{6\Gamma(3/4)} \left( \frac{m_\psi \Omega_{\text{in}}^2 L_0^2}{10K_{\text{EOS}}} \right)^{1/4}, \quad (537)$$

so the previously free bulk inertia scale is now tied directly to the same Family-1 microphysics that defines the parent throat branch.

The same reduction also renormalizes the axial inertia moment. Define

$$c_2(n) = \frac{1}{2} \int_{-1}^1 u^2 (1-u^2)^{1/(n-1)} du, \quad (538)$$

for which

$$\frac{c_2(n)}{c_0(n)} = \frac{n-1}{3n-1}. \quad (539)$$

Therefore

$$\widehat{M}_{LL}^{\text{TF}}(n) = \frac{c_2}{4c_0} = \frac{n-1}{4(3n-1)}. \quad (540)$$

At  $n = 5$ ,

$$\boxed{\widehat{M}_{\text{TF}}(5) = \begin{pmatrix} \frac{3}{5} & 0 \\ 0 & \frac{1}{14} \end{pmatrix}}. \quad (541)$$

So the Family-1  $n = 5$  branch renormalizes the axial breathing moment from the uniform-slice value  $1/12$  to the EOS-sensitive value  $1/14$ .

With this replacement, the exact reduced response becomes

$$Y_{\text{geom}}^{\text{TF}}(s) = \frac{1}{\Sigma} \bar{g}^T (\widehat{H} - s \widehat{M}_{\text{TF}}(5))^{-1} \bar{g}, \quad s = \omega^2 \frac{\rho_{\text{eff}}^{\text{TF}}(5) V_0 a_0^2}{\Sigma}. \quad (542)$$

At the same EM worked point, one finds

$$\lambda_- \approx 5.92556258, \quad \lambda_+ \approx 237.91117494, \quad (543)$$

with residues

$$R_- \approx 0.00262800, \quad R_+ \approx 0.38665771, \quad R_- + R_+ = \frac{109}{280}, \quad (544)$$

and effective Padé pole

$$\lambda_{\text{eff}}^{\text{TF}} \approx 188.17695898. \quad (545)$$

The one-pole reduction remains extremely accurate,

$$\max \text{ rel.err.} \approx 7.10 \times 10^{-5} \quad \text{for } 0 \leq s \leq 0.1\lambda_-. \quad (546)$$

So the bulk part of the monopole dynamics is no longer phenomenological.

## J.6 Wall-completed dynamic monopole closure

Beyond the Thomas–Fermi bulk, the remaining dynamic ingredients are genuine sidewall and endcap inertia corrections. The static closure  $\Delta K_{00} = 109/280$  is unaffected; only the inertia sector is renormalized. That is why the monopole wall response remains of the same structural form,

$$K_{00}(s) = -\frac{757}{2520} + \frac{R_-}{1 - s/\lambda_-} + \frac{R_+}{1 - s/\lambda_+}, \quad R_- + R_+ = \frac{109}{280}, \quad (547)$$

throughout the dynamic completion chain.

The separated-order Family-1 sidewall and endcap reductions of [appendix I](#) already gave a near-final composite wall completion. But the most useful summary datum is the direct coupled full-profile branch, which feeds both effects into the same reduced breathing operator. On the balanced thin-layer reference branch one finds

$$R_{\text{mass}}^{\text{full}} \approx 0.886313972989725, \quad \widehat{M}_{aa}^{\text{full}} \approx 0.563114968953987, \quad \widehat{M}_{LL}^{\text{full}} \approx 0.065829228119349. \quad (548)$$

Using these values in the same geometry Hessian at the EM worked point gives

$$\lambda_- \approx 6.405572392138922, \quad \lambda_+ \approx 254.444968136936126, \quad (549)$$

with positive residues

$$R_- \approx 0.002552474771738, \quad R_+ \approx 0.386733239513976, \quad R_- + R_+ = \frac{109}{280}. \quad (550)$$

Therefore

$$K_{00}(0) = \frac{4}{45} \quad (551)$$

exactly, while the physical pole ratios relative to the sharp-wall Thomas–Fermi baseline are shifted to

$$\frac{\Omega_-^2}{\Omega_{-, \text{TF}}^2} \approx 1.219665554412172, \quad \frac{\Omega_+^2}{\Omega_{+, \text{TF}}^2} \approx 1.206678094538845. \quad (552)$$

The corresponding one-pole Padé reduction is still excellent:

$$\lambda_{\text{eff}} \approx 202.923516367519028, \quad \max \text{ rel.err.} \approx 6.89 \times 10^{-5} \quad \text{on } 0 \leq s \leq 0.1\lambda_-. \quad (553)$$

So even after the wall completion is made explicit, the historical single breathing mode remains a highly accurate low-frequency reduction of the exact two-pole geometry channel.

## J.7 Interpretation and status

The constructive status of the monopole channel is now much cleaner than it was at the start of the 2PN program. There are really three nested statements.

First, at strictly static level, the missing monopole coefficient is not an arbitrary projector but the reduced compressibility of a curvature-completed throat geometry. That is the content of [equations \(496\)](#) and [\(505\)](#).

Second, once the geometry variables  $(a, L)$  are given a reduced inertia metric, the same scalar channel becomes an exact positive two-pole breathing susceptibility. That is the content of [equations \(514\)](#) and [\(516\)](#).

Third, the Family-1 parent PDE and wall program now determine not only the static closure but most of the dynamic normalization as well: the bulk inertia scale  $\rho_{\text{eff}}^{\text{TF}}(5)$ , the EOS-sensitive axial moment  $1/14$ , and the first explicit sidewall/endcap/full-profile corrections. So the dynamic monopole branch is no longer a placeholder pole. It is a concrete low-frequency channel with a known exact static residue, a known reduced geometry origin, and a highly accurate one-pole reduction.

What still remains open is narrower than before. This appendix does *not* derive the non-monopole pole scales of the odd and even  $\ell = 1, 2$  channels, and it does not claim a fully analytic microscopic derivation of the local wall profiles beyond the reduced Family-1 completion. But as far as the scalar breathing branch is concerned, the constructive program has already reached a near-final form:

$$\boxed{\text{exact static monopole closure} + \text{exact reduced two-pole breathing response} + \text{explicit Family-1 bulk and wall}} \quad (554)$$

That is the sense in which the geometry-breathing and dynamic monopole closures are now substantially in hand.

**Claim class.** *Controlled reduction* for the reduced geometry-side model, affine breathing ansatz, and Family-1 wall reductions. **Exact** for the static compressibility formula once the reduced model is declared, for the exact two-pole response of the resulting  $(a, L)$  system, and for the Thomas–Fermi inertia integrals. **Protocol closure** for the specific thin-wall and coupled full-profile wall completion used to package the final breathing branch..

## K Final coupled Family-1 response module

This appendix is the constructive end-point of the throat-side program developed in [appendices G to J](#). Its role is not to add a new ingredient to the conservative 2PN proof, but to show that the previously separated constructive pieces can now be packaged into one coherent low-frequency Family-1 response module. There are two parts to that statement.

First, the exact zero-frequency odd and even channel data already derived in the earlier appendices remain unchanged: the odd dipole wake, the canonical  $P_0 \oplus P_2$  even support/source sector, and the pure-potential geometry-closure channel still reproduce the solved added comparable-mass conservative 2PN cross block exactly. Second, the monopole breathing channel can now be fed not merely with a separated sidewall-plus-endcap inertia estimate, but with a direct *coupled* Family-1 full-profile wall completion. The resulting monopole susceptibility still closes the same static scalar target, but now does so with the best current dynamic normalization of the Family-1 program.

This is therefore the right place to summarize what the constructive appendix program has actually achieved. At zero frequency, the conservative added 2PN cross block is fixed exactly. In the monopole wall channel, the scalar breathing response is fixed to an exact positive low-frequency two-pole form whose inertia data now come from the direct coupled Family-1 wall profile. What remains open after this point is narrower: the genuinely non-monopole dynamic pole scales and a more microscopic derivation of the wall profiles themselves.

### K.1 Static wall law and source profile remain unchanged

The direct coupled wall completion does *not* change the static support/source law already derived in [appendix I](#). The carried local wall profiles remain

$$z_{\text{base}}(\mu) = \frac{17}{56} - \frac{5}{56}\mu^2, \quad t(\mu) = \frac{593}{672} - \frac{1553}{672}\mu^2 + \frac{7}{8}\mu^4, \quad (555)$$

with exact scalar source profile

$$S(\mu) = 10 - \frac{63}{2} z_{\text{base}}(\mu) = \frac{7}{16} + \frac{45}{16}\mu^2. \quad (556)$$

Equivalently, the local static even-wall law is still the same operator already recorded in [appendices I and J](#):

$$\mathcal{O}_{\text{loc}} = z_{\text{base}}(\mu) + \frac{1}{2} \left\{ -\Delta_S - 2, t(\mu) \right\}. \quad (557)$$

So the coupled full-profile step is purely a *dynamic* refinement of the breathing sector. It does not alter the previously solved static support/source data.

## K.2 Exact zero-frequency odd and even channel packaging

The zero-frequency constructive module also remains unchanged. Using the carried odd dipole wake from the 1PN program and the canonical  $P_0 \oplus P_2$  even ports of [appendices G](#) and [H](#), the added conservative comparable-mass 2PN cross block is still reproduced exactly by

$$L_{\text{odd}}^{\text{add}} = \frac{1}{2}(v_A^2 + v_B^2)L_{\text{1PN}}^{\text{wake}} - \frac{15}{4}(U_A + U_B)(\mathbf{v}_A \cdot \mathbf{v}_B), \quad (558)$$

combined with the even block

$$L_{\text{even}}^{\text{add}} = \Pi_A \cdot \Pi_B + U_A J \cdot \Pi_B + U_B J \cdot \Pi_A + (J \cdot J - \Delta_{\text{geom}})U_A U_B, \quad (559)$$

where the six-component real even-port basis is the one already defined in [appendices G](#) and [H](#) and the source vector is

$$J = \left( \frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right), \quad \Delta_{\text{geom}} = \frac{281}{80}. \quad (560)$$

Again,  $J$  denotes the appendix-side throat-response source-weight vector, not a Maxwell current. Thus

$$J \cdot J = \left( \frac{4}{\sqrt{5}} \right)^2 + \left( \frac{5}{4} \right)^2 = \frac{381}{80}, \quad (561)$$

so the support part alone would overproduce the static coefficient, and the direct geometry-closure subtraction  $\Delta_{\text{geom}} = 281/80$  lowers it to the exact solved pure-static added-cross value

$$\frac{381}{80} - \frac{281}{80} = \frac{5}{4}. \quad (562)$$

In other words, the constructive residual against the exact solved added comparable-mass 2PN target still vanishes identically at zero frequency. Nothing in the coupled Family-1 wall completion changes that algebraic closure.

## K.3 Direct coupled full-profile Family-1 wall completion

What *does* change at the present stage is the monopole breathing normalization. The earlier wall-completed breathing branch could be assembled in separated sidewall and endcap order. The present step goes beyond that approximation and evaluates the full coupled Family-1 Thomas–Fermi wall profile directly in the throat variables  $(x, y)$ .

On the balanced thin-layer-consistent reference branch,

$$\alpha_r = 10, \quad \varepsilon_r = 0.05, \quad p_r = 2, \quad (563)$$

$$\chi_{\text{cap}} = 4, \quad \varepsilon_z = 0.05, \quad \alpha_{\text{cap}} = 4\varepsilon_z \chi_{\text{cap}} = 0.8, \quad p_z = 2, \quad (564)$$

use the full coupled profile

$$\tilde{\rho}(x, y) = \left[ 1 - y^2 - \alpha_r S\left(\frac{x-1}{\varepsilon_r}\right)^{p_r} - \alpha_{\text{cap}} S\left(\frac{y-1}{2\varepsilon_z}\right)^{p_z} \right]_+^{1/4}, \quad (565)$$

for  $0 \leq x \leq 1$ ,  $0 \leq y \leq 1$ , with the usual symmetry factor 2 in the axial variable  $y$  and with the soft-wall switch function

$$S(\xi) = \frac{1 + \tanh \xi}{2}. \quad (566)$$

The exact coupled mass and inertia integrals are then

$$I_2 = 2 \int_0^1 \int_0^1 x^2 \tilde{\rho} dx dy, \quad I_4 = 2 \int_0^1 \int_0^1 x^4 \tilde{\rho} dx dy, \quad I_w = 2 \int_0^1 \int_0^1 \frac{y^2}{4} x^2 \tilde{\rho} dx dy. \quad (567)$$

Normalizing against the sharp-wall filled-to-endcap Thomas–Fermi baseline gives the final coupled wall data

$$R_{\text{mass}}^{\text{full}} \approx 0.886313972989725, \quad \widehat{M}_{aa}^{\text{full}} \approx 0.563114968953987, \quad \widehat{M}_{LL}^{\text{full}} \approx 0.065829228119349. \quad (568)$$

These are exactly the values carried into the final monopole breathing closure.

#### K.4 Separated-order approximation was already very accurate

It is useful to record one quantitative closure check. The earlier separated-order completion had already given

$$R_{\text{mass}}^{\text{sep}} \approx 0.8846236634, \quad \widehat{M}_{aa}^{\text{sep}} \approx 0.5623810783, \quad \widehat{M}_{LL}^{\text{sep}} \approx 0.0671965962. \quad (569)$$

Relative to these values, the direct coupled full-profile corrections are only

$$\frac{R_{\text{mass}}^{\text{full}} - R_{\text{mass}}^{\text{sep}}}{R_{\text{mass}}^{\text{sep}}} \approx +1.91 \times 10^{-3}, \quad \frac{\widehat{M}_{aa}^{\text{full}} - \widehat{M}_{aa}^{\text{sep}}}{\widehat{M}_{aa}^{\text{sep}}} \approx +1.30 \times 10^{-3}, \quad (570)$$

$$\frac{\widehat{M}_{LL}^{\text{full}} - \widehat{M}_{LL}^{\text{sep}}}{\widehat{M}_{LL}^{\text{sep}}} \approx -2.03 \times 10^{-2}. \quad (571)$$

So the earlier separated reduction was already excellent for the overall mass scale and radial inertia, and still reasonably good for the axial inertia. That matters because it shows that the constructive program had essentially converged before the final coupled check was performed. The present coupled step is therefore a genuine tightening of the result, not a rescue of an unstable earlier approximation.

#### K.5 Final coupled monopole breathing response

Feeding the direct coupled wall data [equation \(568\)](#) into the same reduced geometry Hessian used in [appendix J](#) gives the final coupled monopole poles

$$\lambda_- \approx 6.405572392138922, \quad \lambda_+ \approx 254.444968136936126, \quad (572)$$

with positive residues

$$R_- \approx 0.002552474771738, \quad R_+ \approx 0.386733239513976, \quad R_- + R_+ = \frac{109}{280}. \quad (573)$$

So the final monopole wall channel is

$$K_{00}(s) = -\frac{757}{2520} + \frac{R_-}{1 - s/\lambda_-} + \frac{R_+}{1 - s/\lambda_+}. \quad (574)$$

At zero frequency this reproduces the required static target exactly:

$$K_{00}(0) = \frac{4}{45}. \quad (575)$$

Relative to the sharp-wall Thomas–Fermi baseline, the physical monopole pole ratios are

$$\frac{\Omega_-^2}{\Omega_{-, \text{TF}}^2} \approx 1.219665554412172, \quad \frac{\Omega_+^2}{\Omega_{+, \text{TF}}^2} \approx 1.206678094538845. \quad (576)$$

The corresponding one-pole Padé reduction remains extremely accurate:

$$\lambda_{\text{eff}} \approx 202.923516367519028, \quad \max \text{ rel.err.} \approx 6.89 \times 10^{-5} \quad \text{on } 0 \leq s \leq 0.1\lambda_-. \quad (577)$$

So the historical single breathing auxiliary still survives as a very controlled low-frequency reduction of the exact coupled two-pole monopole channel.

## K.6 Single-module summary and remaining open items

The cleanest summary is now channel-theoretic rather than coefficient-theoretic. At conservative 2PN order the constructive throat program has effectively reduced to three coupled pieces:

$$\boxed{\text{carried odd dipole wake} \oplus \text{exact } P_0 \oplus P_2 \text{ even support/source sector} \oplus \text{coupled two-pole monopole breathing}} \quad (578)$$

The first two pieces reproduce the solved added comparable-mass conservative 2PN cross block exactly at zero frequency. The third piece closes the scalar monopole wall response with the best current dynamic Family-1 normalization. Together they form the final coupled Family-1 response module in the practical sense relevant for the present paper.

It is also useful to state explicitly what remains open after this packaging. Two unresolved items survive, and both are narrower than the original 2PN problem:

1. the non-monopole dynamic pole scales of the odd and even  $\ell = 1, 2$  channels, which are true inner-throat PDE / DtN observables rather than conservative zero-frequency necessities; and
2. a fully analytic derivation of the local wall profiles from the microscopic soft-wall traction PDE, rather than from the reduced matching and full-profile evaluation used here.

Neither of those open items blocks the conservative 2PN derivation itself. What they block is only the next stage: a fully dynamical first-principles throat PDE realization of the already-solved low-frequency operator.

So the practical finish-line statement of the constructive appendix program is now simply this:

$$\boxed{\text{the conservative 2PN Family-1 throat-response module is effectively in hand.}} \quad (579)$$

That statement should be read in the same disciplined sense as the rest of the paper: not as a claim that every dynamic PDE detail has been solved, but as the statement that the full zero-frequency cross operator is closed exactly and that the scalar breathing branch now has a direct coupled Family-1 wall completion.

**Claim class.** *Controlled reduction* for the use of the specific balanced thin-layer Family-1 wall profile and the reduced geometry breathing model. **Exact** for the unchanged zero-frequency odd/even algebraic reconstruction and for the coupled full-profile integrals once that wall branch is declared. **Protocol closure** for the particular soft-wall / Thomas–Fermi reference branch used to package the final dynamic monopole normalization..

Table 3: Verification archive used for the conservative two-body 2PN paper.

| Artifact                                                                                                | Role in the derivation                                                                                                                                                                                                                                                                                                                                                                                                                     | Reported summary                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4d_gravity_1pn_master_harness.wl<br>vector_wake_rebuild.wl<br>4d_full_1pn_derivation_with_vectorwake.wl | Inherited 1PN referee suite. Freezes the lower-order backbone used unchanged in the present paper: Newtonian mass-dressing coefficient $\kappa_\rho = 1$ , optical matching $n = 5$ , $\kappa_{\text{add}} = 1/2$ , the adiabatic $\kappa_{\text{PV}} = 3/2$ closure, $\beta_{\text{1PN}} = 3$ , the parity-even wake coefficients, and the exact conservative EIH equality.                                                               | Reference transcripts report <b>PASS count = 60, FAIL count = 0</b> for the broad master harness; the wake module recovers $a_H = 0$ , $a_L = \pm\sqrt{3}/2$ , $K_{\text{vec}} = 2/\pi^2$ , $C_{\parallel} = -7/2$ , $C_L = -1/2$ ; and the full 1PN assembly script reports <b>Passes: 37, Fails: 0, Skips: 0</b> . |
| 4d_gravity_2pn_master_harness.wl                                                                        | Compact executable 2PN referee harness. Re-runs the carried 1PN regressions; verifies the exact $n = 5$ Bernoulli continuation; checks the minimal self-sector exactification, the quadratic local-mass-scaling static hierarchy, the generic cubic adiabatic-elimination template, the self/static test-mass reduction, the isotropic Schwarzschild target coefficients, and the universal finite-size gate.                              | Reference transcript reports <b>Passes: 42, Fails: 0, Skips: 0</b> , with an overall all-pass summary.                                                                                                                                                                                                               |
| 4d_2pn_full_verification_harness.wl<br>wl_notes.txt<br>4d_2pn*_prelim.wl                                | Manifest-driven 2PN symbolic derivation suite for the exact comparable-mass ADM lift and for the appendix-side constructive throat-response program. Executes the focused ledger scripts that record the perturbative Legendre identity used for the lift, the compact residual basis, the unique solved coefficients $q_i$ , $t_i$ , $s_1$ , the zero-residual full ADM match, and the exact zero-frequency throat-module reconstruction. | Current suite run passes every manifest entry with no detected FAIL lines. This layer complements the compact master harness: each derivation script remains individually readable, while the suite runner executes the full symbolic ledger in a reproducible order.                                                |

Table 4: Core notation used repeatedly in the main text and appendices.

| Symbol                                                                                        | Meaning                                                                                                                      |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $X^M = (t, x, y, z, w) = (t, \mathbf{x}, w)$                                                  | parent spacetime coordinates                                                                                                 |
| $\mathbf{X} = (x, y, z, w) \in \mathbb{R}^4$                                                  | bulk spatial position                                                                                                        |
| $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$                                                     | brane spatial position                                                                                                       |
| $m_A, m_B$                                                                                    | two-body masses                                                                                                              |
| $\mathbf{x}_A, \mathbf{x}_B$                                                                  | body positions in a generic frame                                                                                            |
| $\mathbf{v}_A, \mathbf{v}_B$                                                                  | coordinate-time velocities, $\mathbf{v}_A = \dot{\mathbf{x}}_A$ , $\mathbf{v}_B = \dot{\mathbf{x}}_B$                        |
| $\mathbf{r} = \mathbf{x}_A - \mathbf{x}_B$ , $r =  \mathbf{r} $ , $\mathbf{n} = \mathbf{r}/r$ | pair separation vector, magnitude, and unit direction                                                                        |
| $v_A^2, v_B^2, v_{AB}, v_{An}, v_{Bn}$                                                        | standard velocity invariants used in the generic-frame 2PN basis                                                             |
| $U_A = Gm_B/r$ , $U_B = Gm_A/r$                                                               | local Newtonian pair potentials                                                                                              |
| $U = GM/r$ , $u = U/c^2$                                                                      | one-body Newtonian potential and its dimensionless PN variable                                                               |
| $L_0, L_1, L_2$                                                                               | Newtonian, 1PN, and 2PN Lagrangian blocks in $L_{\text{cons}} = L_0 + c^{-2}L_1 + c^{-4}L_2 + O(c^{-6})$                     |
| $H_0, H_1, H_2$                                                                               | corresponding Hamiltonian blocks after perturbative Legendre transform                                                       |
| ADM, EIH, Schw, DtN                                                                           | standard abbreviations used throughout the paper                                                                             |
| $C_s(u), G(u)$                                                                                | normalized low-frequency one-body DtN invariants                                                                             |
| $q_i, t_i, s_1$                                                                               | solved ADM-lift coefficients of the added comparable-mass 2PN cross block; unrelated to electric charge                      |
| $R_{1\perp}, R_{10}, R_0, R_{20}, R_{21}, R_{22}, J, \Delta_{\text{geom}}$                    | appendix-side zero-frequency throat-response data; $J$ here is a throat-response source-weight vector, not a Maxwell current |